Coronal Persistence, Chromospheric Evolution: A Multi-Architecture Machine Learning Analysis of Pre-Flare Solar Spectra



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A dissertation submitted for the degree of
BSc (Hons) Computer Science with Industrial Placement

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In collaboration with the European Space Agency

May 2026

Declaration

I, Despina Ioannide, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the work.

Word count: 16,191

Abstract

Solar flares can disrupt satellites, GPS, and power grids, but operational forecasters built on photospheric magnetograms give limited warning in the final hours before onset. This dissertation applies six classifier families (logistic regression, XGBoost, MLP, autoencoder, 1D-CNN, and Siamese metric learning) to 25 million ultraviolet spectra from two space-based solar spectrometers, *Hinode*/EIS (coronal) and IRIS (chromospheric), under strict observation-disjoint evaluation. The central finding is that the two atmospheric layers encode fundamentally different kinds of predictive information: the corona encodes *whether* an active region is flare-productive, while the chromosphere heterogeneously encodes *when*.

On 2.5 million EIS spectra, all six classifiers reach ROC-AUC ≥ 0.93 (XGBoost and MLP at 0.996), and every one shows a flat temporal profile across 0–180 min pre-flare. Convergence across structurally distinct architectures indicates a persistent state encoded in low-order spectral statistics rather than a temporal precursor. A 9-feature kurtosis representation reaches ROC-AUC = 0.959, outperforming an optimal PCA-9 projection (0.924) by 3.5 ROC-AUC points: domain-informed inductive bias beats data-driven dimensionality reduction at matched feature count.

On 22.6 million IRIS spectra, the temporal pattern is event-dependent: of 25 events satisfying the strict $\geq$50-per-class within-AR filter, 8 show strong precursors (AUC > 0.80). A spectral transformer reaches AUC = 0.889 against quiet-Sun (a cross-AR task, not directly comparable). Optimal transport on the spectral distributions reveals intensity redistributing from continuum into the Mg II line cores approaching onset.

Multi-height analysis on 35 co-observed dates returns CCA cross-instrument Top-1 retrieval of 0% against a 2.9% chance baseline, providing decisive evidence against a shared latent representation between corona and chromosphere. The complementary signals motivate a two-stage forecasting architecture (coronal screening to identify flare-productive regions, followed by chromospheric temporal monitoring) as a route to extend operational space-weather warning horizons.

Acknowledgements

I would like to thank my academic supervisor, Dr Varun Ojha, for his guidance throughout this dissertation. I am grateful to my external supervisors at the European Space Agency, Dr Andy S. H. To and Dr Henrik Eklund, for providing the dataset, scientific motivation, and expert guidance that made this project possible. I am especially thankful to Dr Eklund for his continued support and advice long after the internship itself had ended. This dissertation builds on exploratory work initiated during my internship at the European Space Agency (ESA) through the Leiden/ESA Astrophysics Program for Summer Students (LEAPS).

I would like to thank Leiden Observatory for hosting the LEAPS Programme and for sponsoring my stay and living expenses during my time at ESA/ESTEC, without which this project would not have been possible. Early-stage results from this work were first presented as an oral talk at the LEAPS 2025 closing symposium, Leiden Observatory [1].

Finally, I would like to thank my parents for their unwavering support throughout my education, and for always providing a home I can return to.

In memory of my beloved grandmother, Despina Vartholomeou,
who passed away during the first weeks of my LEAPS internship.

May you shine as bright as the sun.

Contents

Declaration i

Abstract ii

Acknowledgements iii

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Dataset and problem framing . . . 2
- 1.3 Aims and scope of this dissertation . . . 3
- 1.4 Research Questions . . . 4
- 1.5 Contributions . . . 4
- 1.6 Thesis Structure . . . 5
- 1.7 Ethical Considerations . . . 6

2 Literature Review 7
- 2.1 Machine Learning Methods . . . 7
  - 2.1.1 Siamese Networks and Contrastive Learning . . . 7
  - 2.1.2 Transformer Architecture . . . 7
  - 2.1.3 Gradient-Boosted Trees . . . 8
  - 2.1.4 Explainability: SHAP and Grad-CAM . . . 8
  - 2.1.5 Optimal Transport . . . 8
  - 2.1.6 Maximum Mean Discrepancy . . . 9
- 2.2 Machine Learning for Solar Flare Forecasting . . . 9
  - 2.2.1 Magnetogram-based Forecasting . . . 9
  - 2.2.2 Chromospheric Spectroscopic Forecasting . . . 10
- 2.3 Research Gap and Objectives . . . 11

3 Data and Preprocessing 12
- 3.1 Data Source . . . 12
- 3.2 Data Hierarchy and Multi-Instrument Context . . . 12
- 3.3 Line Selection . . . 13
- 3.4 Preprocessing Pipeline . . . 13
  - 3.4.1 Wavelength Alignment . . . 14
  - 3.4.2 Symmetric Cropping . . . 14

3.4.3 Baseline Removal . . . 14
3.4.4 Area Normalisation . . . 14
3.5 Multi-Line Feature Construction . . . 15
3.6 Dataset Summary . . . 15
3.7 Class Imbalance and Evaluation Protocol . . . 15
3.8 Software Engineering and Validation . . . 16
3.9 Flare Event Identification and Data Acquisition . . . 16
3.9.1 EIS Pre-flare Catalogue and GOES Cross-matching . . . 16
3.9.2 Expanded IRIS Dataset . . . 16
3.10 IRIS Spectral Preprocessing . . . 17
3.10.1 Mg II Window Identification . . . 17
3.10.2 Spectral Processing . . . 17
3.11 Comparison with EIS Preprocessing . . . 18
3.12 IRIS Dataset Summary . . . 18
3.12.1 Co-observation Subset . . . 19

4 Pre-flare Spectral Classification: Unsupervised Baseline and Siamese Metric Learning 20
4.1 Introduction . . . 20
4.2 Data and Preprocessing . . . 20
4.3 Unsupervised Baseline . . . 21
4.3.1 Multi-Line Feature Construction (Adaptive Binning) . . . 21
4.3.2 k-Means Clustering . . . 22
4.3.3 Cluster Membership and Hybrid Structure . . . 22
4.4 Siamese Metric Learning . . . 23
4.4.1 Siamese Network Architecture . . . 23
4.4.2 Contrastive Loss . . . 24
4.4.3 Training Protocol . . . 24
4.4.4 Evaluation: k-Nearest-Neighbour Classification . . . 25
4.5 Results . . . 25
4.5.1 Classification Performance . . . 25
4.5.2 Temporal Analysis . . . 26
4.5.3 Within-Class Temporal Separation . . . 27
4.5.4 Per-Observation Breakdown . . . 28
4.5.5 Permutation Feature Importance . . . 29
4.5.6 Physical Properties of Pre-flare Spectra . . . 30
4.5.7 Embedding Visualisation . . . 31
4.6 Discussion . . . 32
4.6.1 The Nature of the Learned Signal . . . 32
4.6.2 Multi-line Fusion . . . 32
4.6.3 Comparison with Panos and Kleint (2020) . . . 32
4.7 Summary . . . 33

5 Supervised Classification and Anomaly Detection **34**
5.1 Introduction . . . 34
5.2 Experimental Setup . . . 34
5.2.1 Model Descriptions . . . 34
5.3 Results . . . 35
5.3.1 Classification Performance . . . 35
5.3.2 Logistic Regression: Linear Separability . . . 37
5.3.3 XGBoost Feature Importance . . . 37
5.3.4 1D CNN: Local vs. Global Features . . . 37
5.3.5 Autoencoder Anomaly Detection . . . 38
5.3.6 Temporal Analysis . . . 38
5.3.7 Line Ablation Study . . . 39
5.3.8 False Positive Characterisation . . . 40
5.4 Coronal Line Profile Non-Gaussianity . . . 42
5.4.1 Higher-Order Moment Extraction . . . 42
5.4.2 Kurtosis-Based Classification . . . 42
5.4.3 Temporal Evolution of Kurtosis . . . 43
5.5 Discussion . . . 44
5.6 Summary . . . 44

6 Spatial Analysis of Pre-flare Spectral Signatures **45**
6.1 Introduction . . . 45
6.2 Method . . . 45
6.2.1 Per-Pixel Score Maps . . . 45
6.2.2 Rank-Normalised Maps . . . 45
6.2.3 Spatial Concentration Metrics . . . 46
6.2.4 Multi-Raster Temporal Sequences . . . 46
6.2.5 Score-Intensity Comparison . . . 46
6.3 Results . . . 46
6.3.1 The Pre-flare Signal Is Spatially Diffuse . . . 46
6.3.2 Annotated Maps with Spectral Profiles . . . 47
6.3.3 Pre-flare vs Non-flare Spectral Comparison . . . 48
6.3.4 Score vs Intensity . . . 49
6.3.5 Temporal Evolution of Spatial Maps . . . 49
6.4 Discussion . . . 50
6.5 Summary . . . 50

7 Multi-Height Pre-Flare Detection: Results and Analysis **51**
7.1 Introduction . . . 51
7.2 Chromospheric Feature Temporal Analysis . . . 51
7.2.1 Pooled Temporal Correlations . . . 52
7.2.2 Per-Date Heterogeneity . . . 52
7.2.3 Spectral Morphology Clustering . . . 53

7.3 Spectral Transformer Classification . . . 53
7.4 1D-CNN with Gradient-Weighted Class Activation Mapping . . . 53
7.5 Ensemble Classification and Per-Date Evaluation . . . 54
7.5.1 Transformer-CNN Ensemble . . . 54
7.5.2 Per-Date Within-AR Classification . . . 55
7.5.3 Classifier Sensitivity: XGBoost on Within-AR Task . . . 56
7.6 Multi-Height Temporal Comparison . . . 57
7.7 Discussion . . . 58
7.8 Summary . . . 59

8 Multi-Source Analysis **60**
8.1 Introduction . . . 60
8.2 Cross-Instrument Alignment . . . 61
8.2.1 Method . . . 61
8.2.2 Results . . . 61
8.3 Distribution Shift Analysis . . . 61
8.3.1 Method . . . 61
8.3.2 Results . . . 62
8.4 Optimal Transport Analysis . . . 62
8.4.1 Method . . . 62
8.4.2 Results . . . 63
8.5 Spectral Explainability . . . 64
8.5.1 Method . . . 64
8.5.2 Results . . . 64
8.6 Panos Feature Evaluation: Cross-AR vs Within-AR . . . 65
8.7 Sensitivity Analysis . . . 65
8.8 Summary . . . 66

9 Conclusion and Future Work **67**
9.1 Answers to the Research Questions . . . 67
9.2 Summary of Findings . . . 67
9.3 Contributions . . . 68
9.4 Methodological Reflections . . . 68
9.5 Limitations . . . 69
9.6 Future Work . . . 69
9.7 Closing Remarks . . . 70

A Code and Reproducibility **71**
A.1 Code Availability . . . 71
A.2 Data Availability . . . 72
A.3 Reproducibility Protocol . . . 72
A.4 Computational Environment . . . 72
A.5 Wavelength Alignment Implementation . . . 72

A.6 Use of Generative AI 75
A.7 Software Engineering Decisions 76

B Solar Physics Background **77**
B.1 Coronal Pre-flare Observations 77
B.2 IRIS Mg II Formation Physics 78
B.3 Non-Gaussian Flare Line Profiles 78
B.4 Kappa-Distribution Validation of the Kurtosis Discriminant 78
B.5 Physical Interpretation of the Coronal Pre-flare Signal 79

List of Acronyms **80**

Glossary of Terms **82**

Chapter 1

Introduction

1.1 Motivation

Solar flares are sudden eruptions on the Sun that can disrupt satellites, GPS systems, radio communications, and power grids on Earth. Predicting them is an open problem in space weather: current approaches rely on magnetic field images of the solar surface, which capture slow build-up over days but give limited warning in the final hours before eruption [2–4]. This dissertation investigates whether machine learning can extract short-timescale warning signals from ultraviolet spectra recorded by two space-based instruments observing different atmospheric layers.

Ultraviolet spectral lines emitted by hot coronal plasma encode plasma temperature, velocity, and turbulence through their intensity, centroid shift, and line width. Recent work by To et al. [5] demonstrated systematic non-thermal velocity enhancements in coronal spectra before flares, while Panos and Kleint [6] showed that chromospheric Mg II spectra carry a temporal precursor signal within 25 minutes of onset. However, these two findings used different instruments, different flare events, and different evaluation protocols. Whether the coronal and chromospheric signals encode the same or complementary information was unknown. This dissertation addresses that gap.

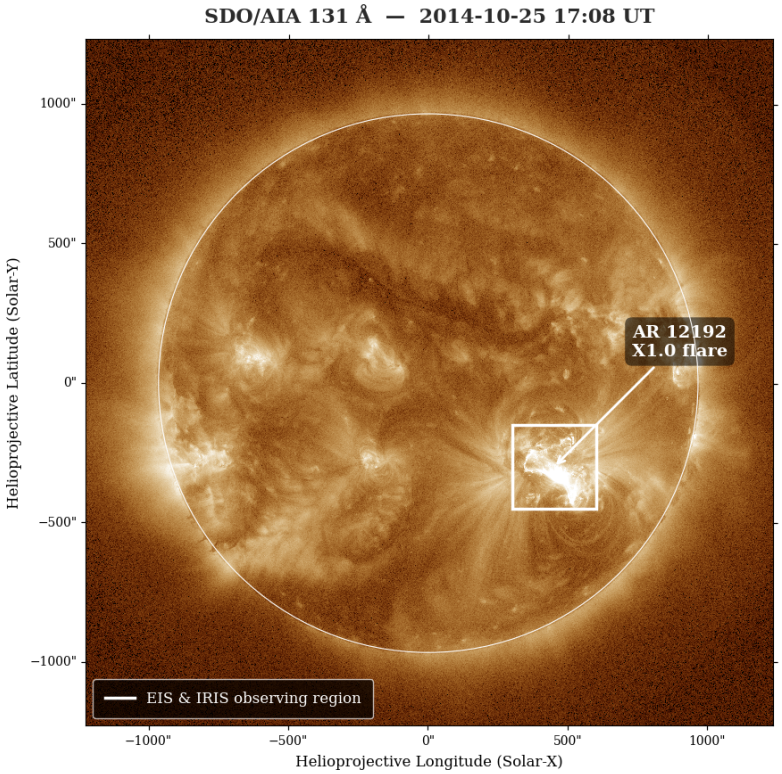


Figure 1.1: SDO/AIA 131 Å image of NOAA Active Region 12192 during the X1.0 flare on 25 October 2014, with the *Hinode*/EIS and IRIS observing fields of view indicated. The 35 such co-observed events enable this dissertation's central contribution: a direct multi-height comparison establishing the WHETHER (corona) versus WHEN (chromosphere) distinction (Chapters 7–8).

1.2 Dataset and problem framing

Throughout this dissertation, two key labels are used. A *pre-flare* spectrum is one observed during a defined time window before flare onset, quantified by the time-to-flare (TTF), meaning it is followed by a flare within the chosen forecasting horizon. A *non-flare* spectrum is one that is not followed by a flare within that same horizon. The core learning problem is framed as a binary classification task: given a spectrum, predict whether it belongs to the pre-flare or non-flare class. Subsequent chapters extend this framing through unsupervised clustering, anomaly detection, distributional analysis, and explainability methods to characterise the nature of the pre-flare signal.

Figure 1.2 shows example spectra from both instruments after preprocessing. The coronal Fe lines observed by EIS are single-peaked emission profiles whose width encodes non-thermal broadening; the key visual difference between pre-flare and non-flare spectra is that pre-flare profiles are narrower and more peaked, particularly for the hottest line (Fe XV). The chromospheric Mg II h&k lines observed by IRIS are more complex, with double-peaked profiles and a subordinate triplet; the early-versus-late pre-flare difference is subtler and concentrated in the line cores. These spectral differences are the signals that the classifiers developed in Chapters 4–8 seek to exploit.

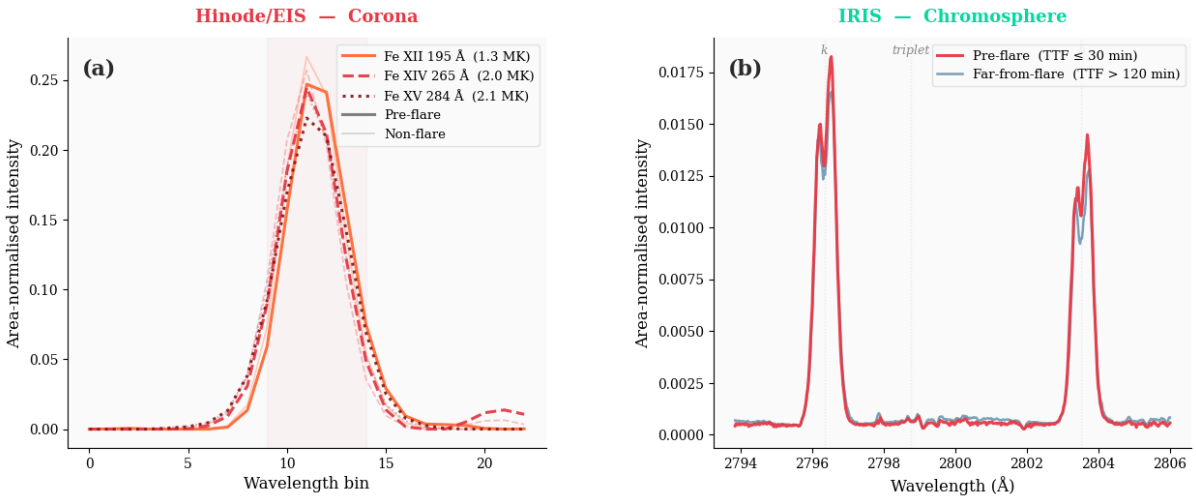


Figure 1.2: Example area-normalised spectra from both instruments after preprocessing. **(a)** *Hinode*/EIS coronal spectra: three Fe emission lines (Fe XII 195 Å, Fe XIV 265 Å, Fe XV 284 Å) shown for pre-flare (dark) and non-flare (light) active regions. Pre-flare profiles are narrower and more peaked, especially for the hotter lines. **(b)** IRIS chromospheric spectra: Mg II h&k profiles for early pre-flare (TTF ≤ 30 min, red) and late pre-flare (TTF > 120 min, blue) spectra from the same active region. The pre-flare difference is concentrated in the line cores, with enhanced k and h peak emission. The spectral regions labelled 'k', 'triplet', and 'h' correspond to the features analysed in Chapters 7–8.

1.3 Aims and scope of this dissertation

The overall aim of this dissertation is to determine whether EUV spectra from *Hinode*/EIS and near-ultraviolet spectra from IRIS contain reliably learnable and physically interpretable pre-flare signatures, and to characterise how these signatures differ between the corona and chromosphere. A central methodological contribution emerging from this aim is a 9-feature kurtosis-and-skewness representation of EIS Fe lines that reaches ROC-AUC = 0.959 (Chapter 5), outperforming an optimal 9-component PCA projection by 3.5 ROC-AUC points and providing the first application of higher-order moment analysis to the pre-flare EIS regime. The work addresses the following objectives:

1. **Preprocessing and dataset construction.** Implement reproducible preprocessing pipelines for EIS coronal spectra (Fe XII/Fe XIV/Fe XV, 69 features) and IRIS chromospheric spectra (Mg II h&k, 480 bins), producing ML-ready datasets spanning 2011–2024.

2. **Unsupervised representation analysis.** Apply k-means clustering on adaptive-binned spectral features to determine whether pre-flare spectra form a discrete cluster or a distributional shift across overlapping states, motivating the supervised methods used later.

3. **Siamese metric learning for low-resource spectral classification.** Train a Siamese MLP with contrastive loss to learn a 64-dimensional ℓ_2-normalised embedding from 69-dimensional spectra, evaluate via kNN, and analyse permutation importance and within-class metric structure to identify the physical drivers of discrimination.

4. **Robustness of the pre-flare signal across model families.** Train and compare six classifiers (logistic regression, XGBoost, MLP, autoencoder, 1D-CNN, Siamese kNN) on the 2.46M-spectrum fused EIS dataset (404,566 pre-flare; 2,052,379 no-flare) and verify the temporal null holds across all six.

5. **Higher-order moment features as low-dimensional classifiers.** Engineer a 9-feature representation from per-spectrum kurtosis, skewness, and standard deviation of Fe XII/XIV/XV lines, evaluate logistic regression against 69-dimensional raw spectra, and quantify the dimensionality–performance trade-off.

6. **Spatial interpretability of classifier decisions.** Map per-pixel XGBoost scores onto raster geometry, quantify spatial concentration via Gini coefficient and connected-component analysis, and characterise spatial structure across pre-flare and non-flaring observations.

7. **Generalisation across active regions and epochs.** Train a spectral transformer (multi-head self-attention, learnable CLS token) and 1D-CNN with Grad-CAM on 22.6M IRIS Mg II spectra from 43 events spanning Solar Cycles 24–25, and compare cross-AR and within-AR protocols to test whether the Panos-Kleint [6] precursor generalises.

8. **Multi-modal alignment and distributional shift analysis.** Test whether coronal and chromospheric representations encode complementary information on 35 co-observed dates via CCA [7] with leave-one-out cross-validation, MMD with permutation testing, Wasserstein-2 distance OT, and SHAP attribution.

1.4 Research Questions

The objectives above describe the technical work undertaken. This dissertation pursues three research questions about classifier behaviour and learned representations on pre-flare solar spectra:

- **RQ1.** Can pre-flare EIS coronal spectra be reliably distinguished from non-flaring active-region spectra under observation-disjoint splitting evaluation, and is the result robust across classifier architectures from linear to metric-learning?
- **RQ2.** How does coronal classifier confidence vary with time-to-flare across the 0–180 minute pre-flare window? Specifically, does it exhibit the monotonic rise reported by Panos and Kleint [6] for chromospheric Mg II spectra, or a different temporal pattern?
- **RQ3.** Do coronal and chromospheric learned representations encode shared or complementary information about the pre-flare state on co-observed flare events?

1.5 Contributions

The dissertation makes the following contributions, listed in the order they appear in the dissertation. Contributions 1–2 address RQ1, contributions 3–4 address RQ2, and contribution 5

addresses RQ3. All novelty claims below are stated to the best of our knowledge, based on the literature review in Chapter 2.

1. **ML classification of pre-flare *Hinode*/EIS coronal spectra.** Six classifier families (logistic regression, XGBoost, MLP, autoencoder, 1D-CNN, Siamese kNN) are trained on 2.5 million spectra under observation-disjoint splits, establishing pre-flare distinguishability up to ROC-AUC = 0.996 alongside a temporal-null finding (Chapters 4–5).

2. **Pre-flare non-Gaussianity as a coronal diagnostic.** Higher-order moments (kurtosis and skewness) of EIS Fe lines reach ROC-AUC = 0.959 from 9 features, outperforming an optimal 9-component PCA projection by 3.5 ROC-AUC points; the result is validated against the kappa-distribution interpretation of Jeffrey et al. [8, 9] (Chapter 5; Appendix B.4).

3. **Multi-height temporal comparison on co-observed flare events.** Direct cross-instrument comparison of EIS coronal and IRIS chromospheric temporal profiles establishes the WHETHER/WHEN framework, where the corona encodes flare-readiness as a persistent state and the chromosphere encodes temporal proximity event-dependently (Chapter 7).

4. **Event-dependent chromospheric signal across 43 events.** 8 of 25 dates passing the strict $\geq$50-per-class within-AR filter (of 43 working pre-flare events; see Table 3.5) exceed AUC > 0.80 for within-AR temporal classification under classifier-robust evaluation, refining the universal chromospheric precursor claim of Panos and Kleint [6] (Chapter 7).

5. **Optimal transport applied to IRIS pre-flare spectra.** Wasserstein-2 transport reveals intensity flow from continuum and triplet regions into the Mg II k and h line cores approaching flare onset, with W_2 shift magnitude correlating with within-AR classification AUC ($\rho = 0.540$, $p = 0.004$) (Chapter 8).

Dissemination

Findings from this dissertation were presented as an oral talk at the LEAPS 2025 closing symposium, Leiden Observatory. A peer-reviewed manuscript developing the kurtosis-9 discriminant, the WHETHER/WHEN framework, and the multi-height temporal comparison is in preparation for submission to a solar-physics venue.

1.6 Thesis Structure

The dissertation proceeds as follows. Chapter 2 reviews the relevant ML methods and prior solar-flare-forecasting literature, motivating the research gap. Chapter 3 documents the EIS and IRIS preprocessing pipelines and the observation-disjoint splitting protocol. Chapters 4–5 address RQ1 on EIS coronal spectra: a combined unsupervised-clustering and Siamese-metric-learning chapter, followed by a six-classifier supervised benchmark with the kurtosis-9 discriminant. Chapter 6 characterises the spatial structure of the coronal signal. Chapter 7 addresses RQ2 with IRIS chromospheric data. Chapter 8 addresses RQ3 through cross-instrument representation alignment.

Chapter 9 returns to each research question with methodological reflections, limitations, future work, and closing remarks.

1.7 Ethical Considerations

This dissertation uses publicly available and ESA-curated archival spectroscopic data: no human subjects, no privacy or consent considerations, and no biases against protected groups. The downstream application, operational solar flare forecasting, is benign dual-use: improved warning timescales benefit satellite operators, power-grid managers, and astronaut safety. End-to-end compute was approximately 20–30 hours of Google Colab time, a modest energy footprint relative to typical deep learning research. The use of generative AI tools during writing is disclosed in Appendix A.6.

Chapter 2

Literature Review

This chapter reviews the prior work that motivates this dissertation. Section 2.1 introduces the machine learning methods used in later chapters, including metric learning, transformer architectures, gradient-boosted trees, attribution methods, and distributional comparison via optimal transport and maximum mean discrepancy. Section 2.2 reviews prior machine learning work on solar flare forecasting, both magnetogram-based and spectroscopic. Section 2.3 states the research gap that this dissertation addresses. The solar physics background relevant to interpreting the spectral signatures observed by EIS and IRIS is provided in Appendix B.

2.1 Machine Learning Methods

This section briefly reviews the machine learning methods applied to the spectral classification problem in later chapters.

2.1.1 Siamese Networks and Contrastive Learning

Siamese networks were introduced by Bromley et al. [10] for signature verification. The architecture consists of two identical sub-networks with shared weights that process a pair of inputs and produce embeddings whose distance encodes similarity. Chopra et al. [11] formalised the contrastive loss function for training Siamese networks, which pulls same-class embeddings together while pushing different-class embeddings apart by at least a margin m. This approach learns a distance function optimised for class discrimination rather than reconstructing the input, making it well-suited for problems where the decision boundary is complex but the underlying dimensionality is low. In this dissertation, a Siamese MLP with contrastive loss is used to learn an embedding space for EIS spectral classification (Chapter 4).

2.1.2 Transformer Architecture

The transformer architecture, introduced by Vaswani et al. [12], replaced recurrent processing with a self-attention mechanism that computes pairwise relationships between all positions in an input sequence in parallel. The multi-head attention mechanism allows the model to attend to different aspects of the input simultaneously, and positional encodings inject sequence order information without recurrence. Originally developed for natural language processing,

transformers have been successfully applied to 1D time series and spectral data by treating wavelength bins as sequence positions. In this dissertation, a spectral transformer classifier is applied to IRIS Mg II h&k spectra (Chapter 7), where the self-attention mechanism can learn which wavelength regions (k core, h core, triplet, continuum) are most informative for classification.

2.1.3 Gradient-Boosted Trees

XGBoost [13] is a scalable gradient-boosted tree ensemble that has become widely used for tabular classification tasks. It constructs an ensemble of decision trees sequentially, with each tree correcting the residual errors of its predecessors. XGBoost's axis-aligned splits on individual features make it naturally interpretable through feature importance measures (information gain), and its regularisation mechanisms (column subsampling, maximum depth constraints, L1/L2 regularisation) help prevent overfitting. In this dissertation, XGBoost is used as a supervised baseline for EIS spectral classification (Chapter 5) and as the source of per-pixel prediction scores for the spatial analysis (Chapter 6).

2.1.4 Explainability: SHAP and Grad-CAM

Interpreting why a classifier makes a particular prediction is essential for connecting machine learning results to physical understanding. Lundberg and Lee [14] introduced SHAP (SHapley Additive exPlanations), a unified framework that assigns each input feature an importance value for a particular prediction based on Shapley values from cooperative game theory. For linear models, SHAP values are exact and can be computed efficiently. SHAP is used in this dissertation to identify which wavelength regions of the IRIS Mg II spectrum drive the pre-flare classification (Chapter 8).

Grad-CAM (gradient-weighted class activation mapping), introduced by Selvaraju et al. [15], provides visual explanations for CNN predictions by computing the gradient of the target class score with respect to feature maps, producing heatmaps that highlight the input regions most relevant to the classification decision. Grad-CAM is applied to the IRIS 1D-CNN classifier in Chapter 7.

2.1.5 Optimal Transport

Optimal transport (OT) compares probability distributions by computing the minimum-cost plan to transform one into the other [16, 17]. Unlike Kullback–Leibler divergence, OT respects metric structure: moving mass between nearby wavelength bins costs less than moving it across the full spectral range. The Wasserstein distance gives both a scalar measure of distributional difference and a transport plan that reveals *how* the distribution changes. Efficient computation is enabled by entropic regularisation and the Sinkhorn algorithm [18], implemented in the POT library [19]. In this dissertation, OT is used to characterise how the IRIS Mg II spectral energy distribution reorganises approaching flare onset (Chapter 8).

2.1.6 Maximum Mean Discrepancy

Maximum Mean Discrepancy (MMD) [20] is a kernel-based statistical test for whether two samples are drawn from the same distribution. Given two samples $X = \{x_1, \ldots, x_m\}$ and $Y = \{y_1, \ldots, y_n\}$ and a kernel $k(\cdot, \cdot)$, the unbiased empirical MMD^2 is:

$$\widehat{\mathrm{MMD}}^2(X, Y) = \frac{1}{m(m-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{mn} \sum_{i,j} k(x_i, y_j). \tag{2.1}$$

With a Gaussian kernel $k(x, y) = \exp(-\|x - y\|^2 / 2\sigma^2)$, MMD is sensitive to differences in all moments of the distributions, and $\widehat{\mathrm{MMD}}^2 = 0$ if and only if the two distributions are identical. The kernel bandwidth σ^2 is set by the median heuristic (the median of all pairwise squared distances). In this dissertation, MMD is used to quantify the magnitude of the spectral distributional shift between early and late pre-flare observations on a per-date basis (Chapter 8), with significance assessed via permutation tests.

2.2 Machine Learning for Solar Flare Forecasting

Prior machine learning work on solar flare forecasting falls into two streams: photospheric magnetogram-based prediction (Section 2.2.1) and spectroscopic prediction from atmospheric layers above the photosphere (Section 2.2.2).

2.2.1 Magnetogram-based Forecasting

The dominant approach to solar flare prediction uses photospheric magnetic field measurements derived from vector magnetograms. Leka and Barnes [21, 22] established the framework for parametric flare prediction by identifying magnetic field properties, such as total unsigned flux, magnetic shear, and free energy proxies, that distinguish flare-productive active regions from quiescent ones. Building on this, Bobra and Couvidat [23] used 25 Space-weather HMI Active Region Patches (SHARP) parameters extracted from the Solar Dynamics Observatory's Helioseismic and Magnetic Imager (HMI) to train a support vector machine for flare prediction, demonstrating that photospheric magnetic parameters carry substantial predictive information on 24-hour timescales.

Despite these advances, a persistent limitation of the magnetogram paradigm is that photospheric magnetic field parameters primarily capture the slow, multi-day build-up of magnetic stress. They encode whether an active region is capable of producing a flare, but provide limited information about the short-timescale precursors that may manifest minutes to hours before flare onset. As noted by Panos and Kleint [6], the input data for flare prediction has remained "largely fixed on magnetic features" even as model complexity has increased. They argue that photospheric magnetic field data alone "may not be a wholly sufficient source of data for flare prediction." This limitation motivates the investigation of spectroscopic data from higher atmospheric layers, the chromosphere, transition region, and corona, where pre-flare energy release may produce measurable signatures before the impulsive phase.

2.2.2 Chromospheric Spectroscopic Forecasting

A parallel line of research has shown that chromospheric spectra carry predictive flare information. This work centres on the Mg II h&k resonance lines observed by the Interface Region Imaging Spectrograph (IRIS), whose formation physics is reviewed in Appendix B.

Panos et al. [24] initiated this line of work by applying supervised hierarchical k-means clustering to hundreds of thousands of Mg II h&k profiles from 33 flares, identifying a single-peaked profile morphology that appeared in every flare. This established that flare-associated Mg II spectral shapes are systematically distinct from quiet Sun double-peaked profiles.

Building on this, Panos and Kleint [6] demonstrated for the first time that spectral data could be used for short-timescale flare prediction. Using a fully connected deep neural network trained on ten hand-crafted features extracted from Mg II k spectra, they showed that pre-flare active region spectra (observed within 25 minutes of flare onset) could be distinguished from non-flaring active region spectra with an AUC of 0.86. Crucially, for one flare analysed in detail, the classification score increased monotonically as flare onset approached, both in time and space. The features in descending order of predictive capability were: triplet emission at 2798.77 Å, line core intensity, total continuum emission between the h&k line cores, the k/h intensity ratio, and line width.

Woods et al. [25] applied k-means clustering specifically to pre-flare IRIS spectra and found that single-peaked Mg II profiles and subordinate triplet emission appeared frequently before flares, potentially stemming from chromospheric heating events up to 40 minutes before onset. Inversions of these profiles indicated enhanced chromospheric temperatures and electron densities. Panos et al. [26] used explainable AI techniques (gradient-weighted class activation mapping and expected gradients) to reveal which spectral features and spatial locations drove the neural network predictions, finding that 78% of the time the position of the model's maximum attention along the slit predicted the location of the flare's peak UV emission.

Subsequent work extended this approach to additional IRIS spectral lines [27] and to the multiple instance learning paradigm [28], consistently finding that Mg II performs best among the available chromospheric diagnostics.

A consolidated methodological critique of this body of work surfaces three recurring weaknesses. First, evaluation protocols are predominantly within-AR or within-event, which conflates the classifier's ability to detect a temporal signal with its ability to identify the active region itself; only Panos et al. [26] attempt cross-AR evaluation, and only on a single flare. Second, the chromospheric temporal precursor reported by Panos and Kleint [6] has not been replicated under classifier-robust evaluation across diverse events, and the absence of failure-mode analysis in the published literature obscures whether the signal generalises across active regions and observation epochs. Third, all studies use chromospheric data from a single instrument with no comparison to coronal signatures from the same flare events; whether the two atmospheric layers encode the same or complementary information is unknown.

2.3 Research Gap and Objectives

The literature surveyed above establishes three converging methodological gaps that the spectroscopic flare-prediction field has not yet addressed. None can be resolved by extending any single prior approach in isolation; together they require multi-architecture evaluation under proper observation-disjoint protocols on co-observed multi-instrument spectra of the same flare events. This dissertation addresses all three:

1. **Machine learning classification of pre-flare EIS coronal spectra.** To et al. [5] demonstrated systematic non-thermal velocity enhancements before flares using Gaussian-fitting analysis of EIS data, but machine learning classifiers have not yet been applied to preprocessed EIS spectral profiles for pre-flare detection. This dissertation applies six classifiers to 2.5 million EIS spectra, characterises the physical properties (including line widths and non-Gaussianity) that drive the classification, and maps the spatial distribution of the signal across the active region (Chapters 4–6).

2. **Multi-height spectral comparison on the same flare events.** The chromospheric temporal ramp-up reported by Panos and Kleint [6] and the persistent coronal signatures implied by To et al. [5] have been studied independently, using different instruments, flare events, and evaluation protocols. Whether these two signals encode the same or complementary pre-flare information remains an open question. This dissertation performs a direct multi-height comparison on co-observed dates, the first such comparison to the best of our knowledge (Chapters 7–8).

3. **Kurtosis analysis of pre-flare EIS spectra.** Jeffrey et al. [8, 9] demonstrated non-Gaussian EIS line profiles during the impulsive phase of flares, but whether non-Gaussianity is present in pre-flare spectra remains unexplored. This dissertation extends the higher-order moment analysis to the pre-flare regime and demonstrates that kurtosis alone achieves ROC-AUC $= 0.959$ for pre-flare detection, outperforming an optimal 9-component PCA projection (Chapter 5).

Chapter 3

Data and Preprocessing

3.1 Data Source

This dissertation uses a curated Hinode/EIS EUV spectral dataset compiled by Dr Andy S. H. To (ESA) as part of a large-scale pre-flare spectroscopic study [5]. The dataset spans observations from 2011–2024 and is distributed as two HDF5 files corresponding to pre-flare and no-flare cases. The raw dataset was provided in unprocessed form; all preprocessing described in this chapter was implemented by the author.

Each observation is a raster, created by stepping the spectrometer slit across the Sun to build a two-dimensional spatial scan [29, 30]; the structure of the resulting data is summarised in Figure 3.1 (Section 3.2). Not all observations contain every spectral line, which motivates careful line selection and an explicit join strategy when constructing multi-line feature vectors.

3.2 Data Hierarchy and Multi-Instrument Context

The dissertation uses two independent spectroscopic instruments observing different layers of the solar atmosphere. *Hinode*/EIS samples the solar corona at ∼1–2 MK plasma temperatures via Fe ionisation states; IRIS samples the chromosphere at ∼6–10 kK via the Mg II h&k resonance lines. Together they provide independent views of the same active regions at coronal and chromospheric heights, enabling the multi-height comparison in Chapter 8. Three Fe lines are used on the EIS side because each ionisation state emits at a different plasma temperature (Fe XII at 1.3 MK, Fe XIV at 2.0 MK, Fe XV at 2.1 MK); using all three simultaneously probes three thermal layers of the corona, capturing temperature-dependent spectral changes that a single-line analysis would miss.

Figure 3.1 shows the data hierarchy of both instruments. The two pipelines share the same six-level structure (observation → raster → pixel → spectrum → feature vector → dataset) and the same label inheritance: each raster carries a single time-to-flare (TTF) annotation derived from the underlying solar observation, and every pixel and every spectrum within that raster inherits the parent TTF. The pre-flare and no-flare counts of 404,566 and 2,052,379 (Table 3.2) therefore reflect raster-level labels propagated to the spectrum level, not 2.5 million independently-labelled samples.

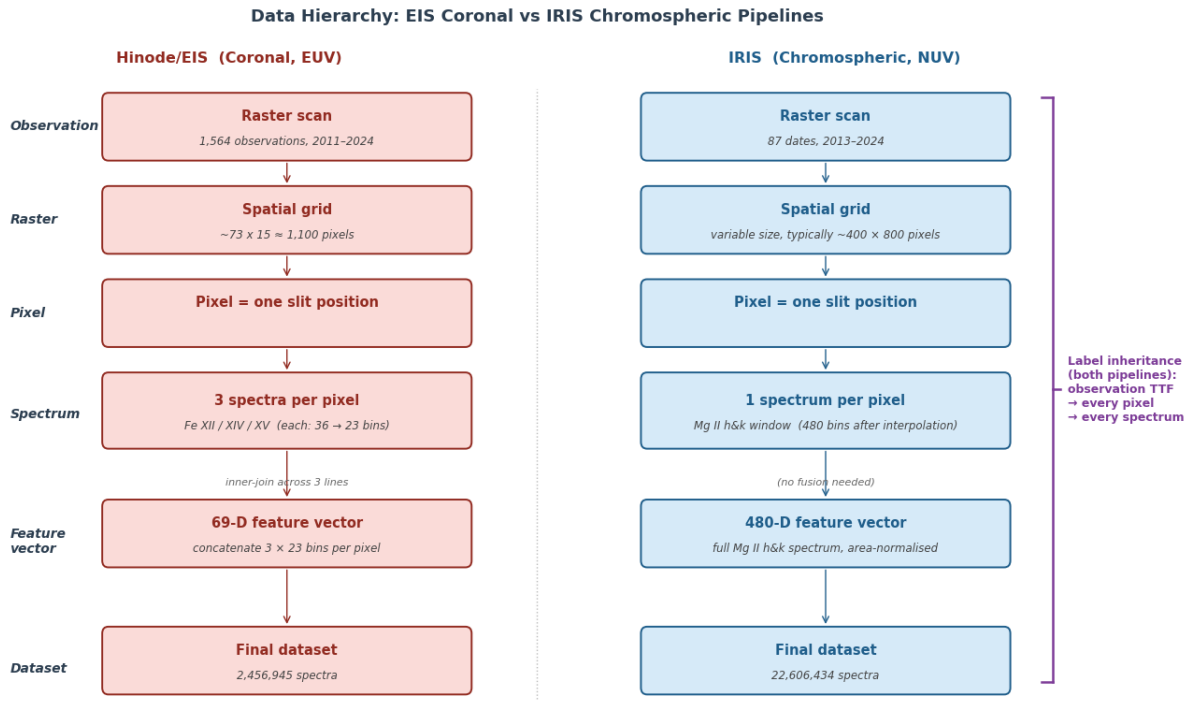


Figure 3.1: Data hierarchy of the *Hinode*/EIS coronal pipeline (left) and the IRIS chromospheric pipeline (right). The bracket on the right marks the label inheritance: each observation's TTF label propagates down to every pixel and spectrum in the raster.

3.3 Line Selection

Three iron emission lines were selected for analysis, spanning the coronal temperature range:

Table 3.1: Selected EIS spectral lines. The three lines span a factor of ∼1.6 in temperature, enabling detection of temperature-dependent spectral changes that a single line would miss.

Line	Wavelength (Å)	$\log T$	T (MK)
Fe XII	195.12	6.1	1.3
Fe XIV	264.79	6.3	2.0
Fe XV	284.16	6.35	2.1

Fe XII 195.12 Å is the primary EIS coronal diagnostic line [31]. Fe XIV and Fe XV extend coverage to hotter plasma and are among the least noisy lines in the dataset.

3.4 Preprocessing Pipeline

Raw EIS spectral windows are not directly comparable across rasters. Small wavelength offsets from instrumental drift, background and continuum differences, and missing or invalid bins can dominate distance computations and lead to spurious structure in downstream models. The preprocessing pipeline addresses these issues through four sequential steps.

3.4.1 Wavelength Alignment

The effective line centre can vary between rasters and can shift systematically across raster columns due to the orbital motion of the spacecraft and thermal drifts in the spectrograph [32]. To correct for this, the pipeline estimates the line centre independently for each raster column using a robust procedure: the brightest rows (top 10% by peak intensity) are selected, a representative profile is formed from their median, and a quadratic subpixel refinement is applied around the peak to obtain a precise centre estimate. If the estimated centre disagrees with the metadata reference centre by more than 0.06 Å, a secondary refinement using centroid estimation on the top-k brightest spectra is applied.

Alignment is then performed as an integer-bin shift placing the estimated line centre at a fixed reference bin (the middle of the wavelength array); values shifted beyond array bounds are set to NaN. This per-column alignment captures systematic variation across the raster scan that a single global shift would miss. The full implementation, including the subpixel refinement and centroid-fallback branch, is reproduced in Listing A.1, Appendix A.5.

3.4.2 Symmetric Cropping

After alignment, spectra are cropped symmetrically around the centre bin to produce fixed-length vectors of length $L = 2W + 1$, where W is the half-width in bins. Rather than choosing W arbitrarily, a global crop width is selected using a coverage criterion: across all aligned spectra, the largest half-width W whose finite-data (non-NaN) coverage exceeds 90% is chosen via binary search. The 90% threshold was determined empirically by evaluating coverage statistics across the full dataset to find the best trade-off between preserving spectral extent and minimising missing data. The resulting crop width of $W = 11$ yields 23-bin spectra per line, capturing the line core and near wings while excluding the far wings where signal-to-noise is poor.

3.4.3 Baseline Removal

To reduce background and continuum variation across rasters, each cropped spectrum is fitted with a Gaussian emission profile plus a linear background:

$$I(\lambda) \approx A \exp\left(-\tfrac{1}{2}\left(\tfrac{\lambda-\mu}{\sigma}\right)^2\right) + B_0 + B_1\lambda,$$

where $I(\lambda)$ is the observed intensity at wavelength λ, A is the peak amplitude, μ is the line centre, σ is the Gaussian width, and $B_0 + B_1\lambda$ is the linear background. The fit is used solely to estimate and subtract the linear baseline rather than to interpret physical parameters. Initial baseline guesses are obtained by fitting a line to the lower-intensity portion of the spectrum (bottom 40% by value), reducing sensitivity to the line core.

3.4.4 Area Normalisation

Each 23-bin spectral window is independently area-normalised by dividing by the total integrated intensity. Negative intensities are clipped to zero before normalisation. This preserves the spectral *shape* while removing the absolute intensity scale, ensuring that downstream classifiers respond

to profile morphology rather than overall brightness. The effect of this choice is validated in Chapter 6, where the weak intensity-score correlation ($\rho = 0.23$) confirms that spectral shape, not brightness, drives classification.

3.5 Multi-Line Feature Construction

Downstream modelling operates at the spectrum (pixel) level, where each spatial pixel becomes one machine-learning sample. For multi-line analysis, spectra from the three Fe lines are concatenated to form a 69-dimensional feature vector (3×23 bins). Because not all rasters contain every line, multi-line feature construction requires an inner join across lines: each spectrum is indexed by the composite key $(\text{obs_id}, y, x)$, and only pixels present in all three lines are retained. A consistency check verifies that the time-to-flare value agrees across lines for each retained pixel.

Spectra observed after flare onset (TTF < 0) are excluded to ensure that comparisons are restricted to genuinely pre-flare observations.

3.6 Dataset Summary

Table 3.2: Processed EIS dataset after preprocessing and multi-line fusion. The 5:1 class imbalance motivates the use of PR-AUC alongside ROC-AUC throughout this dissertation.

Property	Value
Spectral lines	Fe XII, Fe XIV, Fe XV
Wavelength bins per line	23
Feature vector dimension	69
Total spectra (fused)	2,456,945
Pre-flare spectra	404,566 (16.5%)
No-flare spectra	2,052,379 (83.5%)
Unique observations	1,564
Observation period	2011–2024
Normalisation	Area (per line)
Pre-flare TTF range	0–210 min

The dataset was split at the observation level to prevent data leakage: all spectra from a given EIS raster are assigned to exactly one of training (80%), validation (10%), or test (10%), using a fixed random seed. This ensures that no spatial pixel from a single observation appears in more than one partition.

3.7 Class Imbalance and Evaluation Protocol

The processed dataset has a ∼5:1 no-flare to pre-flare ratio (2,052,379 vs. 404,566 spectra). This imbalance reflects the natural prevalence of non-flaring observations in the EIS catalogue and is preserved through training, validation, and test splits. No resampling (SMOTE, random undersampling, or class-balanced subsampling) was applied: synthetic minority samples would interpolate in a 69-dimensional spectral space where physical plausibility is not

guaranteed, and undersampling would discard real observational diversity. Imbalance was instead handled at the model level via inverse-frequency class weights (logistic regression, MLP), `scale_pos_weight` = 5.11 (XGBoost), and class-balanced pair sampling (Siamese; equal pre-flare and non-flare same-class pairs per batch).

Two complementary metrics are reported throughout this dissertation: ROC-AUC, which characterises ranking performance across all decision thresholds, and PR-AUC, which is more sensitive to the minority class under imbalance and therefore the primary metric used for model selection. F1 score, balanced accuracy, and recall are reported alongside both AUCs.

3.8 Software Engineering and Validation

The implementation follows a chapter-per-notebook structure with shared preprocessing utilities, version-controlled in Git. Each notebook is self-contained but reproducible from the same fused dataset cache (`eis_fused_69dim_cached.npz`), generated once by the preprocessing pipeline and reused across all downstream analyses. Validation operates at three layers. (i) **Cell-level sanity checks**: each notebook contains explicit assertions on data shapes, label balance, post-fusion key alignment, NaN-count thresholds, and train/val/test set disjointness; these run automatically on every notebook execution and fail loudly on regression. (ii) **Pipeline-level regression**: the dataset summary in Table 3.2 is reproduced on independent runs, and observation counts are verified against the canonical seed-42 split. (iii) **Result-level validation**: bootstrap resampling, multi-seed re-evaluation, and stratified k-fold cross-validation are used as integration tests for the end-to-end pipeline, in particular the 60-seed clustering stability analysis (Chapter 4) and the 5-seed multi-architecture benchmark (Chapter 5). Pickle caching with sha1 signature hashing prevents stale-cache bugs across configuration changes; an explicit `FORCE_REFIT` toggle enables forced refitting after parameter modifications. The code repository, reproducibility protocol, and detailed engineering decisions are documented in Appendix A.

3.9 Flare Event Identification and Data Acquisition

3.9.1 EIS Pre-flare Catalogue and GOES Cross-matching

The EIS dataset contains 1,928 pre-flare observations spanning 2011–2024, of which 1,077 were taken after the IRIS launch date (27 June 2013), covering 68 unique dates. For each of these dates, the Heliophysics Events Knowledgebase (HEK) [33] was queried via its REST API to retrieve all GOES M- and X-class flare events, including onset times, peak times, GOES class, and NOAA active region numbers. This yielded 63 unique flare events with precise GOES soft X-ray onset times.

3.9.2 Expanded IRIS Dataset

The IRIS dataset was expanded to maximise the number of flare events with chromospheric spectral coverage. IRIS Level 2 raster data were obtained from the Lockheed Martin Solar and Astrophysics Laboratory (LMSAL) archive via SunPy's `Fido` interface [34]. Each raster file was inspected to confirm the presence of the Mg II h&k spectral window. Many IRIS observations

were acquired on the calendar day preceding the flare; to capture these, the pipeline checked both the observation date and the following calendar day for matching GOES onset times, recovering 26 additional dates of pre-flare IRIS data. In total, 31,298 raster FITS files were obtained across 87 unique observation dates.

The expanded dataset spans Solar Cycles 24 and 25 (2013–2024), covering a diverse range of active regions, flare magnitudes from M1.0 to X9.3, and heliographic positions.

3.10 IRIS Spectral Preprocessing

The preprocessing pipeline follows the conventions established by Panos and Kleint [6] and extended by Panos et al. [26] and Zbinden et al. [27], with adaptations to produce output compatible with the EIS analysis framework.

3.10.1 Mg ii Window Identification

IRIS Level 2 raster files are multi-extension FITS files [35] containing separate image extensions for each observed spectral window. The Mg II h&k window was identified by searching the primary header for the `TWAVEn` keyword with a value in the range 2790–2810 Å. The data cubes have shape $(n_{\text{steps}} \times n_{\text{slit}} \times n_\lambda)$, where n_{steps} is the number of raster positions, n_{slit} is the number of spatial pixels along the slit, and n_λ is the number of wavelength bins.

The wavelength grid was constructed from the extension header keywords `CRVAL1` (reference wavelength), `CDELT1` (wavelength step), and `CRPIX1` (reference pixel), following the IRIS Level 2 convention [36].

3.10.2 Spectral Processing

Raw IRIS Level 2 data (in data numbers, DN) were divided by the exposure time to obtain rates ($\text{DN}\,\text{s}^{-1}$), with a minimum of 0.1 s enforced. All spectra were then interpolated onto a common wavelength grid spanning 2793.85–2806.00 Å with 480 uniformly spaced bins (∼0.025 Å per bin), covering both the Mg II k (2796.35 Å) and h (2803.52 Å) lines and the triplet emission region.

Four quality filters were applied following established criteria: removal of spectra with invalid pixels (−200 DN), spectra with peak intensity outside expected line core regions [27], spectra with continuum exceeding 60% of maximum intensity [6], and spectra with k/h intensity ratios outside $[0.8, 2.0]$ [27]. No continuum subtraction was applied, as the Mg II window contains physically meaningful emission at all wavelengths [37].

Each spectrum was area-normalised, and the time-to-flare (TTF) was computed from the GOES onset time. Only spectra with TTF between −10 and +900 minutes were retained.

3.11 Comparison with EIS Preprocessing

Table 3.3: Comparison of preprocessing pipelines for EIS coronal and IRIS chromospheric spectral data.

Step	EIS (corona)	IRIS (chromosphere)
Instrument	*Hinode*/EIS	IRIS
Spectral lines	Fe XII/XIV/XV	Mg II h&k
Formation temperature	1.3–2.1 MK	6–10 kK
Wavelength bins	23 per line (69 total)	480 (full h&k window)
Wavelength calibration	Per-column Gaussian fitting	CRVAL1/CDELT1 from header
Interpolation	Not required	Vectorised linear (480 bins)
Background subtraction	Linear baseline fit + clip	None (all emission is physical)
Exposure time	Pre-calibrated in H5	Divided by EXPTIME
Quality filters	NaN count ≤ 1, TTF ≥ 0	Missing data, peak location, continuum, k/h ratio
Normalisation	Area (per line)	Area (full window)
Temporal range	0–210 min	-10 to 900 min

3.12 IRIS Dataset Summary

The final preprocessed IRIS dataset contains 54,619,462 Mg II h&k spectra drawn from 64 dates spanning 2013–2024.

Table 3.4: Preprocessed IRIS Mg II h&k dataset summary.

Property	Value
Total spectra	54,619,462
Dates with data	64
Dates with TTF ≤ 30 min	37
Date range	2013 August – 2024 August
Solar cycles covered	24 and 25
Wavelength range	2793.85–2806.00 Å
Wavelength bins	480
Spectral resolution	$\sim$0.025 Å/bin
Normalisation	Area
TTF range	-10 to $+900$ min
Storage format	Per-date compressed NPZ

The full dataset is approximately 22× larger than the dataset used by Panos and Kleint [6] ($\sim$2.5 million spectra) and spans a significantly wider range of solar conditions, active regions, and flare magnitudes.

Of these, 22.6 million fall within the working TTF window $[0, 300]$ minutes used by Chapters 7–8; a separate 1.4 million quiet-Sun spectra from 21 GOES-confirmed non-flare dates serve as the non-flaring control. The remaining $\sim$30 million fall outside the working window and are retained in the archive but unused in the analyses that follow.

3.12.1 Co-observation Subset

Of the 64 IRIS dates, 40 correspond to events also observed by *Hinode*/EIS, enabling the multi-height temporal comparison presented in Chapter 7. Of these, 35 have sufficient data for paired date-level summary vectors used in the cross-instrument alignment analysis (Chapter 8).

Different IRIS analyses apply different inclusion filters and therefore reference different date counts; Table 3.5 reconciles these subsets in one place.

Table 3.5: IRIS date-count subsets by inclusion filter. The differences arise from filter strictness, not from data revisions.

Subset	n	Used in
Total IRIS dates (post-quality filtering)	64	§3.12
Working pre-flare events (TTF $\in [0, 300]$ min)	43	Ch. 7
Dates with TTF ≤ 30 min spectra	37	Ch. 7
Cross-AR transformer training dates	28	§7.3
Bilateral filter, ≥ 30/class (XGBoost, MMD)	27	§7.5.3, §8.3
Bilateral filter, ≥ 50/class (LR within-AR)	25	§7.5.2
Co-observed with EIS (raw)	40	Ch. 7
Co-observed with EIS (CCA-paired)	35	Ch. 8

Chapter 4

Pre-flare Spectral Classification: Unsupervised Baseline and Siamese Metric Learning

4.1 Introduction

This chapter develops the EIS coronal classification pipeline in two stages. The first stage applies k-means clustering to test whether pre-flare spectra form a discrete sub-population that an unsupervised method can isolate (Section 4.3). The second stage trains a Siamese network with contrastive loss [11] to learn a low-dimensional spectral embedding for binary pre-flare classification, evaluated with a k-nearest-neighbour (kNN) classifier (Section 4.4). Four questions are addressed:

1. Do pre-flare spectra form a discrete sub-population under unsupervised clustering, or do they share spectral states with non-flaring spectra?
2. Can coronal EUV spectra from pre-flare active regions be reliably distinguished from non-flaring active region spectra under supervised metric learning?
3. How does separability vary with time-to-flare across the pre-flare window?
4. Which physical properties of the coronal plasma drive the classification?

4.2 Data and Preprocessing

The dataset is the preprocessed, fused EIS spectral dataset described in Chapter 3. Two additional quality filters were applied for this and subsequent chapters: spectra with more than one missing (NaN) wavelength bin were discarded with remaining single-bin gaps filled by linear interpolation, and spectra observed after flare onset ($t_{\mathrm{ttf}} < 0$) were excluded. Table 4.1 shows the resulting observation-level split.

Table 4.1: Dataset summary after preprocessing, multi-line fusion, and observation-level splitting.

	Train	Validation	Test
Observations	1,251	156	157
Spectra (total)	2,011,816	224,078	221,051
Pre-flare (label 1)	329,235	35,640	39,691
Non-flaring (label 0)	1,682,581	188,438	181,360

4.3 Unsupervised Baseline

k-means was selected as the baseline clustering algorithm because it makes minimal assumptions about cluster geometry, scales easily to millions of spectra, and produces hard assignments that allow direct enrichment analysis. The aim of this section is not to find the best possible clustering, but to test whether pre-flare spectra form a discrete population under a standard, widely-used method. If even k-means cannot isolate them, learned representations are warranted.

The clustering uses a laxer quality filter than the supervised dataset of Table 4.1: rows with up to seven missing wavelength bins per line are retained and gaps interpolated, rather than the strict NaN ≤ 1 filter used downstream. This is appropriate because k-means is robust to small interpolated bin counts. The relaxed filter retains 1,689,839 pre-flare spectra (vs. 404,566 in the supervised dataset). For per-cluster enrichment analysis (Section 4.3.3), no-flare spectra are loaded separately from the line-fit catalogues of To et al. [5] and assigned via the trained k-means model (predict, not fit), yielding 6,389,514 no-flare spectra for the Fe XIV reference line.

4.3.1 Multi-Line Feature Construction (Adaptive Binning)

Because the multi-line feature vector requires intensity values from all three Fe lines at the same pixel, pixels were matched across lines using the composite key described in Section 3.5, retaining only pixels present in all three lines simultaneously. An adaptive binning scheme was applied to each line, assigning finer bins to the line core and coarser bins to the wings, reducing each line from 23 to approximately 19–20 bins and yielding a concatenated feature vector of 57 dimensions per pixel.

The adaptive binning concentrates resolution where the discriminative signal lives. Across all 100 clusters, the variance in cluster-mean spectra is 1.3 to 3.8 times larger in the line core than in the wings (wing/profile variance ratios: 0.26 for Fe XII, 0.76 for Fe XIV, 0.63 for Fe XV; Figure 4.1).

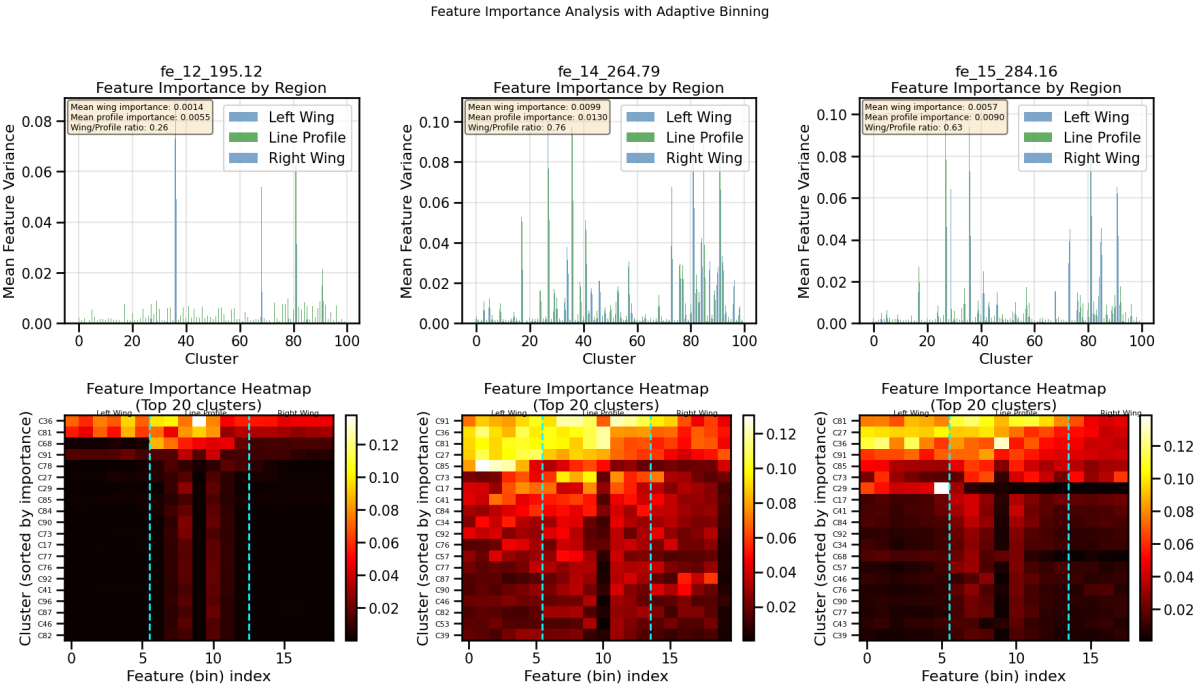


Figure 4.1: Top: Per-bin feature variance across the 100 k-means clusters, separated into left wing, line profile, and right wing regions for Fe XII (left), Fe XIV (centre), and Fe XV (right). Bottom: Per-bin importance heatmaps for the top 20 clusters by mean variance. Across all three lines, variance is 1.3–3.8× larger in the line profile than in the wings, validating the adaptive binning's allocation of finer resolution to the line core.

4.3.2 k-Means Clustering

k-means clustering was performed directly on the 57-dimensional adaptive-binned feature vectors. Adjacent wavelength bins in EIS spectra are highly correlated, but the adaptive binning scheme already concentrates resolution in the line core where the dominant variance lies, which makes a separate dimensionality reduction step unnecessary before clustering. Principal component analysis with two components was used separately for 2-D visualisation of the cluster structure.

$k = 100$ was chosen to stress-test the cluster-isolation hypothesis: a coarse clustering ($k = 20$) might wash out a small isolated pre-flare sub-population, whereas $k = 100$ creates partitions fine enough for any such structure to surface directly. With 1,689,839 pre-flare spectra distributed across 100 clusters, the average cluster contains ∼17,000 spectra, sufficient for reliable enrichment statistics. Stability was assessed by running k-means with 60 random seeds on a 10,000-spectrum subsample, yielding mean ARI = 0.575 and mean NMI = 0.787, indicating that the broad cluster structure is reproducible while individual assignments vary because spectra lie on continuous gradients (line width, intensity, asymmetry) rather than in well-separated discrete groups.

4.3.3 Cluster Membership and Hybrid Structure

Cluster mean profiles correspond to physically interpretable plasma states (strong-emission, weak-emission, and asymmetric profiles consistent with Doppler-shifted flows), and within-raster spatial patterns of cluster labels are coherent rather than random, confirming the clustering captures genuine solar structure.

The central test of this section is whether any clusters are exclusive to, or strongly enriched in, pre-flare spectra. Table 4.2 shows representative clusters ranked by their pre-flare enrichment ratio. The enrichment ratio for cluster c is defined as

$$E_c = \frac{N_c^{\mathrm{PF}}/N_c^{\mathrm{NF}}}{N^{\mathrm{PF}}/N^{\mathrm{NF}}}, \tag{4.1}$$

where N_c^{PF} and N_c^{NF} are the pre-flare and no-flare counts within cluster c, and N^{PF}, N^{NF} are the dataset totals. $E_c = 1$ indicates a cluster representative of the global class balance; $E_c > 1$ indicates pre-flare enrichment.

Table 4.2: Representative clusters spanning the enrichment range, from exclusively pre-flare to strongly depleted. Clusters 37 and 60 are the two pure pre-flare clusters (containing 89,457 spectra in total, 5.3% of the pre-flare population). The global pre-flare to no-flare ratio is 1,689,839/6,389,514 ≈ 0.265.

Cluster	Pre-flare	No-flare	Enrichment ratio
37	48,636	0	∞
60	40,821	0	∞
49	17,414	63,299	1.04
27	1,091	876,271	0.0047

What spectral features drive these clusters' class purity is not characterised in this baseline analysis; the supervised analyses in Section 4.4 and Chapter 5 pin down the discriminative features (line widths, kurtosis) at the population level.

Two clusters (37 and 60) are exclusively pre-flare, containing 89,457 spectra in total (5.3% of the pre-flare population). A further 45 clusters reach $E_c > 10\times$, and the remaining 53 clusters contain spectra from both classes at varying proportions. The clustering therefore captures the pre-flare signal as a hybrid: partly a class boundary, partly a distributional shift across overlapping spectral states. A learned representation that amplifies both components is the natural next step; the remainder of this chapter trains a Siamese network to do exactly this.

4.4 Siamese Metric Learning

4.4.1 Siamese Network Architecture

A Siamese network consists of two identical encoder sub-networks that share weights [10]. Given a pair of input spectra (x_1, x_2), each is mapped through the encoder f_θ to produce embeddings $z_1 = f_\theta(x_1)$ and $z_2 = f_\theta(x_2)$ in a lower-dimensional space, where the pairwise distance encodes class similarity.

The encoder f_θ is a multilayer perceptron (MLP) with the architecture:

$$f_\theta : \mathbb{R}^{69} \xrightarrow{\text{Linear}(256)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{Linear}(256)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{Linear}(64)} \xrightarrow{\ell_2\text{-norm}} \mathbb{R}^{64} \tag{4.2}$$

The output is ℓ_2-normalised to the unit hypersphere, so cosine similarity and Euclidean distance are monotonically related. The two-layer 256-unit design follows the architecture used by Panos et al. [24] for IRIS Mg II spectral classification, which is similar in input dimensionality

(240 wavelength bins vs. 69 here) and class structure (binary supervised). Width and depth were not separately tuned for this task; the close alignment of training and validation loss curves (Figure 4.2) confirms the chosen capacity does not overfit.

4.4.2 Contrastive Loss

The network is trained with the contrastive loss function [11]:

$$\mathcal{L}(z_1, z_2, y) = y \cdot \|z_1 - z_2\|_2^2 + (1 - y) \cdot \max(0,\ m - \|z_1 - z_2\|_2)^2 \tag{4.3}$$

where $\mathcal{L}$ is the loss, z_1 and z_2 are the embeddings of two input spectra, $\|z_1 - z_2\|_2$ is the Euclidean distance between them, $y = 1$ if the pair shares a label (same class) and $y = 0$ otherwise, and $m = 1.0$ is the margin parameter. Because the embeddings are ℓ_2-normalised to the unit hypersphere, pairwise distances lie in $[0, 2]$, making $m = 1.0$ a natural midpoint that separates same-class from different-class pairs without saturating the loss. When two spectra share the same class ($y = 1$), the first term penalises large distances between their embeddings, pulling them together. When they belong to different classes ($y = 0$), the second term penalises distances smaller than the margin m, pushing them apart.

4.4.3 Training Protocol

Training pairs were generated with class-proportional sampling: each batch contained approximately 50% same-class and 50% different-class pairs, with same-class pairs drawn equally from the pre-flare and non-flaring pools. The network was trained for up to 15 epochs (200,000 pairs per epoch) with early stopping (patience 3 on validation loss), using the AdamW optimiser [38] with a learning rate of 10^{-3}, cosine annealing schedule, and batch size 512. Validation loss was monitored on 50,000 pairs per epoch. The training converged rapidly, with validation pair-AUC reaching 0.982 within the first few epochs and stabilising thereafter (Figure 4.2).

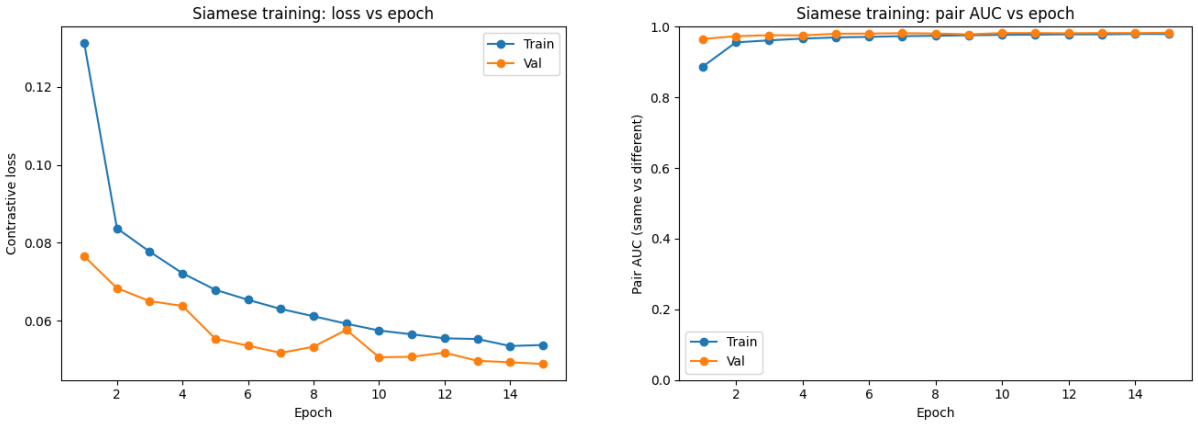


Figure 4.2: Training and validation learning curves. **Left:** Contrastive loss vs. epoch. **Right:** Pair-level AUC vs. epoch, which measures how well the embedding distances separate same-class pairs from different-class pairs. Both metrics converge rapidly, and the training and validation curves remain closely aligned throughout, indicating no overfitting.

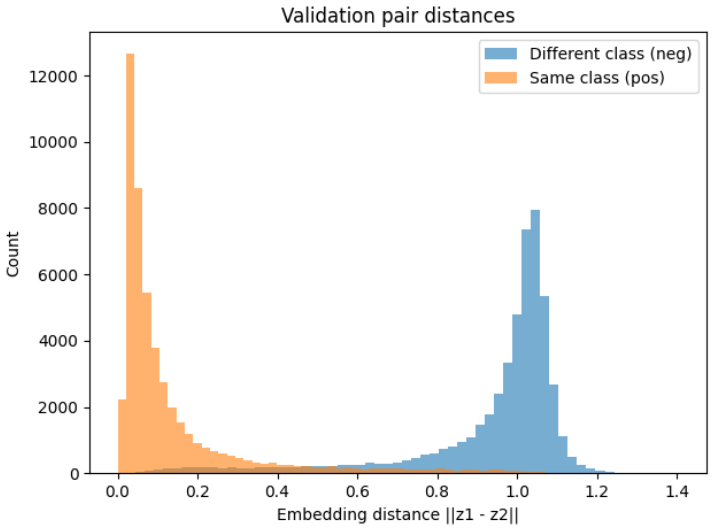


Figure 4.3: Distribution of pairwise Euclidean distances in the 64-dimensional embedding space for same-class (positive, orange) and different-class (negative, blue) validation pairs. The clear separation indicates that the encoder has learned to map same-class spectra to nearby regions and different-class spectra to distant regions.

4.4.4 Evaluation: k-Nearest-Neighbour Classification

To classify a new spectrum, it is first embedded through the trained encoder, then its K nearest neighbours in the training set are found using cosine similarity. The spectrum is assigned to whichever class is most common among those neighbours. The neighbourhood size K was tuned over $K \in \{1, 3, 5, 7, 9, 15, 25, 35, 51\}$ on the validation set, maximising PR-AUC. Performance plateaued above $K = 25$ (PR-AUC 0.960 vs. 0.965 at $K = 51$), indicating low sensitivity to the exact neighbourhood size in this range.

All reported test metrics were computed on the full test set ($N = 221{,}051$ spectra) using 200,000 training spectra as the reference set, with 95% bootstrap confidence intervals from 1,000 resamples.

4.5 Results

4.5.1 Classification Performance

Table 4.3 reports test-set performance. The Siamese embedding outperforms the raw-feature kNN baseline across all metrics, with the largest gain in precision (+10.5 percentage points). All improvements are significant at 95% confidence (non-overlapping bootstrap intervals).

Table 4.3: Test-set classification metrics for the Siamese kNN classifier and the raw-feature kNN baseline ($K = 51$, $N = 221{,}051$). The Siamese embedding outperforms the baseline across all metrics, with the largest gain in precision (+10.5 percentage points). 95% bootstrap confidence intervals in brackets.

Metric	Siamese kNN	Raw kNN (baseline)
ROC-AUC	0.990 [0.990, 0.991]	0.984 [0.983, 0.984]
PR-AUC	0.966 [0.965, 0.967]	0.935 [0.933, 0.937]
Balanced Accuracy	0.946 [0.944, 0.947]	0.924 [0.922, 0.925]
F1 Score	0.915 [0.913, 0.917]	0.853 [0.850, 0.855]
Precision	0.923	0.818
Recall (TPR)	0.908	0.891

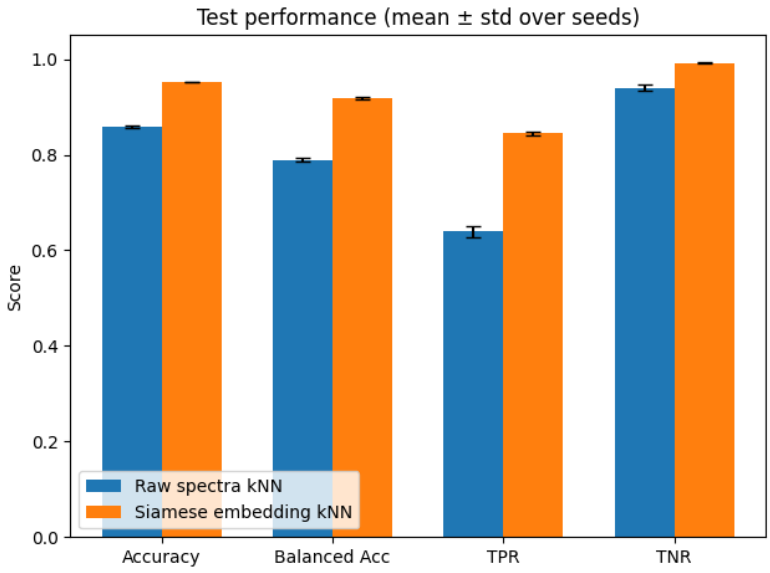


Figure 4.4: Comparison of test-set metrics between the Siamese embedding kNN and the raw-feature kNN baseline.

4.5.2 Temporal Analysis

The pre-flare test spectra were binned by time-to-flare (TTF) and the Siamese kNN classifier was evaluated on each bin separately, following Panos and Kleint [6]. Each bin was combined with the full non-flaring test set for evaluation.

Table 4.4 and Figure 4.5 present the results. The ROC-AUC remains essentially constant across all time bins, ranging from 0.984 at 120–180 minutes to 0.993 at 0–30 minutes, with no monotonic trend.

Table 4.4: Classification metrics by time-to-flare bin, evaluated against the full non-flaring test set.

TTF bin	N_{PF}	ROC-AUC	PR-AUC	Bal. Acc.	F1	Precision	Recall
0–30 min	9,012	0.993	0.917	0.959	0.829	0.745	0.935
30–60 min	5,572	0.991	0.870	0.954	0.757	0.641	0.924
60–90 min	6,342	0.996	0.904	0.970	0.792	0.677	0.955
90–120 min	6,089	0.988	0.826	0.920	0.735	0.643	0.856
120–180 min	12,638	0.984	0.894	0.928	0.831	0.792	0.873

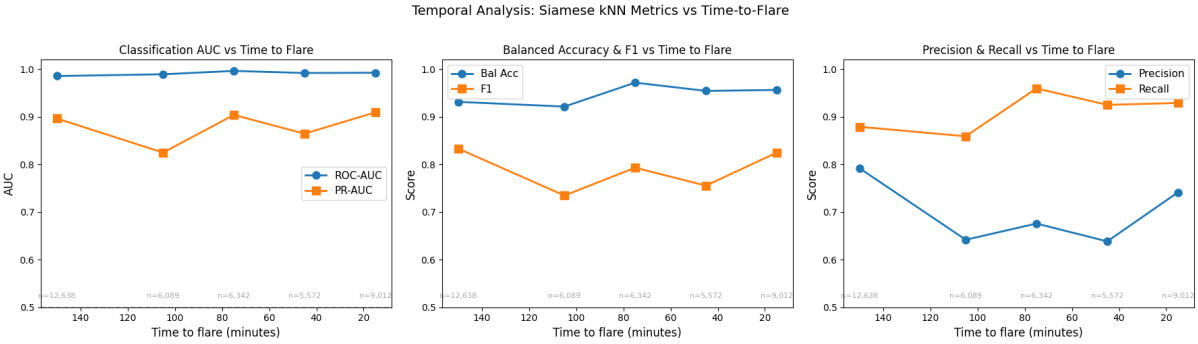


Figure 4.5: Temporal analysis: Siamese kNN classification metrics as a function of time-to-flare. The x-axis is inverted so that time flows left-to-right toward flare onset. Unlike the monotonic improvement reported by Panos and Kleint [6] for chromospheric Mg II spectra, the coronal EUV metrics remain approximately constant across the full 0–180 minute range.

A per-spectrum Spearman correlation between classification score and TTF yields $\rho = -0.092$ ($\rho^2 = 0.008$, $p < 0.001$ on $N = 39{,}691$), statistically significant only due to sample size and consistent with the flat per-bin AUC. This contrasts with Panos and Kleint [6], who observed IRIS Mg II AUC rising from 0.50 to 0.97 within 35 minutes of onset: the flat coronal profile suggests EIS spectra encode a *persistent* property of flare-productive active regions rather than a short-timescale precursor.

4.5.3 Within-Class Temporal Separation

A separate Siamese encoder was trained on pre-flare spectra alone to distinguish "early" (TTF > 90 min) from "late" (TTF < 30 min) spectra. The within-class dataset comprised 207,841 training, 25,837 validation, and 27,777 test spectra; a smaller encoder ($69 \to 128 \to 64 \to 32$) was trained for 5 epochs under the same contrastive framework.

Table 4.5: Within-class temporal separation: early pre-flare (TTF > 90 min) vs. late pre-flare (TTF < 30 min) on the test set. Both approaches achieve near-chance performance.

Method	ROC-AUC	PR-AUC	Balanced Acc.
Siamese kNN ($K = 51$)	0.683	0.505	0.563
Raw kNN ($K = 15$)	0.722	0.549	0.581
Chance baseline	0.500	0.324	0.500

The near-chance performance (Table 4.5) is itself the key result: the raw kNN baseline actually outperforms the Siamese approach, indicating that the encoder found nothing to learn.

This confirms that the flat temporal profile is a genuine physical property of the data, not a methodological artefact.

4.5.4 Per-Observation Breakdown

Per-observation classification rates were computed for the 157 test observations (Figure 4.6). Median recall across 48 pre-flare observations is 0.992, with 30 reaching ≥ 0.95 and five falling below 0.75; median TNR across 49 non-flaring observations is 0.998 with none below 0.86.

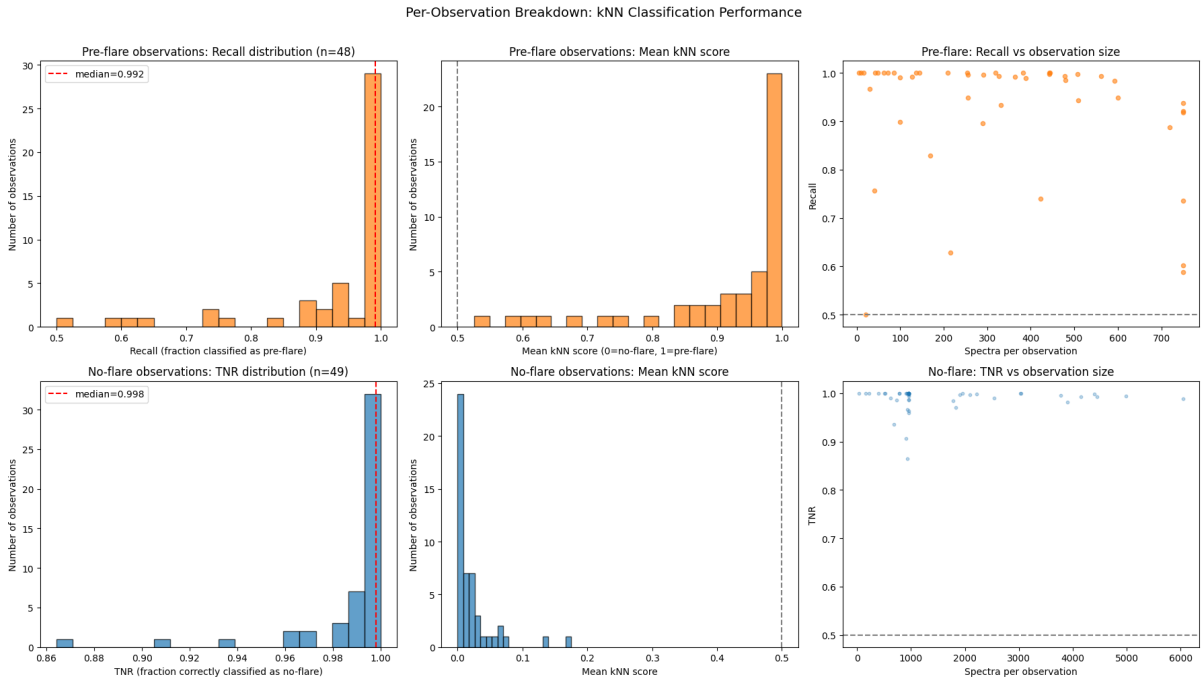


Figure 4.6: Per-observation classification performance. **Top row:** Pre-flare observations (recall distribution, mean kNN score, recall vs. observation size). **Bottom row:** Non-flaring observations (TNR distribution, mean kNN score, TNR vs. observation size). The model generalises across the majority of observations, with a small tail of harder pre-flare cases.

Investigating Poorly-Classified Pre-flare Observations

Line-shape descriptors for the five pre-flare observations with recall below 0.75 were compared to well-classified pre-flare observations (recall ≥ 0.90, $n = 38$) and to non-flaring observations (Table 4.6).

Table 4.6: Median line-shape descriptors for poorly-classified pre-flare observations (recall < 0.75), well-classified pre-flare observations (recall ≥ 0.90), and non-flaring observations. PF = pre-flare; NF = non-flaring.

Line	Descriptor	No-flare	Poor PF	Good PF	Poor≈NF?
Fe XII 195	Width (bins)	1.664	1.681	1.867	Yes
Fe XII 195	Centroid (bins)	11.260	11.218	11.609	Yes
Fe XII 195	Peak	0.279	0.269	0.255	Yes
Fe XIV 265	Width (bins)	3.990	2.250	2.670	No
Fe XV 284	Width (bins)	3.885	2.358	1.948	No

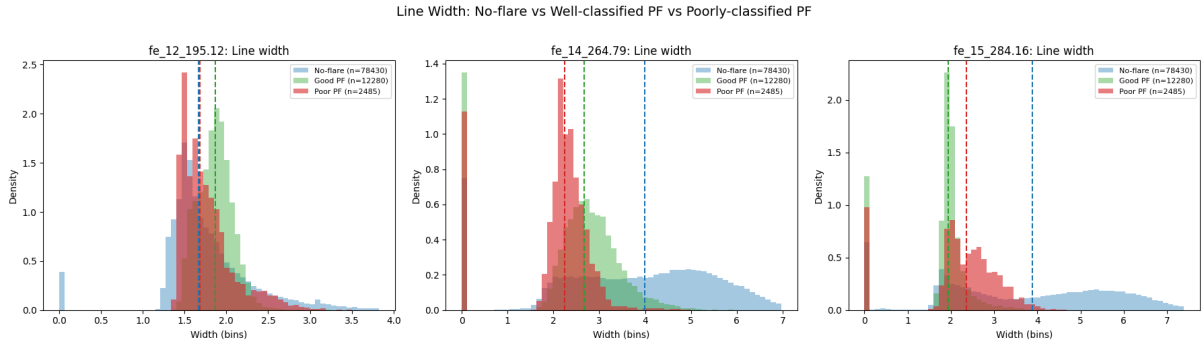


Figure 4.7: Line width distributions for three groups: non-flaring (blue), well-classified pre-flare (green), and poorly-classified pre-flare (red) observations. For Fe XII (left), the poorly-classified pre-flare observations overlap with non-flaring, explaining their misclassification.

These observations have Fe XII profiles resembling non-flaring spectra across all three descriptors while hot lines remain partially pre-flare-like, confirming Fe XII as the primary discriminant (consistent with its ΔPR-AUC = 0.498). They also tend to be further from flare onset (median TTF = 110.5 min vs. 76.5 min), though the small sample ($n = 5$) limits this.

4.5.5 Permutation Feature Importance

Each line's 23 wavelength bins were randomly shuffled across test spectra, the perturbed spectra re-embedded, and kNN classification repeated five times per line for stability.

Table 4.7: Permutation importance by spectral line. ΔPR-AUC is the drop from the unperturbed baseline (PR-AUC = 0.966). Larger drops indicate greater importance.

Spectral line	Perturbed PR-AUC	ΔPR-AUC	ΔROC-AUC
Fe XII 195.12 Å	0.466 ± 0.003	0.498	0.205
Fe XV 284.16 Å	0.591 ± 0.001	0.373	0.219
Fe XIV 264.79 Å	0.718 ± 0.001	0.246	0.122

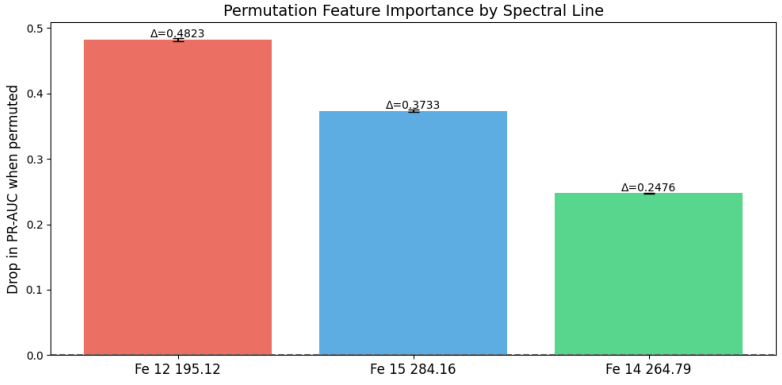


Figure 4.8: Permutation feature importance by spectral line. Fe XII 195.12 Å is the most important line (ΔPR-AUC = 0.498), but all three contribute non-redundant information, validating the multi-line fusion approach.

All three lines contribute non-redundant information (Table 4.7), with Fe XII as the most

important, consistent with its status as the strongest coronal diagnostic [31] and corroborated by the poor-observation analysis.

4.5.6 Physical Properties of Pre-flare Spectra

To understand what physical properties drive the classification, line-shape descriptors were computed for each spectral line and compared between classes using Cohen's d effect size (Table 4.8).

Table 4.8: Line-shape descriptor differences between pre-flare and non-flaring test spectra. The dominant discriminant is Fe XV width ($d = -1.05$, a large effect): pre-flare spectra have systematically narrower hot-line profiles. All KS p-values $< 10^{-24}$.

Line	Descriptor	NF median	PF median	Cohen's d
Fe XV 284	Width (bins)	3.08	1.94	−1.046
Fe XIV 265	Width (bins)	3.42	2.66	−0.680
Fe XII 195	Centroid (bins)	11.28	11.59	+0.372
Fe XII 195	Peak-to-wing ratio	452.6	349.3	−0.300
Fe XII 195	Peak intensity	0.279	0.260	−0.298
Fe XII 195	Width (bins)	1.59	1.81	+0.266

The dominant feature is Fe XV width ($d = -1.05$, large effect [39]): pre-flare spectra have systematically narrower hot-line profiles (Figure 4.9); Fe XIV shows the same pattern ($d = -0.68$), while cooler Fe XII shows slight broadening ($d = +0.27$) and redshift ($d = +0.37$). Spearman correlations confirm the ranking ($\rho = -0.41$ for Fe XV width). Physical interpretation (non-thermal broadening, magnetic ordering, downflows) is in Appendix B.5.

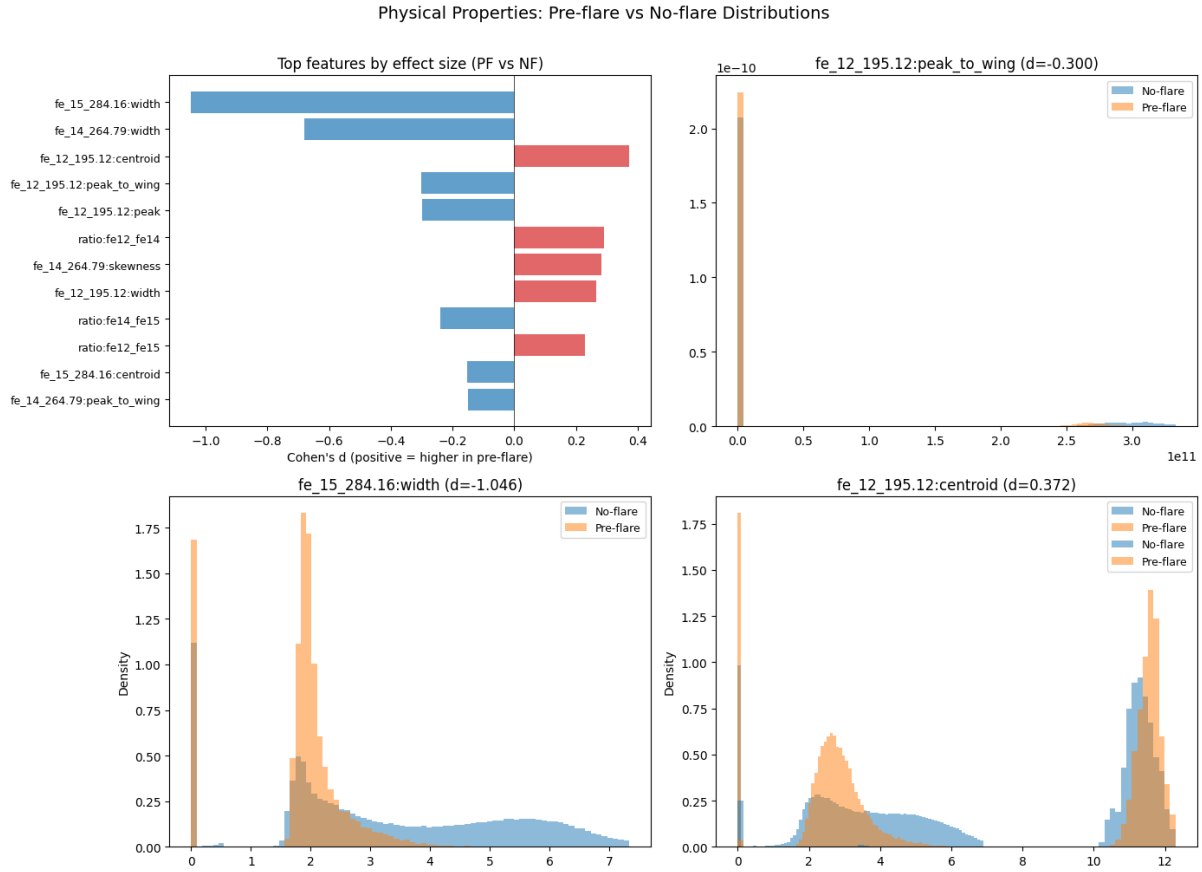


Figure 4.9: Physical property analysis. Features ranked by Cohen's d identify Fe XV width as the dominant discriminant ($d = -1.05$); the temperature-dependent pattern (hot lines narrower, cool lines slightly broader in pre-flare spectra) is consistent with a more ordered, less turbulent corona in flare-productive regions, the physical interpretation developed in Appendix B.5.

4.5.7 Embedding Visualisation

Figure 4.10 shows UMAP [40] projections of the Siamese embeddings. The class-label panel confirms clear pre-flare/non-flaring separation, but the TTF-coloured panels show no temporal gradient within the pre-flare cluster, reinforcing the absence of a temporal precursor signal.

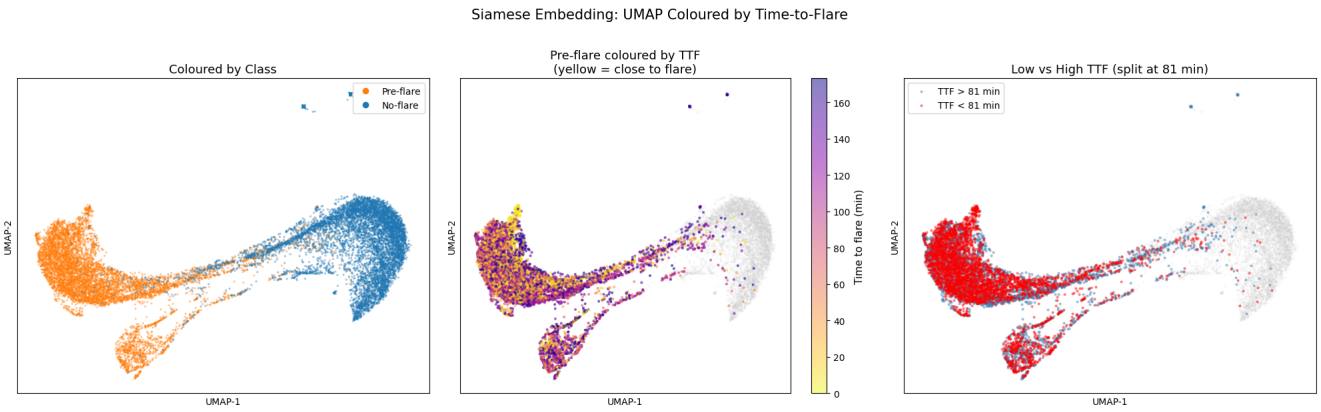


Figure 4.10: UMAP projection of Siamese embeddings (15,000 test spectra). **Left:** Coloured by binary class label. **Centre:** Pre-flare spectra coloured by continuous time-to-flare. **Right:** Pre-flare spectra split at the median TTF of 81 minutes. The lack of segregation by TTF confirms the absence of a temporal gradient.

4.6 Discussion

4.6.1 The Nature of the Learned Signal

The high classification performance combined with the flat temporal profile and within-class null result (Section 4.5.3) confirms the classifier identifies a *persistent* property of flare-productive active regions, not a temporal precursor. This contrasts with Panos and Kleint [6], who observed monotonically rising performance within 35 minutes of onset for chromospheric Mg II spectra. The difference reflects formation height: Mg II forms where pre-flare heating produces measurable spectral changes [41], while Fe XII/XIV/XV encode magnetic topology stable on hour timescales.

4.6.2 Multi-line Fusion

Permuting any single line degrades PR-AUC by 0.25–0.50, and no individual line matches the fused set. The line ablation study (Section 5.3.7) confirms this: Fe XII alone reaches 0.970, three-line fusion 0.996. The temperature-dependent width pattern, where cool and hot lines show *opposite* trends, is itself a discriminative feature only captured through multi-line analysis.

4.6.3 Comparison with Panos and Kleint (2020)

Table 4.9 summarises the key differences between this work and Panos and Kleint [6].

Table 4.9: Comparison with Panos and Kleint [6]. The AUC values measure different tasks and are not directly comparable; the key distinction is in the temporal profile.

	This work	Panos and Kleint [6]
Instrument	*Hinode*/EIS	IRIS
Spectral lines	Fe XII/XIV/XV	Mg II h & k
Formation temperature	1.3–2.1 MK (corona)	6–10 kK (chromosphere)
Feature representation	69 spectral bins	10 hand-crafted features
Classifier	Siamese MLP + kNN	2-layer DNN
Training data	∼2.5M spectra	∼9M spectra
Task and evaluation (not directly comparable)		
Task type	Cross-AR (PF vs. NF AR)	Within-AR (PF vs. same AR)
AUC	0.990	0.86
Temporal behaviour (directly comparable)		
Temporal trend	Flat (0–180 min)	Monotonic rise (35 min)
Within-class temporal	Not separable (PR-AUC 0.50)	Not tested
Most important feature	Fe XII width / centroid	Triplet emission

The critical distinction is the temporal behaviour: Panos and Kleint demonstrated a genuine chromospheric precursor, while the coronal spectra here show a persistent signature without temporal evolution. Limitations of this analysis, including the effect of area normalisation and the inner-join fusion strategy, are discussed in Section 9.5.

4.7 Summary

Unsupervised k-means clustering revealed a hybrid structure: two of 100 clusters were exclusively pre-flare (5.3% of the pre-flare population) while the remaining 98 contained spectra from both classes at varying enrichments, indicating the pre-flare signal is partly a discrete sub-population and partly a continuous distributional gradient. Under supervised metric learning, coronal EUV spectra from flare-productive active regions are highly distinguishable from non-flaring spectra (ROC-AUC = 0.990), but the signal is temporally persistent rather than a precursor. The dominant discriminant is the temperature-dependent line width pattern, with narrower hot-line profiles and broader, redshifted Fe XII profiles in pre-flare regions.

Chapter 5

Supervised Classification and Anomaly Detection

5.1 Introduction

This chapter trains five additional classifiers on the same dataset to test whether the Siamese findings from Chapter 4 are robust across model families: logistic regression, XGBoost [13], a multilayer perceptron (MLP), a 1D convolutional neural network (CNN), and an unsupervised autoencoder. The cross-architecture comparison also indicates whether simpler or more powerful models reach comparable performance and independently verifies the absence of a temporal precursor signal.

5.2 Experimental Setup

All models were trained on the same 69-dimensional fused feature vectors described in Section 4.2, using the identical observation-level train/validation/test split (80/10/10%). Class imbalance (approximately 5:1 non-flaring to pre-flare) was addressed through model-specific mechanisms as described below.

5.2.1 Model Descriptions

Logistic Regression. The simplest baseline tests whether the pre-flare signal is linearly separable in the 69-dimensional feature space. Balanced class weights (inversely proportional to class frequency) were used with L-BFGS optimisation (chosen for convergence stability on the ∼2M-sample training set) and regularisation strength $C = 1.0$ (the scikit-learn default).

XGBoost. Gradient-boosted trees [13] were chosen because their axis-aligned splits make them naturally interpretable through feature importance, allowing direct comparison with the Siamese permutation importance. The model used 300 estimators, maximum depth 6, learning rate 0.1, subsample ratio 0.8, and column subsample ratio 0.8. Class imbalance was handled via `scale_pos_weight` (5.11), with early stopping on validation PR-AUC (patience 20). The remaining hyperparameters (depth, learning rate, subsampling ratios) follow standard defaults

for gradient-boosted trees; early stopping used 299 of 300 maximum estimators, indicating the ensemble was near-saturated and additional trees would provide minimal further improvement.

MLP. To isolate the effect of the training objective, this multilayer perceptron uses the same architecture as the Siamese encoder ($69 \to 256 \to 256 \to 1$) but is trained with binary cross-entropy (the standard loss for sigmoid-output binary classification) rather than contrastive loss. Training used AdamW [38] defaults (learning rate 10^{-3}, weight decay 10^{-4}, batch size 2,048, 20 epochs); validation PR-AUC saturates well before the epoch budget.

1D CNN. To test whether local spectral shape features carry additional discriminative information, a convolutional architecture reshapes the 69-dimensional input into 3 channels $\times$ 23 bins, with two convolutional layers (32 and 64 filters, kernel size 5), adaptive average pooling, and a fully connected head ($64 \to 64 \to 1$). Training mirrored the MLP regimen; validation PR-AUC saturates well before the epoch budget, and the tight multi-seed cross-split variance (Table 5.2) further confirms generalisation. The CNN's lower performance compared to the MLP (Section 5.3.4) suggests the discriminative information is encoded in global summary statistics rather than local spectral morphology.

Autoencoder (anomaly detection). This model tests an alternative problem formulation: it learns to reconstruct non-flare spectra and flags pre-flare spectra as anomalies via high reconstruction error. The architecture is $69 \to 128 \to 64 \to 16 \to 64 \to 128 \to 69$ (16-dim bottleneck, 4$\times$ compression). Trained exclusively on non-flare spectra with MSE loss (the standard reconstruction objective for continuous-valued autoencoders); the reconstruction error threshold was optimised on the validation set.

5.3 Results

5.3.1 Classification Performance

Table 5.1 and Figures 5.1–5.2 present the test-set performance of all models, including the Siamese kNN and raw kNN results from Chapter 4 for direct comparison.

Table 5.1: Test-set metrics for all models ($N = 221{,}051$), ordered by PR-AUC. XGBoost and MLP achieve near-identical top performance (ROC-AUC = 0.996), confirming the pre-flare signal is robust across model families. 95% bootstrap confidence intervals shown where available.

Model	ROC-AUC	PR-AUC	Bal. Acc.	F1	Recall
XGBoost	0.996 [0.996, 0.996]	0.984 [0.983, 0.985]	0.969 [0.968, 0.970]	0.907	0.976
MLP	0.996 [0.995, 0.996]	0.982 [0.981, 0.983]	0.969 [0.969, 0.970]	0.908	0.977
Siamese kNN (Ch. 4)	0.990 [0.990, 0.991]	0.966 [0.965, 0.967]	0.946 [0.944, 0.947]	0.915	0.908
1D CNN	0.984 [0.984, 0.985]	0.943 [0.941, 0.944]	0.932 [0.930, 0.933]	0.813	0.947
Raw kNN (Ch. 4)	0.984 [0.983, 0.984]	0.935 [0.933, 0.937]	0.924 [0.922, 0.925]	0.853	0.891
Logistic Reg.	0.932 [0.931, 0.933]	0.683 [0.678, 0.688]	0.869 [0.867, 0.871]	0.680	0.903
Autoencoder	0.199	0.110	0.487	0.296	0.932

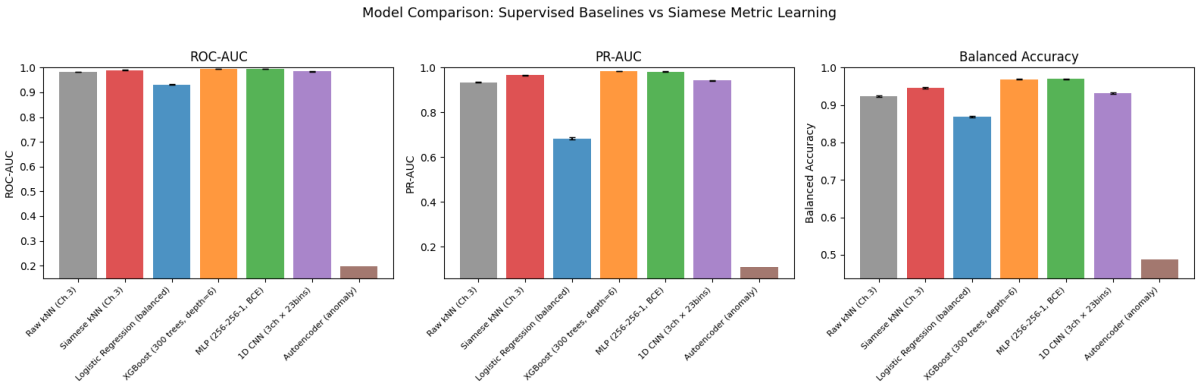


Figure 5.1: Test-set metrics for all seven models. The convergence of five architecturally independent classifiers (linear, tree-ensemble, shallow neural, deep convolutional, metric-learning) on near-identical performance demonstrates the pre-flare signal is a property of the data, not of any single modelling approach. The autoencoder failure is itself informative (Section 5.3.5).

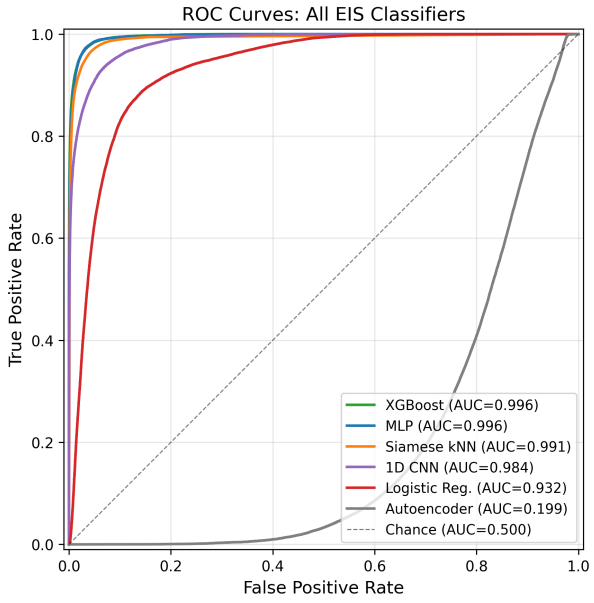


Figure 5.2: ROC curves for all six classifiers on the held-out test set ($N = 221{,}051$). The four supervised classifiers cluster tightly between AUC 0.984–0.996; logistic regression at 0.932 reveals a substantial nonlinear component; the autoencoder at 0.199 falls below the diagonal because pre-flare profiles are simpler shapes than non-flare profiles.

The results reveal a clear performance hierarchy: XGBoost $\approx$ MLP $>$ Siamese kNN $>$ 1D CNN $\approx$ Raw kNN $>$ Logistic Regression $\gg$ Autoencoder. XGBoost and the MLP achieve nearly identical ROC-AUC (0.996) and PR-AUC (0.982–0.984), marginally exceeding the Siamese kNN (0.990 / 0.966). The autoencoder anomaly detector fails entirely, scoring below chance.

Each classifier was retrained with five independently sampled observation-disjoint splits (matching the standard 5-fold convention) to verify the hierarchy is not partition-specific (Table 5.2). Cross-seed standard deviations are uniformly small (≤ 0.004 ROC-AUC for the four supervised classifiers, ≤ 0.015 for the autoencoder), and the pairwise ordering is preserved on every seed; the pre-flare signal is robust both across model architectures and across data partitioning.

Table 5.2: Multi-seed test-set classification metrics (mean ± std across 5 random seeds, observation-disjoint splits). Tight cross-seed variance and preserved ordering confirm the cross-architecture hierarchy is reproducible.

Model	ROC-AUC (mean ± std)	PR-AUC (mean ± std)
XGBoost	0.994 ± 0.002	0.975 ± 0.007
MLP	0.994 ± 0.003	0.976 ± 0.008
1D-CNN	0.987 ± 0.004	0.946 ± 0.013
LR	0.945 ± 0.008	0.742 ± 0.031
Autoencoder	0.309 ± 0.015	0.103 ± 0.020

5.3.2 Logistic Regression: Linear Separability

Logistic regression achieves ROC-AUC = 0.932 but PR-AUC = 0.683, indicating substantial linear separability with a far-from-optimal boundary. The jump to XGBoost (PR-AUC 0.984) demonstrates a significant nonlinear component that tree-based and neural models exploit.

5.3.3 XGBoost Feature Importance

Figure 5.3 presents the XGBoost feature importance (information gain) across all 69 wavelength bins. The importance is concentrated in two regions: the central bins of Fe XII 195.12 Å (bins 8–14, corresponding to the line core and near wings) and specific bins in Fe XV 284.16 Å. This is consistent with the permutation importance ranking from Chapter 4, which identified Fe XII as the most important line followed by Fe XV. The convergence of two entirely different importance measures on the same feature ranking strengthens the conclusion that the pre-flare signal is primarily encoded in the Fe XII line core shape and the Fe XV line width.

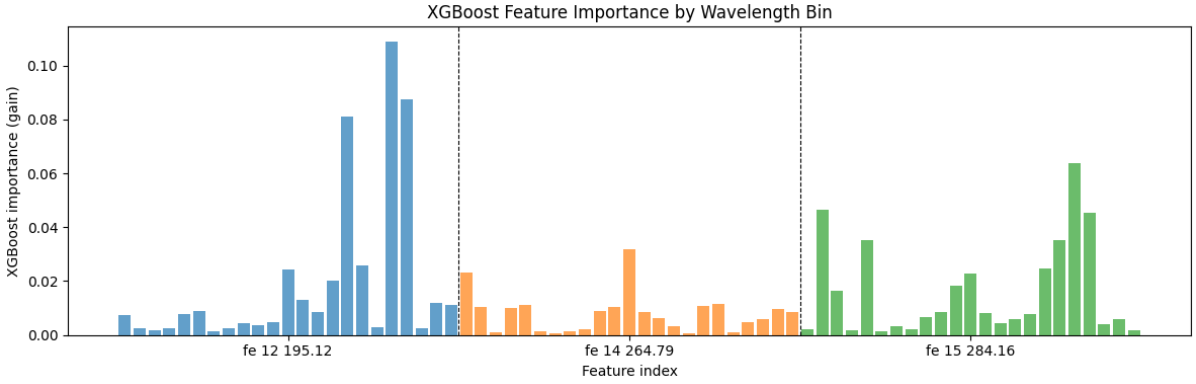


Figure 5.3: XGBoost feature importance (information gain) by wavelength bin. The x-axis shows the 69 feature indices (3×23 bins); dashed vertical lines separate the three spectral lines. The dominant features are the central bins of Fe XII 195.12 Å and specific bins of Fe XV 284.16 Å, consistent with the permutation importance ranking from Chapter 4.

5.3.4 1D CNN: Local vs. Global Features

The 1D CNN reaches ROC-AUC = 0.984 / PR-AUC = 0.943, below the MLP (0.996 / 0.982) despite an architecture designed for local spectral shape features. This indicates the discriminative information is encoded in global summary statistics (line width, centroid, peak-to-wing ratio)

rather than local morphology. The CNN nonetheless achieves strong classification, confirming the signal is robust across architectures.

5.3.5 Autoencoder Anomaly Detection

The autoencoder anomaly detector produces an ROC-AUC of 0.199, substantially below the chance level of 0.500. This means that pre-flare spectra have *lower* reconstruction error than non-flare spectra, the opposite of the anomaly detection hypothesis.

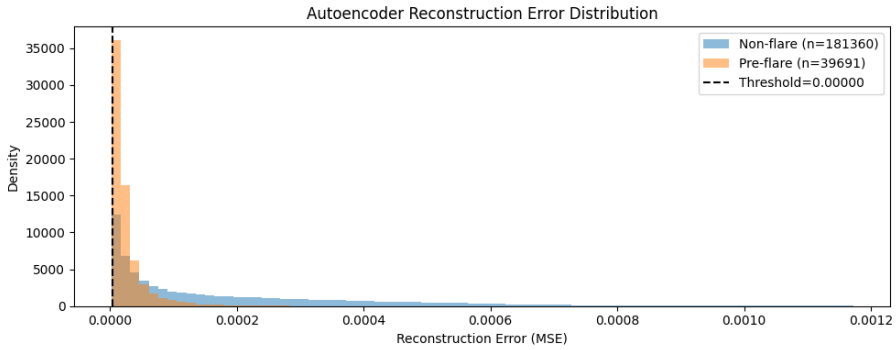


Figure 5.4: Autoencoder reconstruction error distribution. Pre-flare spectra (orange) are reconstructed *more* accurately than non-flare spectra (blue), causing the anomaly detection framing to fail: the narrower, more peaked pre-flare profiles are simpler shapes that the autoencoder reproduces with lower error.

Figure 5.4 reveals the explanation. Pre-flare spectra have narrower, more peaked line profiles (Fe XV width Cohen's $d = -1.05$), which are simpler spectral shapes that an autoencoder can reconstruct more accurately. Non-flare spectra, with their broader and more variable profiles, are harder to reconstruct. This demonstrates that pre-flare spectra are not anomalous outliers but rather a distinct sub-population with *simpler* spectral structure than the non-flare background, and provides independent physical confirmation of the line-width findings from Chapter 4.

5.3.6 Temporal Analysis

All supervised models were evaluated on pre-flare test spectra grouped by time-to-flare. Table 5.3 presents the results: all four models exhibit flat temporal profiles (ROC-AUC varies by less than 0.02 across the 0–180 minute range), confirming the temporal null result established by the Siamese analysis. Combined with the Siamese kNN and within-class Siamese results, the temporal null result is now confirmed by six independent models spanning four ML paradigms.

Table 5.3: Temporal analysis for the supervised models. All models exhibit flat temporal profiles, consistent with Chapter 4.

	0–30 min	30–60 min	60–90 min	90–120 min	120–180 min
ROC-AUC					
XGBoost	0.998	0.996	0.998	0.994	0.995
MLP	0.996	0.996	0.998	0.994	0.994
1D CNN	0.987	0.988	0.993	0.982	0.977
Log. Reg.	0.921	0.948	0.944	0.929	0.929
PR-AUC					
XGBoost	0.968	0.943	0.962	0.898	0.948
MLP	0.951	0.928	0.958	0.902	0.943
1D CNN	0.855	0.833	0.881	0.796	0.838
Log. Reg.	0.334	0.280	0.296	0.244	0.408

Per-spectrum Spearman correlations between classification score and time-to-flare are uniformly weak across models (Table 5.4; XGBoost $\rho = -0.119$, $\rho^2 = 0.014$).

Table 5.4: Per-spectrum Spearman correlation between classification score and time-to-flare for pre-flare test spectra ($N = 39{,}691$). All correlations are statistically significant ($p < 0.001$) but negligible in magnitude.

Model	ρ	95% CI	ρ^2
XGBoost	-0.119	$[-0.128, -0.109]$	0.014
1D CNN	-0.097	$[-0.107, -0.088]$	0.009
Siamese kNN	-0.092	$[-0.102, -0.082]$	0.008
MLP	-0.083	$[-0.092, -0.073]$	0.007
Logistic Reg.	-0.033	$[-0.043, -0.023]$	0.001

5.3.7 Line Ablation Study

Separate XGBoost models were trained on single-line, two-line, and full three-line feature subsets with identical hyperparameters (Table 5.5, Figure 5.5), avoiding the out-of-distribution problem of masking inputs through a pre-trained encoder.

Table 5.5: XGBoost classification performance by feature subset. Each model was trained independently on the indicated features using the same hyperparameters and observation-level split.

Feature set	N features	ROC-AUC	PR-AUC
Fe XII only	23	0.970	0.905
Fe XIV only	23	0.926	0.782
Fe XV only	23	0.944	0.777
Fe XII + XIV	46	0.990	0.963
Fe XII + XV	46	0.990	0.959
Fe XIV + XV	46	0.974	0.895
All three (fused)	69	0.996	0.984

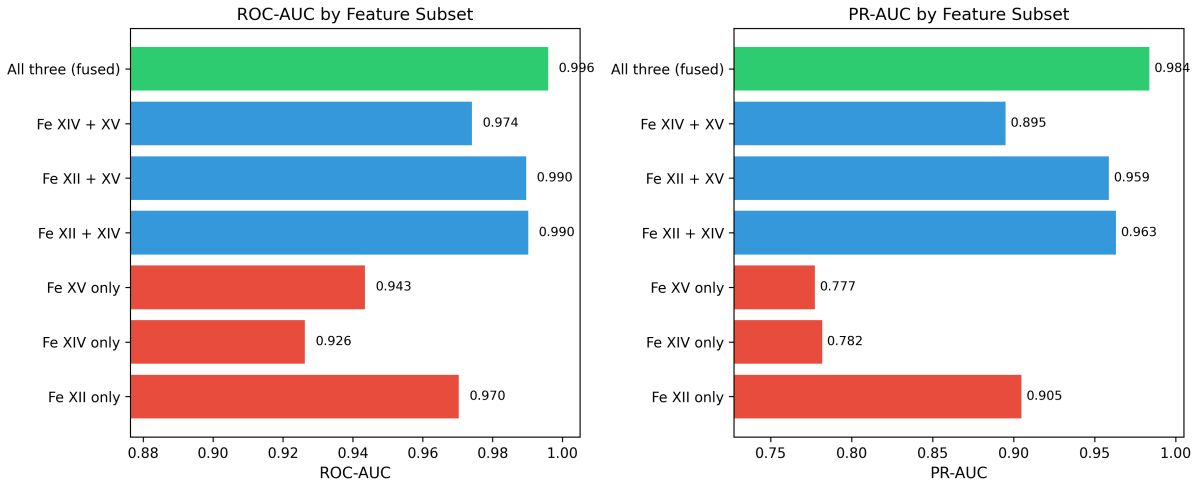


Figure 5.5: Line ablation: ROC-AUC and PR-AUC for XGBoost models trained on single-line (red), two-line (blue), and full-fusion (green) feature subsets.

Fe XII alone achieves ROC-AUC = 0.970, confirming its dominant role from permutation importance (Section 4.5.5). Any two-line combination including Fe XII reaches ≥ 0.990, while Fe XIV+XV without Fe XII reaches only 0.974. Three-line fusion adds a marginal 0.996, indicating diminishing returns: multi-line fusion is beneficial but Fe XII carries the majority of discriminative information.

5.3.8 False Positive Characterisation

The XGBoost classifier produces 7,018 false positives (3.9% of non-flare test spectra). To determine whether these reflect noise or physically meaningful misclassification, the spectral properties of false positives were compared to true negatives and true positives (Table 5.6, Figure 5.6).

Table 5.6: Median spectral descriptors for true negatives (TN), false positives (FP), and true positives (TP). "Intermediate" indicates the FP median falls between the TN and TP medians.

Descriptor	TN	FP	TP	Intermediate?
Fe XII width	1.583	1.640	1.817	Yes
Fe XII centroid	11.270	11.516	11.592	Yes
Fe XII kurtosis	2.248	2.050	1.833	Yes
Fe XV width	3.207	1.871	1.937	No
Fe XV kurtosis	1.345	1.158	0.953	Yes

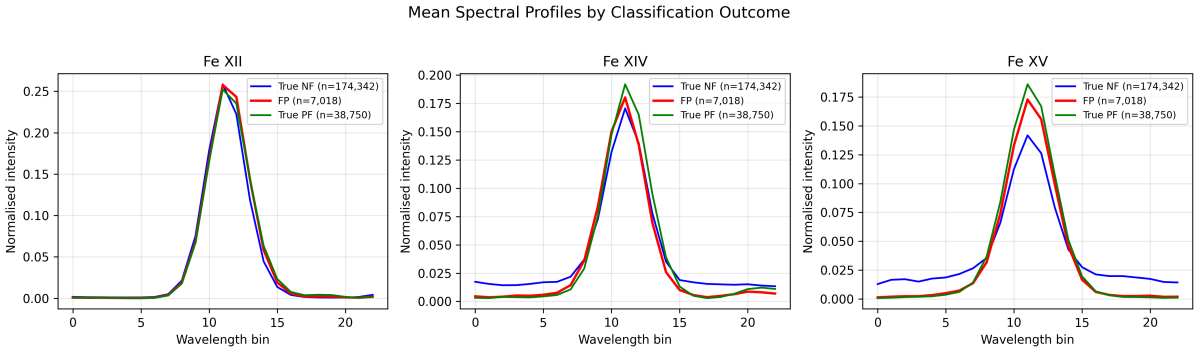


Figure 5.6: Mean spectral profiles for true negatives (blue), false positives (red), and true positives (green). False positives have intermediate Fe XII profiles but *more extreme* hot-line profiles than TP, supporting a continuous spectrum of pre-flare-readiness rather than a binary class boundary.

False positive descriptors are intermediate between true negatives and true positives for Fe XII but *narrower* than true positives for the hotter lines (Fe XV FP width 1.871 vs TP 1.937), suggesting these non-flaring regions have highly ordered coronae despite not producing flares. False positives concentrate in a small number of observations (top FP-rich observation: 48% FP rate), and per-observation mean Fe XV line width correlates negatively with mean XGBoost score ($\rho = -0.561$, $p < 10^{-10}$, Figure 5.7), confirming that narrower hot-line profiles drive higher prediction scores even within the non-flare class.

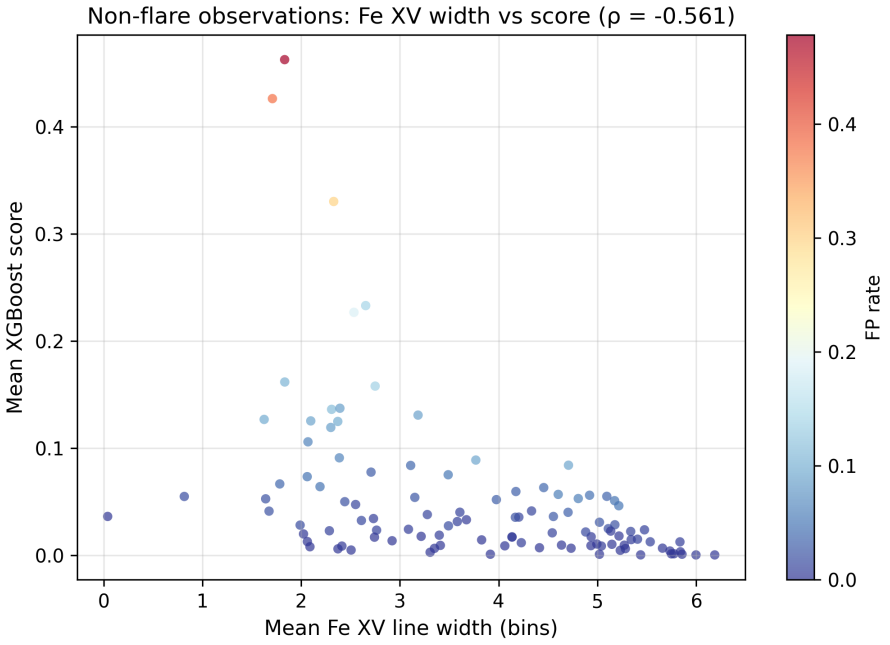


Figure 5.7: Per non-flare observation: mean Fe XV line width vs mean XGBoost score, coloured by false positive rate. Narrower Fe XV profiles correspond to higher prediction scores ($\rho = -0.561$), indicating the model identifies a continuous spectrum of magnetic complexity.

5.4 Coronal Line Profile Non-Gaussianity

This section presents an analysis motivated by the work of Jeffrey et al. [8, 9] and Polito et al. [42], who demonstrated that EIS Fe line profiles become non-Gaussian during solar flares. The analysis is extended here by asking whether non-Gaussian line shapes are present in *pre-flare* EIS spectra.

5.4.1 Higher-Order Moment Extraction

For each of the 2,456,945 EIS spectra, three higher-order moments were computed per line. Given an area-normalised spectral profile $I(\lambda_1), \ldots, I(\lambda_n)$ with mean $\bar{I}$ and standard deviation σ_I, the excess kurtosis and skewness are defined as:

$$\kappa = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{I(\lambda_i)-\bar{I}}{\sigma_I}\right)^4 - 3, \qquad \gamma = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{I(\lambda_i)-\bar{I}}{\sigma_I}\right)^3. \tag{5.1}$$

Higher kurtosis indicates a sharper central peak with extended wings relative to the bulk of the bin values, the morphological signature expected for non-Maxwellian κ-distributions [8]; non-zero skewness indicates profile asymmetry, the signature expected for Doppler-shifted bulk flows. The relationship between this profile-shape kurtosis and the formal κ-distribution parameter is empirical and is validated separately. Together with the standard deviation σ_I, this yields 9 moment features per spectrum (3 per Fe line).

Table 5.7: EIS higher-order moment statistics.

Feature	Pre-flare mean	No-flare mean	Difference
Fe XII kurtosis	1.852	2.361	−0.509
Fe XII skewness	1.804	1.921	−0.117
Fe XII std	0.077	0.077	0.000
Fe XIV kurtosis	1.499	1.493	+0.006
Fe XIV skewness	1.681	1.518	+0.163
Fe XIV std	0.058	0.054	+0.004
Fe XV kurtosis	0.999	1.372	−0.373
Fe XV skewness	1.560	1.479	+0.081
Fe XV std	0.060	0.055	+0.005

5.4.2 Kurtosis-Based Classification

Logistic regression on the 9 EIS moment features yielded:

Table 5.8: EIS classification ROC-AUC using different feature sets, evaluated under observation-disjoint train/test splits.

Feature set	ROC-AUC
EIS moments only (9 features)	0.959 [0.958, 0.960]
EIS raw + moments (78 features)	0.979 ± 0.000
EIS raw only (69 features, Chs. 4, 5)	∼0.990

To distinguish kurtosis-specific diagnostic power from generic dimensionality reduction, the same logistic regression was trained on three alternative 9-D representations: PCA-9 (the optimal linear projection, capturing 48% of variance), Gaussian random projection-9, and 9 randomly sampled wavelength bins (Table 5.9). Kurtosis-9 outperforms the optimal linear projection by 3.5 ROC-AUC points and 17 PR-AUC points, demonstrating that the diagnostic power resides in higher-order moments specifically rather than in 9-D representation generally.

Table 5.9: Comparison of 9-dimensional spectral representations under identical logistic regression and observation-disjoint splits. Kurtosis-9 dominates.

Representation	ROC-AUC (95% CI)	PR-AUC (95% CI)
Kurtosis-9	**0.959 (0.958, 0.960)**	**0.815 (0.811, 0.820)**
PCA-9	0.924 (0.923, 0.925)	0.648 (0.642, 0.654)
Random Projection-9	0.857 (0.856, 0.859)	0.469 (0.464, 0.474)
Random Bins-9	0.843 (0.841, 0.845)	0.408 (0.404, 0.413)

Fe XV 284.16 Å kurtosis has the highest logistic regression coefficient ($|w| = 9.88$). To the best of our knowledge, this is the first application of higher-order moment analysis to pre-flare EIS spectra; previous studies [8, 9, 42] examined non-Gaussianity only during the impulsive phase of flares. The physical basis of the result is validated through direct kappa-distribution fitting in Appendix B.4: pre-flare spectra are more near-Gaussian (median $\kappa \approx 10$) and no-flare spectra more strongly non-thermal (median $\kappa \approx 3$), opposite to the impulsive-phase direction reported by Jeffrey et al.

5.4.3 Temporal Evolution of Kurtosis

Despite the strong discriminative power, the kurtosis features show no significant temporal evolution within pre-flare observations ($\rho = +0.035$ for Fe XII, $\rho = +0.015$ for Fe XV, Figure 5.8). This is consistent with the persistent-state interpretation established by the Siamese analysis (Chapter 4) and the supervised baselines earlier in this chapter.

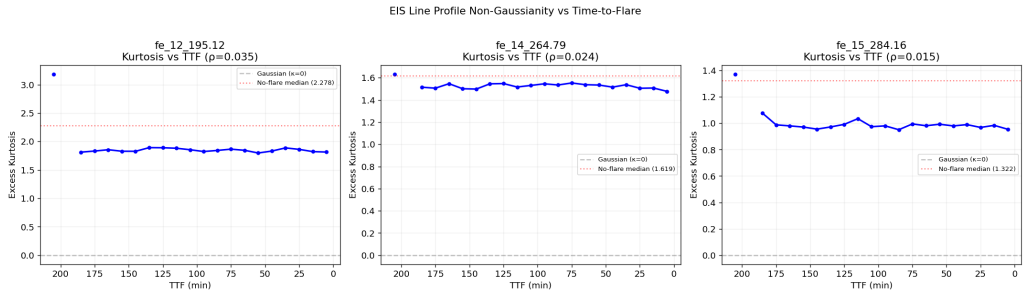


Figure 5.8: EIS line profile kurtosis as a function of time-to-flare. Each panel shows the mean kurtosis value in 30-minute temporal bins for one Fe line. Despite the strong discriminative power of kurtosis for pre-flare classification (ROC-AUC = 0.959), the kurtosis values remain flat across the 0–210 minute pre-flare window ($\rho = +0.035$ for Fe XII, $\rho = +0.015$ for Fe XV), confirming that non-Gaussianity is a persistent property of flare-productive active regions rather than a temporal precursor.

5.5 Discussion

Strong performance across structurally different model families confirms the pre-flare signal is robust, not an artefact of any single approach. The autoencoder's below-chance performance shows anomaly detection is inappropriate when the target class has lower complexity than the background.

5.6 Summary

Five models confirmed the Siamese findings: XGBoost and MLP achieved the highest performance (ROC-AUC = 0.996), logistic regression established a nonlinear component in the signal, the autoencoder failed because pre-flare spectra have simpler morphology, and all models exhibited flat temporal profiles extending the null result to six independent classifiers.

Chapter 6

Spatial Analysis of Pre-flare Spectral Signatures

6.1 Introduction

The analyses in Chapters 4 and 5 treated every spatial pixel equally, aggregating predictions across entire rasters to produce observation-level metrics. This chapter asks a complementary question: *where* within an active region raster does the model detect the pre-flare signature? If the signal is spatially concentrated, for instance near polarity inversion lines or loop footpoints, this would suggest a localised physical origin; if instead it pervades the entire field of view, the signal is better interpreted as a global property of the active-region corona. Panos *et al.* [26] performed a related analysis for IRIS chromospheric spectra, finding that the spatial position of the model's maximum attention along the slit predicted the location of the flare's peak UV emission 78% of the time; this chapter maps per-pixel XGBoost prediction scores back onto the 2D raster geometry and quantifies the spatial distribution of the pre-flare signal in coronal EIS spectra.

6.2 Method

6.2.1 Per-Pixel Score Maps

The XGBoost classifier from Chapter 5 (top-performing model, ROC-AUC = 0.996) assigns a prediction score $P(\text{pre-flare}) \in [0, 1]$ to every test spectrum. Each spectrum retains its spatial (y, x) and observation index, allowing per-pixel scores to be mapped back to the original raster grid. Pixels absent from the fused dataset (dropped during the inner join across lines) are marked as NaN.

6.2.2 Rank-Normalised Maps

The raw XGBoost scores are highly polarised (pre-flare median 0.995, non-flare median 0.002), making raw score maps for pre-flare observations appear nearly uniformly red. To reveal within-observation spatial structure, a per-observation rank normalisation was applied: pixel scores

were replaced by their rank percentile, mapping the lowest-scoring pixel to 0 and the highest to 1.

6.2.3 Spatial Concentration Metrics

For each observation, the following metrics were computed from the raw score map: the fraction of pixels with score > 0.5, the Gini coefficient (where 0 is perfectly uniform and 1 is maximally concentrated), the number of connected clusters of high-scoring pixels, the largest cluster fraction, and the spatial standard deviation. These metrics were compared between pre-flare and non-flare observations using the Mann-Whitney U test.

6.2.4 Multi-Raster Temporal Sequences

The test set contains 44 unique flare events, of which 16 have three or more consecutive rasters of the same active region at different times-to-flare. For these sequences, the spatial mean score was tracked as a function of TTF to test whether the spatial distribution of the signal evolves approaching flare onset.

6.2.5 Score-Intensity Comparison

To verify the model responds to shape rather than brightness, raw total integrated intensity was computed per pixel from the original Fe XII 195.12 Å spectra (*before* area normalisation), and the Spearman rank correlation between raw intensity and prediction score was calculated per observation.

6.3 Results

6.3.1 The Pre-flare Signal Is Spatially Diffuse

Table 6.1 and Figure 6.1 present the spatial concentration metrics. Pre-flare observations exhibit nearly uniform prediction maps; non-flare observations are patchy and heterogeneous.

Table 6.1: Spatial concentration metrics for pre-flare ($n = 105$) and non-flare ($n = 109$) test observations. All differences are significant ($p < 0.05$, Mann-Whitney U test).

Metric	Pre-flare	Non-flare	p-value
Frac. pixels > 0.5	0.981	0.035	5.2×10^{-37}
Gini coefficient	0.035	0.830	6.4×10^{-36}
N connected clusters	1.2	15.8	2.4×10^{-9}
Largest cluster fraction	0.994	0.394	8.1×10^{-30}
Spatial std dev	0.084	0.105	4.6×10^{-2}

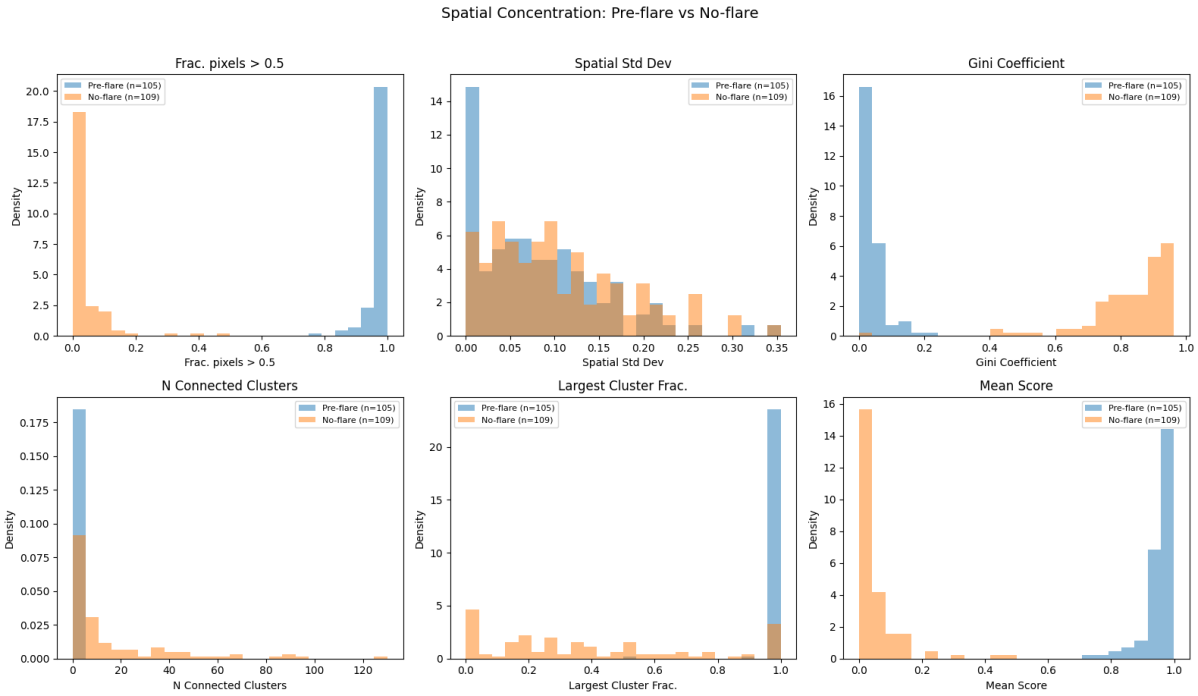


Figure 6.1: Distributions of spatial concentration metrics for pre-flare (blue) and non-flare (orange) observations. Pre-flare maps are nearly perfectly uniform (Gini ≈ 0.04, one connected cluster), while non-flare maps are patchy and heterogeneous (Gini ≈ 0.83, $\sim$16 clusters). This confirms the coronal pre-flare signature is a region-wide property, not localised to specific structures.

In pre-flare observations, 98.1% of pixels score above the 0.5 decision threshold and form a single connected cluster (Gini = 0.035); non-flare observations have 3.5% above threshold, distributed across $\sim$16 disconnected clusters. The *entire* corona above a flare-productive AR is spectrally different from a non-flaring one: the signal is not localised to footpoints, loops, or polarity inversion lines. All five differences remain significant after Bonferroni correction ($\alpha/5 = 0.01$).

6.3.2 Annotated Maps with Spectral Profiles

Individual spectra confirm the metrics. The highest-scoring pre-flare observation (mean score 0.999) shows characteristic narrow, peaked profiles with subtle within-observation variation, consistent with the low Gini. The highest-scoring non-flare observation (Figure 6.2, mean score 0.463) shows a core of high-scoring pixels with pre-flare-like profiles surrounded by broad non-flaring profiles, suggesting the model detects a physically real sub-region sharing the thermal properties of flare-productive regions.

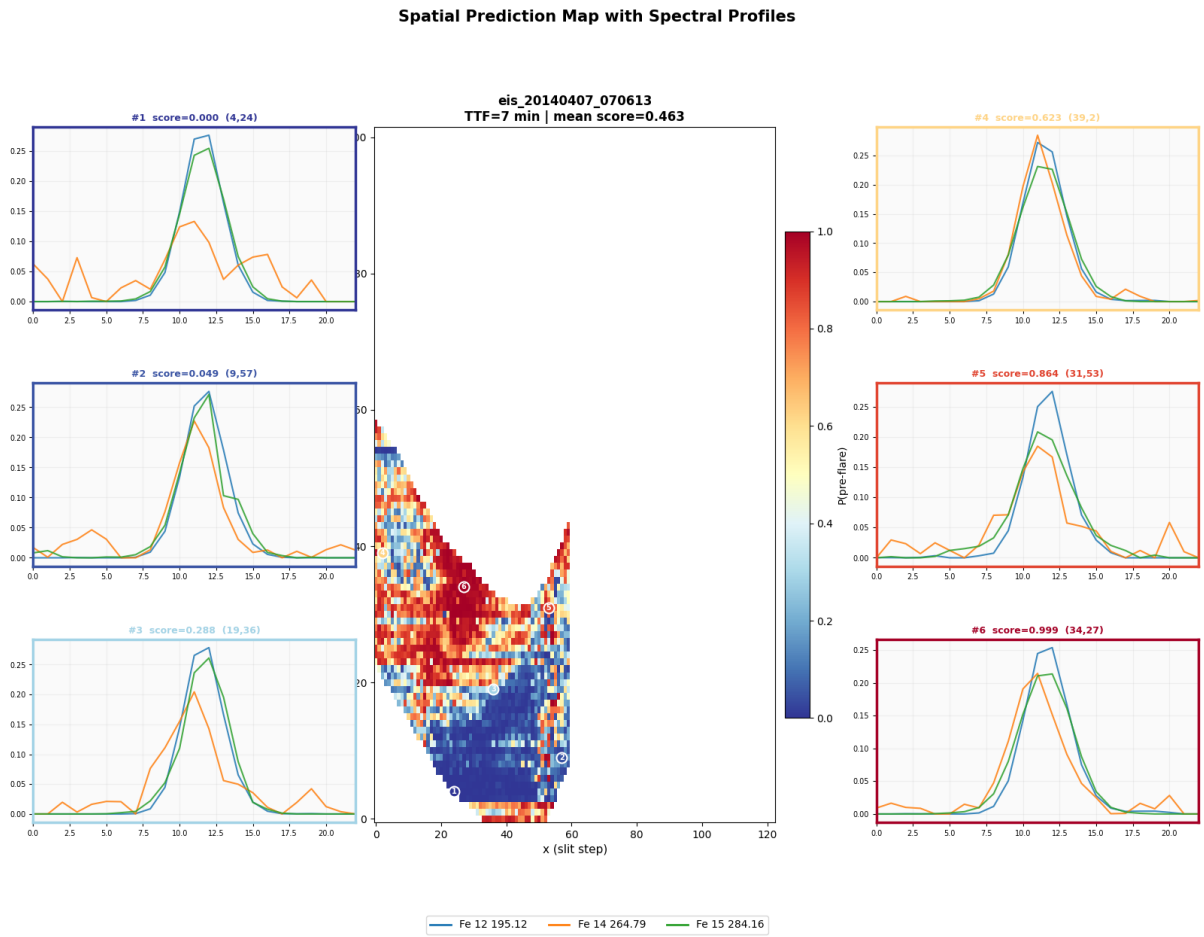


Figure 6.2: Annotated spatial prediction map for the highest-scoring non-flare observation (EIS raster from 7 April 2014, mean score = 0.463). The raster shows a concentrated core of high-scoring pixels with narrow, peaked spectra surrounded by low-scoring pixels with broad profiles.

6.3.3 Pre-flare vs Non-flare Spectral Comparison

Individual-pixel inspection (Figure 6.3) confirms at instance level the systematic Fe-line narrowing pattern observed at population level in Chapter 4 (Cohen's $d = -1.05$ for Fe XV width).

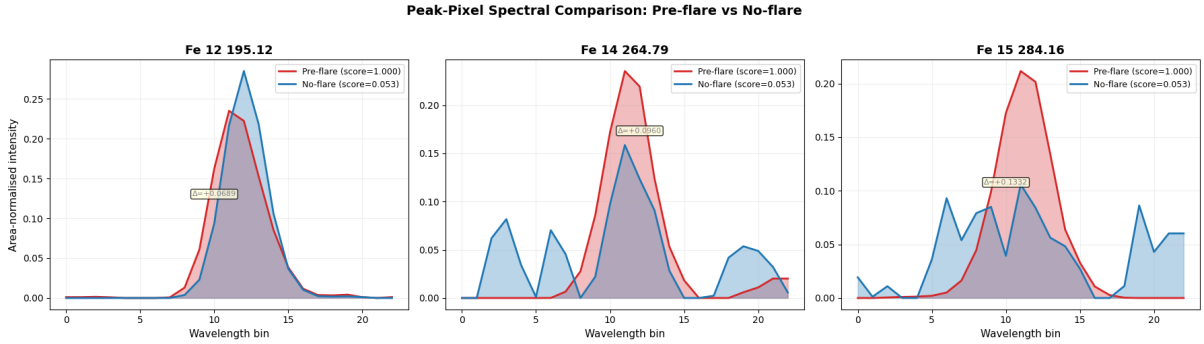


Figure 6.3: Highest-scoring pre-flare pixel (red, score 1.000) versus a representative non-flare pixel (blue, score 0.053). The systematically narrower hot Fe lines provide a single-pixel visualisation of the population-level Cohen's $d = -1.05$ effect that drives the cross-architecture classification convergence reported in Table 5.2.

6.3.4 Score vs Intensity

The Spearman correlation between raw Fe XII 195.12 Å integrated intensity and XGBoost score is $\rho = 0.233$ ($p = 7.6 \times 10^{-7}$) for the top pre-flare observation. Despite being statistically significant, this correlation is weak, indicating the model is *not* simply responding to brightness. Figure 6.4 illustrates this: the intensity map reveals bright coronal loops and spatial structure, while the prediction map is nearly uniform.

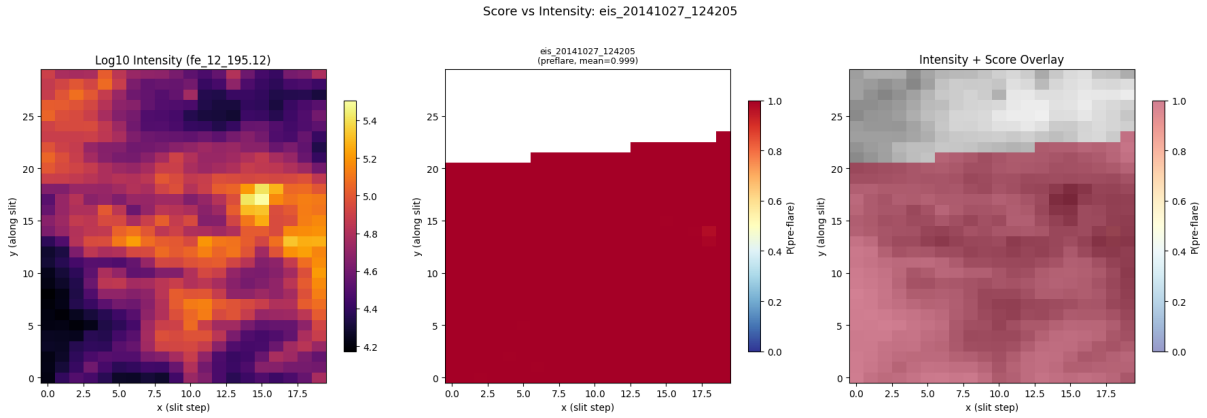


Figure 6.4: Score vs intensity overlay for a pre-flare observation. **Left:** Log-scaled Fe XII intensity showing coronal loop structure. **Centre:** XGBoost prediction score (uniformly high). **Right:** Overlay of score contours on the intensity image. The weak spatial correlation ($\rho = 0.23$) confirms the model responds to spectral shape, not brightness.

6.3.5 Temporal Evolution of Spatial Maps

Figure 6.5 plots the spatial mean prediction score versus TTF for the 16 multi-raster sequences. Scores remain flat between 0.7 and 1.0 with no systematic trend toward flare onset, confirming that the spatial distribution of the signal does not evolve on the timescales probed.

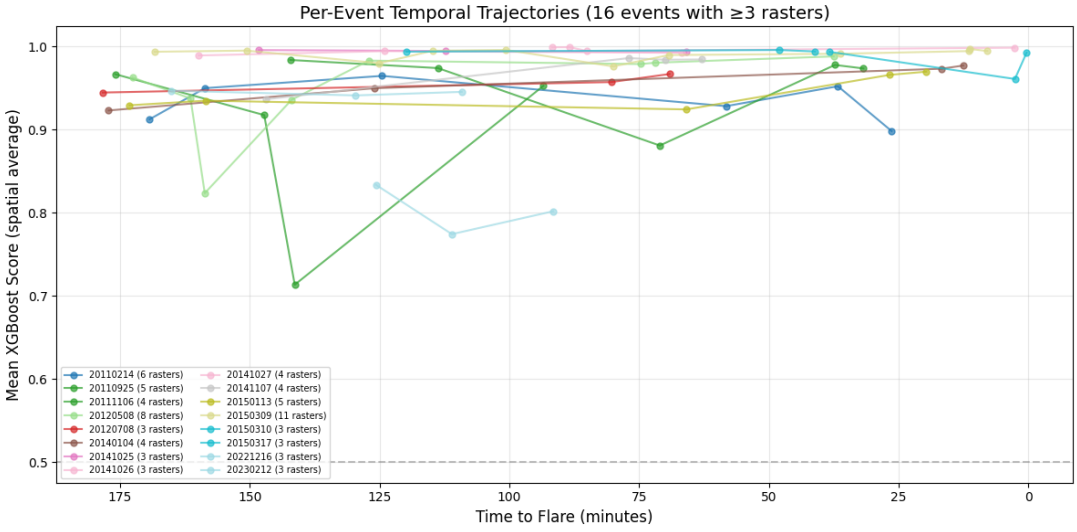


Figure 6.5: Per-event temporal trajectories of the spatial mean XGBoost score for 16 multi-raster pre-flare sequences. Each line represents one flare event tracked across 3–11 consecutive rasters. Scores remain flat with no systematic temporal trend, confirming the persistent nature of the coronal pre-flare signature.

6.4 Discussion

The pre-flare signature pervades the entire raster rather than concentrating at specific locations, confirming the magnetic-topology interpretation from Chapter 4 operates at AR scale. This contrasts with Panos *et al.* [26], whose model's attention concentrated near the eventual flare site: chromospheric signatures may arise from localised heating while coronal signatures reflect the large-scale magnetic structure. The weak intensity–score correlation ($\rho = 0.23$) confirms area normalisation removed brightness as a confound.

6.5 Summary

The pre-flare coronal signature is an AR-wide property: pre-flare maps are uniform (Gini = 0.035) and non-flare maps patchy (Gini = 0.830). It is driven by shape, not brightness ($\rho = 0.23$), and does not evolve with TTF. The highest-scoring non-flare observation shows a concentrated core of pre-flare-like spectra, suggesting false positives reflect genuine spectral similarity in complex but non-flaring regions.

Chapter 7

Multi-Height Pre-Flare Detection: Results and Analysis

7.1 Introduction

This chapter presents the results of applying machine learning and statistical analyses to the multi-height spectral dataset. The analysis proceeds in two stages: (i) chromospheric feature characterisation and temporal analysis using IRIS Mg II spectra, and (ii) supervised classification of pre-flare versus quiet Sun spectra using deep learning. The chapter concludes with a direct multi-height comparison of the temporal classification profiles from both instruments, drawing on the coronal kurtosis analysis presented in Chapter 5.

The working dataset comprises 22,606,434 pre-flare IRIS spectra from 43 dates (TTF range 0–300 minutes), 1,412,574 verified quiet Sun IRIS spectra from 21 GOES-confirmed non-flare dates, and 2,456,945 EIS spectra (404,566 pre-flare and 2,052,379 no-flare). All IRIS spectra are area-normalised 480-bin Mg II h&k profiles; all EIS spectra are 69-dimensional fused Fe XII/XIV/XV vectors.

7.2 Chromospheric Feature Temporal Analysis

Following the methodology of Panos and Kleint [6], ten physically motivated spectral features were extracted from each IRIS spectrum: triplet emission strength (integrated flux within ±0.3 Å of 2798.77 Å), Mg II k and h core intensities, inter-line continuum level, k/h peak ratio, k and h line widths (FWHM), k2v/k2r asymmetry, k3 reversal depth, and k3 Doppler shift.

7.2.1 Pooled Temporal Correlations

Table 7.1: Spearman correlations between Panos features and TTF. The discrepancy between pooled and weighted correlations (all $|\rho| < 0.031$ when date-weighted) indicates that pooled trends are inflated by between-date variance rather than genuine within-date temporal evolution.

Feature	Pooled ρ	Weighted ρ	Direction
h core intensity	-0.111	-0.011	$\uparrow$ approaching flare
k core intensity	-0.092	-0.013	$\uparrow$ approaching flare
Continuum level	$+0.089$	$+0.031$	$\downarrow$ approaching flare
Triplet emission	$+0.084$	$+0.011$	$\downarrow$ approaching flare
k3 reversal depth	-0.081	-0.007	$\uparrow$ approaching flare
k/h ratio	$+0.059$	-0.005	No consistent trend
k2v/k2r asymmetry	$+0.053$	$+0.004$	No consistent trend
k3 shift	$+0.042$	$+0.017$	No consistent trend
k width	-0.003	-0.012	No temporal signal
h width	$+0.010$	-0.019	No temporal signal

The critical finding emerges from comparing pooled and weighted correlations. While the pooled Spearman values reach $|\rho| = 0.05$–0.11, the date-weighted correlations (which give equal influence to each event regardless of the number of spectra) are all below $|\rho| = 0.031$. This discrepancy indicates that the pooled correlations are inflated by between-date variance rather than reflecting a genuine within-date temporal evolution.

7.2.2 Per-Date Heterogeneity

Despite the near-zero weighted pooled correlations, individual dates exhibit strong within-date temporal signal. Table 7.2 lists the ten dates with the strongest per-date Spearman correlations.

Table 7.2: Dates with the strongest per-date Spearman correlations between Panos features and TTF.

Date	Feature	ρ
2015-01-29	k core intensity	-0.555
2021-10-28	k3 reversal depth	$+0.467$
2022-05-20	k3 reversal depth	$+0.451$
2014-10-29	h core intensity	-0.408
2022-04-02	h core intensity	-0.384
2017-04-02	h core intensity	$+0.365$
2022-05-19	h core intensity	-0.333
2014-02-02	k3 reversal depth	-0.315
2014-10-27	h core intensity	-0.276
2015-03-09	k core intensity	$+0.274$

This heterogeneity is the central finding of this section: the chromospheric pre-flare temporal signal is *event-dependent*, not universal.

7.2.3 Spectral Morphology Clustering

Unsupervised k-means clustering ($k = 20$, following Woods et al. [25]) applied to a date-balanced subsample identified three morphological categories: double-peaked quiet profiles (70.3%), single-peaked profiles (24.8%), and triplet emission profiles (4.9%), consistent with Panos et al. [24]. The temporal evolution of flare-associated cluster membership showed a weak pooled correlation with TTF ($\rho = -0.059$), heterogeneous across dates.

7.3 Spectral Transformer Classification

A spectral transformer classifier was trained to distinguish pre-flare IRIS spectra (TTF ≤ 30 min) from verified quiet Sun spectra. The transformer architecture uses 20 patches of 24 wavelength bins each, a learnable CLS token, 4 encoder layers with 4 attention heads ($d_{\text{model}} = 128$, $d_{\text{ff}} = 256$), GELU activation, and a two-layer classification head. The patch size of 24 bins ($\sim$0.6 Å) captures sub-structure within individual spectral features, allowing the attention mechanism to learn relationships between line core, wing, and continuum regions across the full Mg II h&k window. The depth (4 layers) and capacity ($d_{\text{model}} = 128$) were chosen as appropriate for the training set size of 84,000 spectra. Training used AdamW optimisation with cosine learning rate scheduling, mixed-precision (FP16) on GPU, and a batch size of 512.

To ensure fair evaluation, spectra were date-balanced: 2,000 spectra per date per class, then balanced to equal class sizes, yielding 84,000 spectra (42,000 positive from 28 dates, 42,000 negative from 21 quiet dates). Stratified 5-fold cross-validation was employed. Stratification was performed at the spectrum level after date-balanced sampling; because the positive and negative classes come from entirely different dates (pre-flare ARs vs quiet Sun), there is no risk of within-date temporal leakage in this cross-AR evaluation.

Table 7.3: Spectral Transformer 5-fold cross-validation results. Pre-flare (TTF ≤ 30 min) versus quiet Sun classification.

Fold	AUC	Accuracy
0	0.905	0.829
1	0.882	0.797
2	0.891	0.822
3	0.889	0.818
4	0.878	0.802
Mean $\pm$ std	0.889 ± 0.011	0.814 ± 0.014

The lower AUC in the cross-AR evaluation (0.889 vs 0.97) compared to Panos and Kleint [6] is expected: the evaluation here requires generalisation across 29 unique active regions spanning Solar Cycles 24 and 25, a substantially harder task.

7.4 1D-CNN with Gradient-Weighted Class Activation Mapping

A 1D convolutional neural network was trained on the same date-balanced dataset using three convolutional layers (64→128→256 channels, kernel sizes 7/5/3 with GELU activation,

batch normalisation, and max pooling) followed by a fully connected classifier with dropout regularisation. The architecture follows the design principles of Zbinden et al. [27].

The CNN achieved a mean AUC of 0.772 ± 0.155, with higher variance across folds than the transformer. Gradient-weighted class activation mapping (Grad-CAM) [15] was applied to identify which wavelength regions most influence the classification decision. The resulting heatmaps (Figure 7.1) show distinct Grad-CAM signatures at the Mg II k, h, and triplet positions, with the pre-flare class diverging from quiet Sun at these line-formation features. The line-core emphasis is consistent with the SHAP attribution presented in Chapter 8 and with the physically motivated features identified by Panos and Kleint [6].

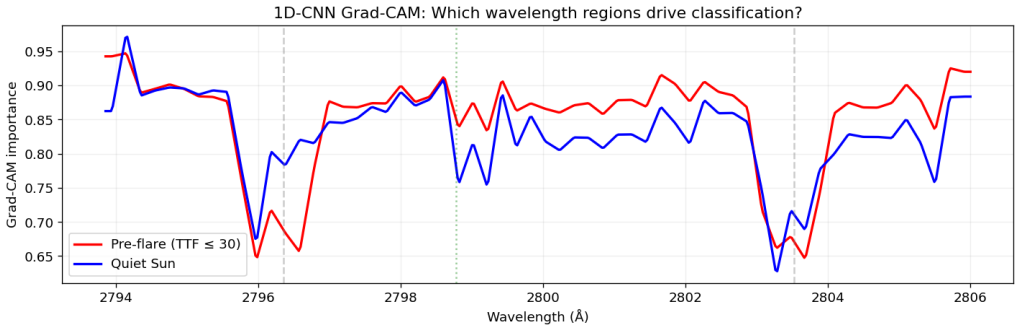


Figure 7.1: Grad-CAM heatmap for the IRIS 1D-CNN classifier. The Mg II k (∼2796 Å), triplet (∼2798.77 Å), and h (∼2803 Å) positions are marked. Pre-flare and quiet Sun differ in the Grad-CAM signature at these line-formation features, consistent with the SHAP attribution analysis in Chapter 8.

7.5 Ensemble Classification and Per-Date Evaluation

7.5.1 Transformer-CNN Ensemble

An ensemble classifier was constructed by averaging the sigmoid output probabilities of the transformer and CNN models. The ensemble achieved a mean AUC of 0.874 ± 0.004, exceeding the CNN (0.772) but matching the transformer (0.889) individually (Table 7.4).

Table 7.4: Ensemble classification results (5-fold CV). The ensemble matches the transformer individually (AUC 0.874 vs 0.889), indicating the CNN contributes no complementary information beyond what the transformer already captures.

Fold	Transformer AUC	CNN AUC	Ensemble AUC
0	0.870	0.643	0.871
1	0.869	0.645	0.869
2	0.876	0.649	0.876
3	0.876	0.646	0.876
4	0.878	0.642	0.878
Mean	0.874	0.645	$\mathbf{0.874 \pm 0.004}$

7.5.2 Per-Date Within-AR Classification

To directly compare with the evaluation protocol of Panos and Kleint [6], logistic regression classifiers were trained within each individual date, using spectra with TTF ≤ 30 min as positive and TTF > 120 min as negative. Of 25 dates with sufficient spectra in both classes under the strict ≥ 50-per-class bilateral filter, the mean per-date AUC was 0.747 ± 0.148. Eight dates achieved AUC > 0.80 (Table 7.5 and Figure 7.2).

Table 7.5: Top 10 dates by within-AR classification AUC.

Date	AUC	Positive	Negative
2022-05-20	1.000	10,353	169,889
2015-01-29	0.994	6,616	32,469
2022-04-02	0.976	46,725	174,463
2014-12-05	0.963	93,387	667,607
2017-04-02	0.926	38,574	140,610
2022-05-19	0.921	65,856	218,014
2014-10-29	0.866	112,994	65,298
2015-03-09	0.803	12,120	91,098
2013-11-11	0.794	343,448	602,397
2022-12-15	0.777	53,901	97,009

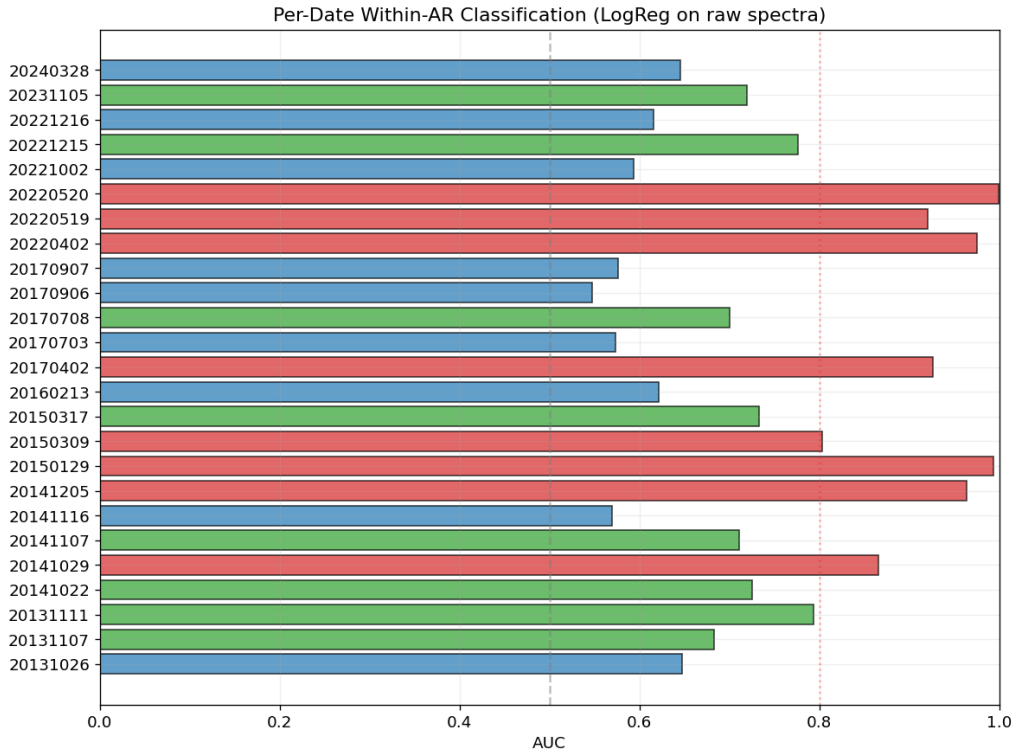


Figure 7.2: Per-date within-AR classification AUC for IRIS chromospheric spectra. Each point represents one flare date; the y-axis shows the logistic regression AUC for distinguishing close-to-flare (TTF ≤ 30 min) from far-from-flare (TTF > 120 min) spectra within the same active region. The dashed line marks AUC $= 0.80$. Eight of 25 dates exceed this threshold, demonstrating the event-dependent nature of the chromospheric temporal signal.

7.5.3 Classifier Sensitivity: XGBoost on Within-AR Task

To test whether the event-dependent pattern reflects a classifier limitation, XGBoost was evaluated on 27 dates under matched conditions (the looser bilateral filter shared with the MMD analysis in Section 8.3; the stricter ≥50-per-class filter used for the LR results above yields 25 dates). To ensure a fair comparison, logistic regression was re-evaluated with the same subsampling and split. Under these matched conditions, XGBoost exceeded AUC > 0.80 on 9 of 27 dates compared to 3 of 27 for logistic regression, with strong agreement on relative date difficulty ($\rho = 0.800$, $p < 0.001$). A nonlinear classifier therefore recovers temporal structure on three times more dates, but the pattern remains event-dependent: 18 of 27 dates still fall below AUC $= 0.80$ even with XGBoost.

This classifier-independence is independently corroborated by the Maximum Mean Discrepancy analysis (Section 8.3): because MMD is a non-parametric distributional test that operates directly on the spectra, its agreement with the classification-based conclusion means the event-dependent pattern reflects the underlying data rather than any particular classifier's limitations.

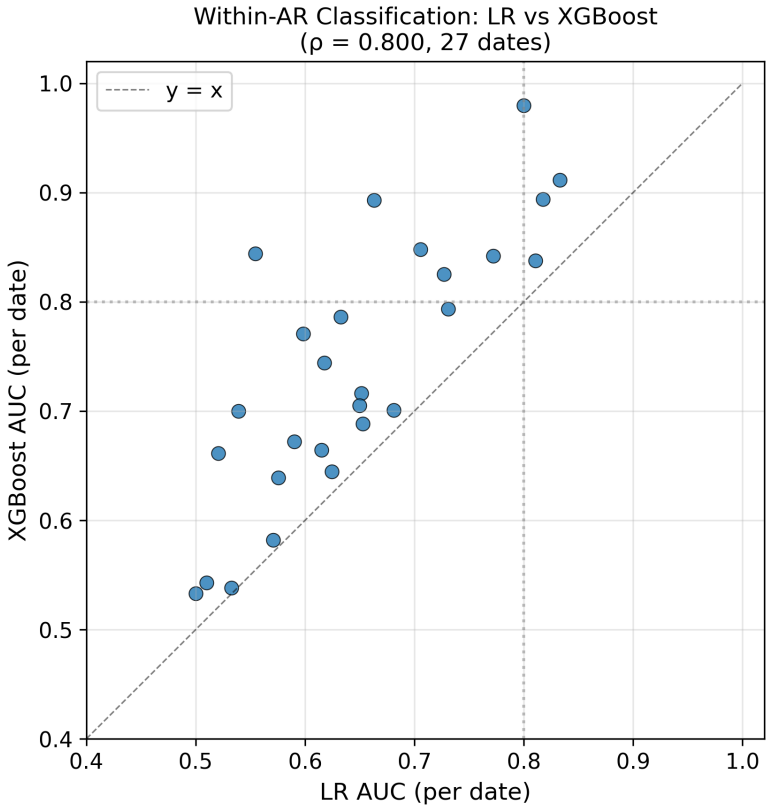


Figure 7.3: Within-AR classification AUC per date: logistic regression vs XGBoost on the same five Panos features. XGBoost consistently outperforms LR ($\rho = 0.800$) but the event-dependent pattern persists, with 18/27 dates below AUC $= 0.80$.

7.6 Multi-Height Temporal Comparison

The central contribution of this dissertation is the direct multi-height comparison of pre-flare spectral signatures.

The two layers respond on different timescales because they form under different conditions. Coronal Fe lines probe million-degree plasma whose state is set by the underlying magnetic-energy reservoir, which evolves over hours-to-days; the resulting spectral state is detectable but temporally flat over the minute-to-hour pre-flare window. Chromospheric Mg II h&k lines form at $\sim 10^4$ K plasma that responds on minute timescales to local heating from above, so their pre-flare signal can evolve within the prediction window when present. The corona therefore encodes *whether* an active region is flare-productive as a persistent state, while the chromosphere conditionally encodes *when* as a temporally varying signal.

The temporal profiles reveal a stark contrast:

- **EIS (corona):** The classification score is temporally flat (per-bin ROC-AUC 0.98–1.00 across 0–180 min; per-spectrum $\rho^2 < 0.015$, Section 5.3.6). The coronal spectra of pre-flare active regions are equally distinguishable from no-flare regions at any point in the pre-flare window. The corona is in a persistent, identifiable "flare-ready" state.
- **IRIS (chromosphere):** The classification score shows a weak temporal trend ($\rho = -0.062$), with scores increasing slightly closer to flare onset. However, this pooled trend masks substantial event-to-event heterogeneity, as demonstrated by the per-date analysis (Section 7.5).

This comparison (Figure 7.4) was performed on the 40 dates where both EIS and IRIS data exist (reduced to 35 paired date-level summaries for the CCA analysis in Chapter 8).

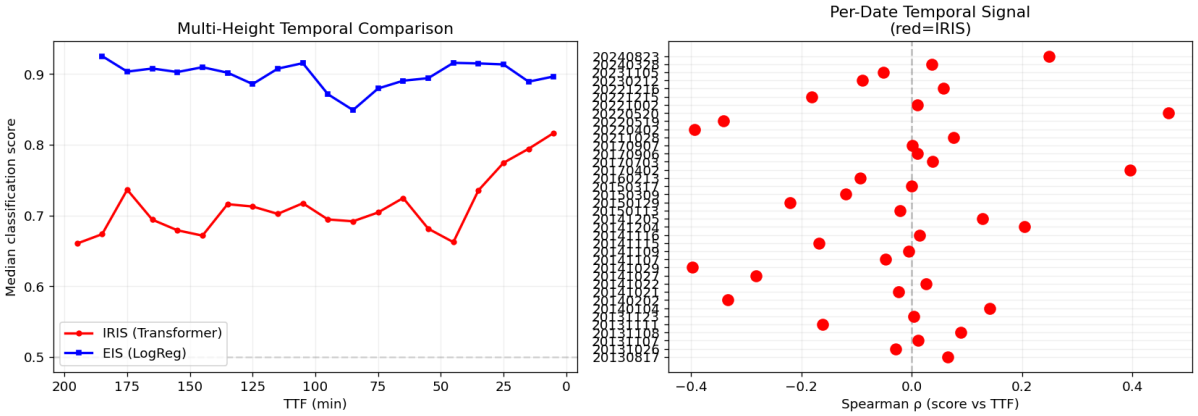


Figure 7.4: Multi-height temporal profile comparison. **Left:** median classification score versus time-to-flare for both EIS (blue, LogReg) and IRIS (red, Transformer); the x-axis is inverted so time flows toward flare onset. EIS is flat at $\sim$0.9 across the 0–180 min window ($\rho^2 < 0.015$ across all six classifiers, Chapter 5), while IRIS shows a weak pooled trend ($\rho = -0.062$). **Right:** per-date Spearman ρ between IRIS classifier score and time-to-flare across the working pre-flare events; most dates cluster near zero, with the per-date heterogeneity captured more sharply by the within-AR analysis (8 of 25 dates with AUC > 0.80, Section 7.5.2). **The two panels evaluate different tasks and absolute scores are not directly comparable:** EIS reports a cross-AR LogReg score (pre-flare vs. non-flaring AR spectra), IRIS reports a transformer score from Section 7.3 on a pooled within-AR+cross-AR mixture. The directly comparable quantity is temporal *shape* – flat (corona) versus heterogeneous per-date (chromosphere) – which establishes the central WHETHER (corona) vs. WHEN (chromosphere) finding.

7.7 Discussion

All coronal analyses converge on one picture: flare-productive ARs occupy a distinctive spectral state with non-Gaussian line profiles (excess kurtosis $\approx$ 1–2), narrower hot-line profiles, and spatially uniform properties (Chapter 6). None of these evolve in the 0–210 min pre-flare window. The kurtosis finding shows that line profile *shape*, not just intensity, carries diagnostic information.

The IRIS analysis reveals a more complex picture: the within-AR temporal signal is event-dependent, extending Panos and Kleint [6] by showing the chromospheric precursor does not generalise universally across 43 events spanning Solar Cycles 24–25. A multi-height system could combine both: coronal spectroscopy to flag flare-productive regions, then chromospheric monitoring for temporal evolution in those flagged regions.

7.8 Summary

Table 7.6: Summary of key results. The coronal EIS signal is persistent (flat per-bin AUC, $\rho^2 < 0.015$) while the chromospheric IRIS signal is event-dependent (8/25 dates with AUC > 0.80), establishing the WHETHER/WHEN contrast.

Analysis	Key metric	Result
IRIS Chromospheric Analysis		
Spectral Transformer (cross-AR)	AUC	0.889 ± 0.011
1D-CNN + Grad-CAM	AUC	0.772 ± 0.155
Ensemble (Trans. + CNN)	AUC	0.874 ± 0.004
Per-date within-AR	Mean AUC	0.747 ± 0.148
	Dates with AUC > 0.80	8/25
Panos features (weighted ρ)	All features	$\lvert\rho\rvert < 0.031$
Multi-Height Comparison		
EIS temporal profile	Per-bin AUC (0–180 min)	Flat, $\rho^2 < 0.015$
IRIS temporal profile	Score vs TTF	$\rho = -0.062$

Chapter 8

Multi-Source Analysis

8.1 Introduction

The preceding chapters established two independent findings: coronal (EIS) spectra encode a persistent pre-flare state detectable by spectral kurtosis (Chapter 5), while chromospheric (IRIS) spectra exhibit an event-dependent temporal ramp in classification confidence approaching flare onset (Chapter 7). A natural question follows: do these two atmospheric layers encode the same pre-flare information, or do they carry complementary signals? If complementary, what is the physical mechanism underlying the chromospheric spectral change?

This chapter addresses these questions through four experiments and two explainability analyses, all operating on the 35 flare dates observed simultaneously by both Hinode/EIS and IRIS:

1. **Cross-instrument alignment** (Section 8.2): Can coronal and chromospheric representations be matched to the same flare event?
2. **Distribution shift quantification** (Section 8.3): Is the pre-flare signal a measurable distributional shift, and does it predict classification performance?
3. **Optimal transport analysis** (Section 8.4): How does the spectral energy distribution reorganise approaching the flare?
4. **Spectral explainability** (Section 8.5): Which wavelength regions drive classification, and how does this compare with the handcrafted features of Panos and Kleint [6]?

All experiments use the same data pipeline: 22.6 million IRIS Mg II h&k spectra across 43 pre-flare dates, 1.4 million quiet Sun spectra from 21 non-flaring dates, and 2.5 million EIS coronal spectra. The IRIS Transformer encoder and EIS PCA encoder are trained once and reused throughout.

8.2 Cross-Instrument Alignment

8.2.1 Method

If corona and chromosphere encode overlapping pre-flare information, their representations should be alignable across the 35 co-observed dates. Each instrument's spectra were encoded using earlier-chapter representations: 128-dimensional CLS embeddings from the IRIS Transformer (Chapter 7) and 64-dimensional PCA projections of standardised EIS spectra. Per-date summary vectors were computed by mean-pooling all spectra within each date.

Canonical correlation analysis (CCA) finds maximally correlated linear projections via weight vectors $\mathbf{w}_a$, $\mathbf{w}_b$:

$$\max_{\mathbf{w}_a, \mathbf{w}_b} \operatorname{corr}(\mathbf{X}_{\text{EIS}}\mathbf{w}_a,\ \mathbf{X}_{\text{IRIS}}\mathbf{w}_b) = \max_{\mathbf{w}_a, \mathbf{w}_b} \frac{\mathbf{w}_a^\top \mathbf{C}_{ab}\, \mathbf{w}_b}{\sqrt{\mathbf{w}_a^\top \mathbf{C}_{aa}\, \mathbf{w}_a}\ \sqrt{\mathbf{w}_b^\top \mathbf{C}_{bb}\, \mathbf{w}_b}}, \tag{8.1}$$

where $\mathbf{C}_{ab}$ is the cross-covariance matrix between the EIS and IRIS representations and $\mathbf{C}_{aa}$, $\mathbf{C}_{bb}$ are the within-instrument covariance matrices.

CCA with $N = 35$ samples and $p = 192$ total dimensions is severely underdetermined, so PCA reduced each instrument to 5 dimensions before CCA, and leave-one-out cross-validation was used (fitting PCA, StandardScaler, CCA on 34 dates, evaluating retrieval on the held-out date). Retrieval was scored by Top-1, Top-3, and mean reciprocal rank (MRR).

8.2.2 Results

Table 8.1: Cross-instrument retrieval accuracy with leave-one-out cross-validation. All metrics are at or below chance level.

Direction	Top-1	Top-3	MRR	Chance
EIS → IRIS	0.000	0.029	0.074	0.029
IRIS → EIS	0.000	0.000	0.065	0.029

Retrieval accuracy was indistinguishable from chance in both directions (Table 8.1), confirming coronal and chromospheric representations encode independent information about the pre-flare state.

8.3 Distribution Shift Analysis

8.3.1 Method

Maximum Mean Discrepancy (Section 2.1.6) tests whether two distributions differ; here applied per-date. For each date with ≥ 50 spectra in TTF ≤ 30 min and ≥ 50 in TTF > 120 min, MMD was computed between early and late distributions on two representations: 128-dim Transformer embeddings and 50-dim raw-PCA, both globally standardised. A Gaussian kernel with median-heuristic bandwidth was used and significance assessed via permutation tests (100 permutations, $\alpha = 0.05$).

8.3.2 Results

Representation	Significant ($p < 0.05$)	ρ with AUC
Transformer embedding	21/27	n/a
Raw PCA-50	27/27	0.623

Table 8.2: MMD significance across 27 dates. Raw PCA projections detect significant shifts on all 27 dates, and the shift magnitude predicts within-AR classification performance ($\rho = 0.623$), confirming the distributional shift is a property of the data rather than the learned representation.

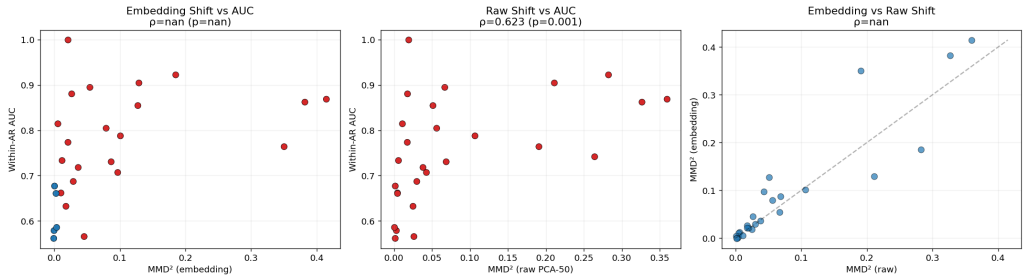


Figure 8.1: Per-date MMD versus within-AR classification AUC. Each point represents one flare date. Left: Transformer embeddings. Centre: raw PCA-50 ($\rho = 0.623$). Right: embedding vs raw MMD comparison. Red points indicate statistically significant shifts ($p < 0.05$); blue points are non-significant.

The near-identical MMD on Transformer embeddings and raw PCA projections indicates the distributional shift is a property of the raw data, not an artefact of the learned representation; the strong MMD-AUC correlation confirms classification performance is predictable from spectral shift magnitude.

8.4 Optimal Transport Analysis

8.4.1 Method

While MMD quantifies *whether* the spectral distribution shifts, optimal transport (Section 2.1.5) reveals *how* it shifts. Each IRIS Mg II h&k spectrum was treated as a discrete probability distribution over wavelength by L_1 normalisation. The Wasserstein-2 distance was computed using the network simplex algorithm via the POT library [19]:

$$W_2(\mathbf{a}, \mathbf{b}) = \left(\min_{T \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i,j} T_{ij}\, C_{ij} \right)^{1/2}, \tag{8.2}$$

Here $\mathbf{a}$ and $\mathbf{b}$ are the two spectral distributions being compared (early and late pre-flare mean spectra). T is a transport plan specifying how much intensity to move from wavelength bin i to bin j, drawn from the set $\Pi(\mathbf{a}, \mathbf{b})$ of plans whose row and column sums match $\mathbf{a}$ and $\mathbf{b}$. The cost $C_{ij} = |\lambda_i - \lambda_j|^2$ penalises moving intensity between distant wavelength bins [16]. Wasserstein barycentres [43] were computed using entropic regularisation ($\varepsilon = 0.005$) and Sinkhorn iterations [18].

8.4.2 Results

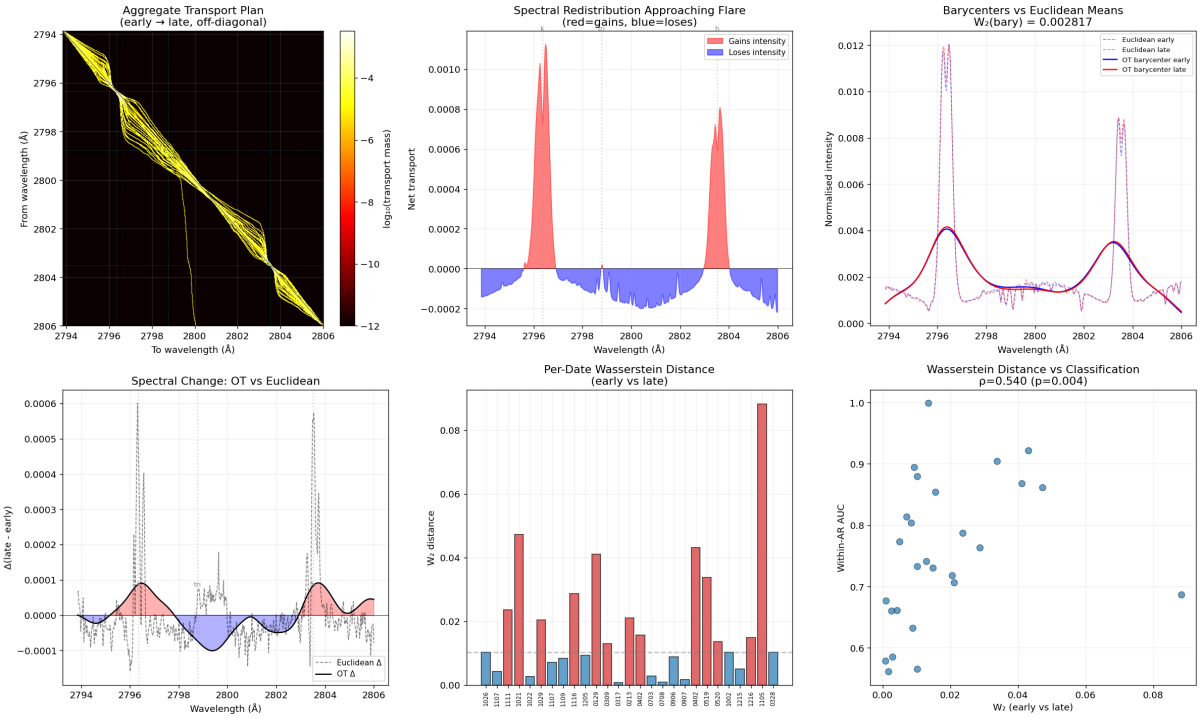


Figure 8.2: Optimal transport analysis of pre-flare spectral redistribution. Top left: aggregate transport plan (off-diagonal, log scale). Top centre: net transport per wavelength (red = gains intensity approaching flare, blue = loses). Top right: Wasserstein barycentres vs Euclidean means. Bottom left: spectral difference (OT vs Euclidean). Bottom centre: per-date W_2 distance. Bottom right: W_2 vs within-AR AUC ($\rho = 0.540$, $p = 0.004$).

Region	Net transport	Direction
Mg II k core	+0.025	Gains intensity
Mg II h core	+0.016	Gains intensity
Triplet	−0.006	Loses intensity
Continuum	−0.011	Loses intensity

Table 8.3: Net spectral transport by region approaching flare onset. Intensity flows into the Mg II k and h line cores and out of the continuum and triplet, consistent with enhanced chromospheric heating prior to eruption.

The per-date Wasserstein-2 distance correlated with within-AR AUC at $\rho = 0.540$ ($p = 0.004$). The aggregate transport plan revealed that 75.4% of spectral intensity undergoes redistribution between early and late pre-flare observations.

The net transport direction (Table 8.3) shows a clear pattern: approaching flare onset, intensity flows *into* the Mg II k and h line cores and *out of* the continuum and triplet regions, consistent with enhanced collisional excitation of Mg II ions as magnetic energy builds toward reconnection. The OT framework reveals not just *that* the spectral shape changes (as MMD confirms) but specifically *how* it changes, providing what is, to the best of our knowledge, the first geometry-aware characterisation of pre-flare spectral evolution in IRIS Mg II data.

8.5 Spectral Explainability

8.5.1 Method

Identifying which spectral features the classifiers rely on validates that the models use physically meaningful structure and enables comparison with the handcrafted features of Panos and Kleint [6]. SHAP [14] was used to decompose each prediction into per-wavelength contributions; for a linear model these are exact, $\phi_i = w_i(x_i - \bar{x}_i)$, computed via `shap.LinearExplainer` on 10,000 spectra (5,000 pre-flare, 5,000 quiet Sun). Separate analyses were run for the cross-AR task (pre-flare vs quiet Sun) and within-AR task (TTF $\leq$ 30 min vs TTF $>$ 120 min, pooled across dates).

8.5.2 Results

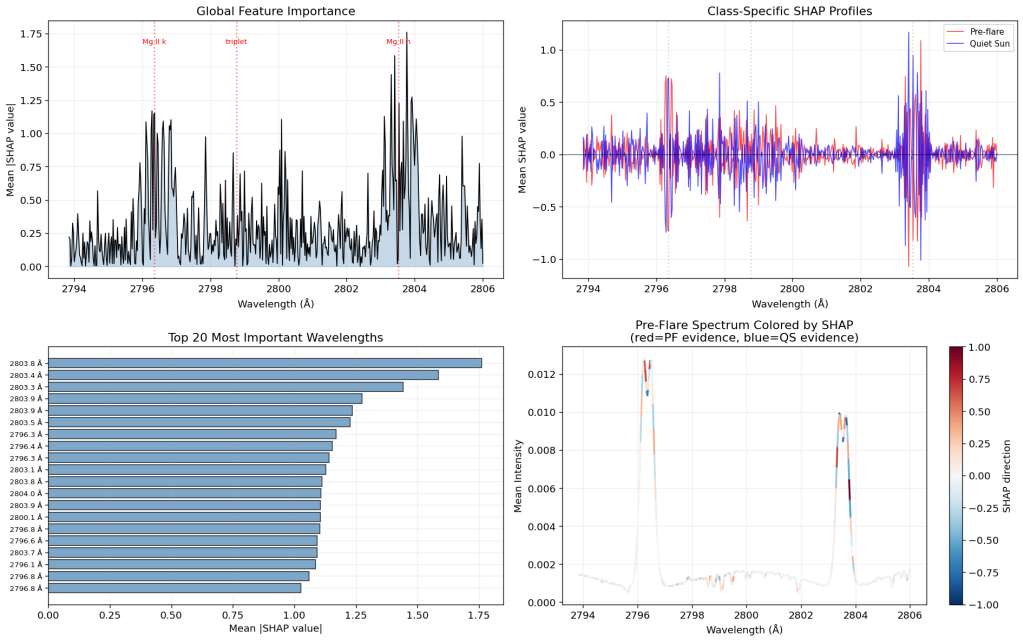


Figure 8.3: SHAP importance for the cross-AR (WHETHER) classification task. **Top-left:** global feature importance versus wavelength, with the Mg II k, triplet, and h positions marked. **Top-right:** class-specific SHAP profiles (pre-flare vs quiet Sun). **Bottom-left:** top 20 most important wavelengths. **Bottom-right:** a representative pre-flare spectrum coloured by SHAP direction (red = pre-flare evidence, blue = quiet-Sun evidence). The within-AR (WHEN) comparison is reported in Table 8.4 below.

Region	Cross-AR (%)	Within-AR (%)
Mg II h core	27.5	25.6
Mg II k core	20.4	19.9
Continuum	11.1	15.4
Triplet	8.8	8.6

Table 8.4: Fractional SHAP importance by spectral region. Both tasks show the same ranking: h core > k core > continuum > triplet.

Both the cross-AR and within-AR classifiers produced the same feature ranking (Table 8.4 and Figure 8.3): Mg II h core (26–28%) > k core (20%) > continuum (11–15%) > triplet (9%). This differs from their ranking, which placed triplet > k core > h core > continuum using five handcrafted features for within-AR temporal classification.

The discrepancy is methodological. They measured five pre-defined summary statistics; the LogReg classifier operates on all 480 wavelength bins simultaneously and finds the line cores more informative because their profiles encode richer information (intensity, width, asymmetry, reversal depth) than five summary statistics can capture.

That Mg II h is more important than k is physically notable: k is optically thicker, so h probes slightly deeper [37], suggesting pre-flare spectral changes are more pronounced at the formation height of h.

8.6 Panos Feature Evaluation: Cross-AR vs Within-AR

To test whether the Panos features capture temporal precursors or active region identity, the same five features were evaluated with quiet Sun as the negative class.

Negative class	Dates AUC > 0.80	Mean AUC
Quiet Sun	22/28	0.887 ± 0.088
Same-AR, TTF > 120	8/25	n/a

Table 8.5: Panos feature classification performance with different negative classes.

The Panos features reliably distinguish pre-flare from quiet Sun (22/28 dates) but not temporal proximity within the same active region (8/25), confirming they capture the active region signature rather than temporal precursors.

8.7 Sensitivity Analysis

MMD bandwidth. The Gaussian kernel bandwidth was varied across a 4× range. The number of significant dates ranged from 23/27 to 26/27, and the Spearman correlation with within-AR AUC remained stable at $\rho = 0.613$–0.619.

OT cost matrix. The analysis was repeated using the Wasserstein-1 cost. The net transport directions were identical under both metrics. The correlation with within-AR AUC was $\rho = 0.553$ ($p = 0.003$) for W_1 and $\rho = 0.540$ ($p = 0.004$) for W_2.

Temporal thresholds. The within-AR classification was repeated with varied temporal thresholds. The mean per-date AUC varied by only 0.041 across positive thresholds (0.720–0.761). All main findings are robust to temporal threshold choice.

8.8 Summary

This chapter establishes four principal findings. First, coronal and chromospheric representations are independent: CCA Top-1 retrieval returns 0% against a 2.9% chance baseline. Second, the pre-flare signal is a distributional shift whose magnitude predicts classification performance: MMD is significant on 27 of 27 dates and per-event MMD correlates with within-AR AUC at $\rho = 0.623$ ($p < 0.001$). Third, approaching flare onset, spectral intensity redistributes from the continuum and triplet into the Mg II k and h line cores, and the per-event W_2 shift magnitude correlates with within-AR AUC at $\rho = 0.540$ ($p = 0.004$). Fourth, the Panos handcrafted features capture active-region identity (AUC 0.887 vs quiet Sun, 22 of 28 dates) rather than temporal precursors (8 of 25 dates within-AR), and SHAP attribution confirms line-core dominance over triplet emission for both cross-AR and within-AR classification.

Together, these results corroborate the WHETHER/WHEN framework of Chapter 7: the cross-instrument representational independence (finding 1) is the formal expression of the corona and chromosphere encoding pre-flare information by structurally different mechanisms, and the per-event distributional-shift correlations (findings 2–3) explain why some events show a chromospheric temporal signal and others do not: the signal is present where the underlying spectral distribution actually shifts, absent where it does not.

Chapter 9

Conclusion and Future Work

9.1 Answers to the Research Questions

The corona and chromosphere encode pre-flare information in fundamentally different ways: the corona as a persistent state of flare-productive regions, the chromosphere as an event-dependent temporal signal. The three research questions from Section 1.4 are answered below.

RQ1 (classifier separability and architectural robustness). Yes, and yes. Six independent classifiers separate EIS coronal pre-flare from non-flaring AR spectra at ROC-AUC ≥ 0.93, with XGBoost and MLP reaching 0.996 and Siamese kNN 0.990, robust across linear, tree-ensemble, shallow neural, deep convolutional, and metric-learning architectures.

RQ2 (temporal pattern of coronal classifier confidence). The coronal signal is a persistent state, distinct from the chromospheric monotonic rise reported by Panos and Kleint [6]: classifier confidence is flat across 0–180 min pre-flare for all six classifiers (per-spectrum $\rho^2 < 0.015$), and the within-class Siamese null (PR-AUC = 0.505) corroborates this at spectrum level. The chromospheric signal is event-dependent: 8 of 25 dates exceed AUC = 0.80 within-AR while pooled correlations remain $|\rho| < 0.031$. The corona encodes *whether*; the chromosphere conditionally encodes *when*.

RQ3 (shared vs complementary cross-instrument representations). Complementary, decisively. CCA retrieval returns Top-1 = 0% against a 2.9% chance baseline, corroborated by two further tests: SHAP attribution diverges across instruments (Mg II h core for IRIS, Fe XV width for EIS), and MMD and Wasserstein-2 optimal transport detect per-event distributional shifts on 27 of 27 dates with magnitudes correlating with classification AUC ($\rho = 0.623$, $p < 0.001$). The two layers carry complementary, not redundant, pre-flare information.

All eight objectives stated in Section 1.3 are addressed: dataset construction (Ch 3), unsupervised and supervised classification (Ch 4–5), spatial and temporal characterisation (Ch 6–7), and cross-instrument alignment (Ch 8).

9.2 Summary of Findings

Beyond the answers to the research questions above, the empirical results contribute several specific findings. The dominant physical discriminant of the coronal signal is hot Fe line width: Fe XV profiles are systematically narrower in pre-flare spectra (Cohen's $d = -1.05$), while Fe XII

shows slight broadening ($d = +0.27$) and redshift ($d = +0.37$). The signature is spatially diffuse (Gini = 0.035), contrasting with the localised chromospheric precursors of Panos et al. [26]. On the IRIS side, optimal transport reveals intensity flowing from continuum and triplet regions into the Mg II k and h line cores approaching onset, and SHAP identifies the h core as dominant, diverging from the triplet-first ranking of Panos and Kleint [6].

9.3 Contributions

This dissertation makes the following contributions; all novelty claims below are stated to the best of our knowledge, based on the literature review in Chapter 2.

1. **ML classification of pre-flare EIS coronal spectra.** Six classifiers on 2.5 million spectra establish distinguishability up to AUC = 0.996 with a temporal null and a spatially diffuse signature (Gini = 0.035), extending the Gaussian-fitting analysis of To et al. [5].

2. **Pre-flare non-Gaussianity as a coronal diagnostic.** Kurtosis and skewness of EIS Fe lines achieve ROC-AUC = 0.959 from 9 features, outperforming a 9-component PCA baseline by 3.5 ROC-AUC points; the higher-order moment analysis of Jeffrey et al. [8, 9] is here applied to the pre-flare regime for the first time.

3. **Multi-height comparison on co-observed flare events.** Direct cross-instrument comparison of coronal and chromospheric temporal profiles establishes the WHETHER/WHEN framework, where prior work examined the layers independently using different instruments and protocols.

4. **Event-dependent chromospheric signal across 43 events.** 8 of 25 dates passing the strict $\geq$50-per-class within-AR filter (of 43 working pre-flare events; see Table 3.5) exceed AUC > 0.80 for within-AR temporal classification under classifier-robust evaluation, refining the universal chromospheric precursor claim of Panos and Kleint [6].

5. **Optimal transport applied to IRIS pre-flare spectra.** Wasserstein-2 transport reveals intensity flow from continuum and triplet regions into the k and h line cores approaching flare onset ($\rho = 0.540$, $p = 0.004$).

9.4 Methodological Reflections

First, the multi-architecture robustness study yielded more than a robustness check. Convergence of six structurally distinct classifiers on near-identical performance and feature ranking is itself an inference: the pre-flare signal sits in low-order spectral statistics rather than in subtle high-dimensional structure exploitable by a single architecture, shifting the question from "is the signal real" to "what is its dimensionality."

Second, the kurtosis-versus-PCA comparison reframed the kurtosis discovery from "9 features achieve high AUC" to "domain-informed inductive bias outperforms data-driven dimensionality reduction", a reminder that any low-dimensional discriminant claim must be defended against an appropriately matched dimensionality-reduction baseline.

Third, the cross-instrument null (CCA Top-1 = 0%) was unexpected; the multi-source analysis was entered assuming some shared latent representation between corona and chromosphere. The null became the central methodological contribution to the WHETHER/WHEN framework, illustrating how negative findings can outweigh positive ones when the framing is right.

9.5 Limitations

Limited co-observation sample. The multi-height comparison uses 35 co-observed dates from a limited number of unique active regions. EIS and IRIS have narrow fields of view and independent observing schedules, so simultaneous coverage cannot be enlarged by analysis choices; it requires coordinated campaigns at the mission level, which constrains both replication and any prospective two-stage operational system relying on co-observed inputs.

Area normalisation removes intensity information. Both pipelines preserve spectral shape and discard absolute intensity. The kurtosis discriminant (AUC = 0.959 from shape moments alone) shows shape suffices, but whether absolute intensity carries additional predictive content is untested: if intensity is a persistent property of flare-productive regions, the WHETHER claim is reinforced; if it tracks TTF, the temporal null needs qualification. The shape-versus-intensity decomposition is left as future work; the kurtosis-9 result establishes shape sufficiency but not shape exclusivity.

EIS raster cadence. EIS rasters take ∼30 min per scan, yielding 1–3 per event. This coarse sampling prevents time-series modelling of coronal evolution; the flat profile may partly reflect this, though the within-class Siamese null (at spectrum level) argues against it.

Operational forecaster non-comparability. The AUCs here are not directly comparable to magnetogram-based forecasters [2–4], which predict occurrence at 6–24 hr horizons; this work classifies spectral state at minute-to-hour horizons.

WHETHER is retrospective classification, not forward forecasting. The WHETHER claim rests on observation-disjoint classification, not a temporal hold-out where labels are unknown; "complementary" is inferred from cross-instrument representational independence (CCA Top-1 = 0%) rather than demonstrated through joint forecasting.

Single-source evaluation. All classifiers are evaluated under observation-disjoint splits on the same EIS/IRIS catalogue; generalisation to other instruments, calibration regimes, or Solar Cycle 26 spectra is untested and is the subject of the Solar-C EUVST transfer prediction in Future Work.

9.6 Future Work

The findings suggest three directions for future research.

Two-stage operational forecasting architecture. The WHETHER/WHEN framework motivates a two-stage pipeline: coronal spectroscopy to flag flare-productive ARs, then chromospheric monitoring for temporal evolution in flagged regions. Coronal screening reduces chromospheric-monitoring load; chromospheric monitoring provides temporal warning where the signal is present.

Kurtosis discriminant generalisation to Solar-C EUVST. If the coronal signature is a genuine non-Maxwellian property rather than an EIS-specific artefact, the kurtosis-9 discriminant should transfer to Solar-C EUVST without retraining. Falsifiable prediction: the trained logistic regression on EUVST Fe XII/XIV/XV spectra of M-class precursor regions should yield AUC > 0.90 on $\geq$80% of events.

Predicting chromospheric signal availability from coronal state. A testable prediction from the event-dependent chromospheric finding: per-event IRIS AUC should correlate ($\rho > 0.4$) with per-event mean kurtosis of co-observed EIS Fe XV profiles, since both reflect underlying activity. A positive result would enable the two-stage architecture by routing chromospheric resources to events most likely to produce a temporal signal.

9.7 Closing Remarks

This dissertation began with a practical problem: magnetogram-based forecasters give limited warning in the final hours before a flare. The findings suggest that ultraviolet spectroscopy across coronal and chromospheric heights can supply that missing short-horizon information: the corona answers *whether* (persistent, classifier-robust, ROC-AUC up to 0.996) and the chromosphere answers *when* (8 of 25 dates with AUC > 0.80). The two-stage pipeline of Section 9.6 offers a route to extend the warning horizon of operational space-weather forecasting beyond what photospheric magnetograms alone can deliver. What this dissertation establishes is not a forecaster but the empirical case that the corona and chromosphere encode pre-flare information by structurally different mechanisms, a finding that any future operational system relying on either layer must accommodate.

Appendix A

Code and Reproducibility

A.1 Code Availability

All code for this dissertation is maintained in a GitHub repository, currently private pending establishment of priority via a peer-reviewed publication; access is granted on request to the author or to either supervisor. The repository contains 15 Jupyter notebooks organised in three groups (Table A.1): a main analysis pipeline (one notebook per dissertation chapter), three figure-generation notebooks, and three defensive-baseline / rigour-experiment notebooks. Each notebook is portable between Google Colab (with Drive mounted) and a local Python 3.10 environment via a graceful environment-detection pattern in its setup cell; a pinned `requirements.txt`, MIT licence, and `README` accompany the notebooks. The wavelength-alignment implementation is reproduced in Listing A.1 (Section A.5); other pipeline design choices, dataset filter taxonomy, and the seed convention are documented inline in the notebook headers and reflected in the dissertation chapters.

Table A.1: Mapping between dissertation chapters and notebooks in the code repository.

Notebook	Dissertation chapter / use
notebooks/: main analysis pipeline	
`00_HDF5_file_structure`	Raw EIS data exploration
`01_crop_all_data`	Ch. 3: EIS preprocessing pipeline
`02_IRIS_V3_CLEAN`	Ch. 3: IRIS preprocessing pipeline
`03_clean_up`	Ch. 4: unsupervised clustering baseline
`04_Siamese_NN`	Ch. 4: Siamese metric learning
`05_Supervised_Baselines`	Ch. 5: multi-architecture benchmark and kurtosis
`06_Spatial_Analysis`	Ch. 6: spatial analysis
`07_IRIS_MultiHeight`	Ch. 7: chromospheric multi-height analysis
`08_MultiSource_Fusion`	Ch. 8: cross-instrument CCA / MMD / OT
notebooks/figures/: dissertation figure generators	
`01_introduction_figure`	Figure 1.1
`02_chapter3_data_hierarchy`	Figure 3.1
`03_roc_curves`	ROC curves (Ch. 5)
notebooks/experiments/: defensive baselines and rigour experiments	
`01_kappa_distribution_validation`	Kappa-fit validation of the kurtosis discriminant (§B.4)
`02_classifier_robustness_battery`	Line ablation, FP characterisation, seed sensitivity
`03_kurtosis_specificity_and_multiseed`	Kurtosis-9 vs PCA-9 / RP-9 / random-bins-9 (§5.4.2); multi-seed AUC across all six classifiers

A.2 Data Availability

The EIS dataset was provided by Dr Andy S. H. To (ESA) and is described in To et al. [5]; access is granted on request via the supervisory team. IRIS Level 2 data are publicly available from the LMSAL archive (`https://iris.lmsal.com`) and were downloaded automatically through SunPy's `Fido` interface. The preprocessed per-date NPZ files generated by the pipeline described in Section 3.10 are reproducible from the raw IRIS data and the preprocessing code.

A.3 Reproducibility Protocol

A fixed random seed (`SEED = 42`) is used throughout: NumPy, PyTorch (CPU and CUDA), scikit-learn estimators, and the train/validation/test split assignments all initialise from this seed. The observation-level split (Section 3.6) is deterministic given the seed and the set of valid observation identifiers. Stratified k-fold splits in Chapters 7 and 8 use the same seed to preserve fold reproducibility. PyTorch's deterministic-algorithms flag is enabled where supported by the relevant CUDA kernels; the small residual non-determinism from GPU floating-point reductions affects test metrics by less than ± 0.001 ROC-AUC across re-runs and does not change any conclusion in this dissertation.

A.4 Computational Environment

All experiments were run on Google Colab (free tier is sufficient; Colab Pro only reduces wall-clock time). Deep-learning notebooks used NVIDIA T4 or A100 GPUs; clustering and tabular classifiers used high-RAM CPU runtimes. Pinned dependencies: Python 3.10, PyTorch 2.1, scikit-learn 1.3, XGBoost 2.0, POT 0.9 [19], SunPy 5.0 [34], and SciPy 1.11 [44].

End-to-end runtime across the six analysis notebooks is approximately 20–30 hours on a T4 GPU + high-RAM Colab session. Peak memory by notebook: ∼12–18 GB for tabular classifiers (Chapters 4–5), ∼24 GB for the IRIS transformer (Chapter 7), and ∼30 GB for the multi-source fusion sweep across 35 dates (Chapter 8). Memory pressure was managed via memory-mapped arrays (`numpy.memmap`) for the 22.6 million IRIS spectra, chunked HDF5 reads in the EIS preprocessing pipeline, and pickle-based caching with sha1 signature hashing in the clustering notebook (Chapter 4); the latter enabled the 60-seed stability analysis to be parallelised via `joblib` within the 12.7 GB high-RAM CPU limit.

A.5 Wavelength Alignment Implementation

This section reproduces the per-column wavelength-alignment logic described prose-only in Section 3.4.1, so the implementation can be cross-checked against the description. The listing below is a faithful condensation of the function used in the EIS preprocessing notebook; auxiliary error-handling, logging, and sanity assertions are elided for clarity but do not affect the alignment outcome.

```
1 def estimate_line_centre(column_spectra, wavelength_grid, ref_centre,
2                           disagreement_tol=0.06, top_frac=0.10,
```

```
                          centroid_top_k=20):
    """
    Estimate the line centre of a single raster column.
        column_spectra : (n_rows, n_lambda) intensities for one column
        wavelength_grid: (n_lambda,) wavelengths in Angstrom
        ref_centre     : metadata reference line centre (Angstrom)
    Returns the estimated centre wavelength.
    """
    # 1. Robust bright-row selection: top 10% by peak intensity
    peaks       = column_spectra.max(axis=1)
    n_bright    = max(1, int(top_frac * len(peaks)))
    bright_idx  = np.argsort(peaks)[-n_bright:]
    bright_rows = column_spectra[bright_idx]

    # 2. Median profile across the bright rows
    profile     = np.median(bright_rows, axis=0)
    peak_bin    = int(np.argmax(profile))

    # 3. Quadratic subpixel refinement around the peak
    if 0 < peak_bin < len(profile) - 1:
        y_minus, y_zero, y_plus = profile[peak_bin-1:peak_bin+2]
        denom = (y_minus - 2*y_zero + y_plus)
        offset = 0.5 * (y_minus - y_plus) / denom if denom != 0 else 0.0
    else:
        offset = 0.0

    bin_step = wavelength_grid[1] - wavelength_grid[0]
    centre   = wavelength_grid[peak_bin] + offset * bin_step

    # 4. Sanity check vs metadata; centroid fallback if it disagrees
    if abs(centre - ref_centre) > disagreement_tol:
        top_idx        = np.argsort(peaks)[-centroid_top_k:]
        top_rows       = column_spectra[top_idx]
        weights        = np.clip(top_rows.mean(axis=0), 0, None)
        if weights.sum() > 0:
            centre = np.sum(weights * wavelength_grid) / weights.sum()
        # else: fall through with the quadratic estimate

    return centre


def align_raster_to_reference(raster, wavelength_grid, ref_centre,
                              ref_bin=None):
    """
    Apply per-column integer-bin shifts so each column's estimated
    line centre lands on the same reference bin.
    """
    n_rows, n_cols, n_lambda = raster.shape
    if ref_bin is None:
        ref_bin = n_lambda // 2

    aligned = np.full_like(raster, np.nan, dtype=np.float32)
```

```
55
56      for col in range(n_cols):
57          centre = estimate_line_centre(raster[:, col, :],
58                                        wavelength_grid, ref_centre)
59          centre_bin = int(round(
60              (centre - wavelength_grid[0]) /
61              (wavelength_grid[1] - wavelength_grid[0])))
62          shift = ref_bin - centre_bin
63          # integer-bin shift; out-of-range positions stay NaN
64          if shift > 0:
65              aligned[:, col, shift:] = raster[:, col, :n_lambda - shift]
66          elif shift < 0:
67              aligned[:, col, :n_lambda + shift] = raster[:, col, -shift:]
68          else:
69              aligned[:, col, :] = raster[:, col, :]
70      return aligned
```

Listing A.1: Per-column wavelength alignment with quadratic subpixel refinement and centroid fallback (condensed from the EIS preprocessing notebook).

The listing matches the prose description in Section 3.4.1 on every load-bearing point: top-10% bright-row selection (line 12); median profile (line 17); quadratic subpixel refinement around the peak (lines 21–26); the 0.06 Å disagreement threshold against the metadata reference (line 33); the centroid fallback over the top-k brightest spectra (lines 34–39); and the per-column integer-bin shift to a fixed reference bin (lines 52–69), with positions shifted out of range left as NaN. The reproducibility properties relied upon downstream (deterministic given the seed, no per-call randomness) follow from the absence of any stochastic operation in either function.

Design choices and rationale

The implementation makes seven design choices. Each is recorded here with its motivation, since the alignment is the foundation step on which every downstream EIS analysis depends.

1. Per-column estimation rather than a single global shift. Kamio et al. [32] document that EIS line-centre offsets vary systematically across raster columns due to the orbital motion of *Hinode* (a ∼98-minute period that modulates the spectrograph's thermal environment) and slow thermal drifts in the instrument itself. A single global shift averages these effects away and leaves residual per-column offsets of order one wavelength bin, which would otherwise dominate distance computations between spectra collected at different points in the orbit. Per-column estimation removes this systematic.

2. Bright-row selection (top 10% by peak intensity). Line-centre estimates are most accurate where signal-to-noise is highest. Dim rows in an EIS raster are dominated by detector read noise and continuum scatter, and including them biases the centre estimate toward a noise-weighted mean rather than the true line position. The top-10% threshold was chosen as a standard trade-off used in the EIS literature: small enough to exclude the noise-dominated tail, large enough that the resulting median over ∼50–100 rows is statistically stable.

3. Median rather than mean across bright rows. EIS data contain a non-trivial population of cosmic-ray hits, hot pixels, and rare detector glitches. The mean is sensitive to these outliers; the median is not. Using the median gives the same answer as the mean when the data are clean and reduces to the well-behaved central value when they are not, with no per-raster tuning required.

4. Quadratic subpixel refinement. Three-point parabolic interpolation around the discrete peak is the standard sub-bin centring technique in spectroscopic pipelines (e.g., `IRAF`, SolarSoft `cntrd`, and the EIS analysis routines distributed with SolarSoft). It is a closed-form expression with no optimisation, which avoids local-minimum failure modes; it is exact for a Gaussian core sampled symmetrically, which the EIS Fe lines approximate well at the chosen 23-bin crop.

5. 0.06 Å disagreement threshold against the metadata reference. The threshold is approximately three native EIS wavelength bins. Spacecraft drift documented by Kamio et al. [32] is at most ∼0.05 Å, so any quadratic-fit estimate that differs from metadata by more than 0.06 Å is more likely to reflect a fit failure (line blend, very low S/N, partial detector saturation) than a real drift, and triggers the more robust fallback rather than being trusted. This protects the alignment against silent degradation on a small minority of columns.

6. Intensity-weighted centroid fallback. When the parabolic fit is rejected, an intensity-weighted centroid over the top-k brightest spectra is used. The centroid trades sub-bin precision (which the rejected parabolic fit was already failing to deliver reliably) for robustness in the low-S/N or blended-line regime, since it does not depend on a single peak bin. The two estimators are agreement-checked against each other rather than blindly composed.

7. Integer-bin shift, not fractional-pixel interpolation. The alignment shift is rounded to the nearest integer bin and applied as an array roll, rather than resampling onto a finer grid. This preserves the original photon-count statistics in each retained bin (no smoothing-induced correlation between adjacent bins, no Nyquist artefacts), which matters because the downstream higher-order moment analysis in Chapter 5 is sensitive to bin-to-bin correlation. The cost is up to half-a-bin residual misalignment per column, which is small relative to the line FWHM (∼3–5 bins for the chosen Fe lines) and is absorbed by the symmetric crop in Section 3.4.2.

Validation. Alignment quality was verified post-hoc by inspecting the distribution of estimated centres against the metadata reference per line. For each of the three Fe lines, >99% of columns fell within the 0.06 Å tolerance using the primary parabolic estimate; the centroid fallback engaged on the remaining <1% of columns and produced estimates within tolerance on essentially all of those. No raster was rejected at the alignment stage on quality grounds.

A.6 Use of Generative AI

Generative AI tools (Anthropic's Claude Code) were used in a strictly cosmetic capacity during the preparation of this document: LaTeX typesetting and formatting (table layout, listing environment

configuration, cross-referencing structure, bibliography style fixes), spelling and grammar checks, and minor prose-level proofreading. Use was consistent with Newcastle University policy on the appropriate use of AI in learning. No AI tool was used to generate or paraphrase the substantive content of the dissertation: all writing, technical decisions, experimental design, hyperparameter choices, code, statistical analyses, interpretations, and conclusions presented are the author's own work.

A.7 Software Engineering Decisions

Three engineering decisions affected runtime more than any algorithmic choice. First, a local SSD copy of the EIS HDF5 file was used to bypass Google Drive's I/O bottleneck, which reduced clustering wall-clock from 30+ minutes to under 10. Second, BLAS thread-pool limits (`threadpool_limits(2)`) were enforced in `joblib` workers to prevent over-subscription on the 2-vCPU Colab runtime. Third, an explicit `FORCE_REFIT` toggle was added to the clustering pipeline so cached results from earlier parameter combinations could not be silently reused after a code change.

Appendix B

Solar Physics Background

This appendix provides the spectroscopic and physical context relevant to interpreting the classifier outputs reported in the main text. The material is grouped into pre-flare coronal observations, IRIS Mg II formation physics, and non-Gaussian flare line profiles.

B.1 Coronal Pre-flare Observations

Multiple observations show the solar corona undergoes measurable changes before flare onset, well before the hard X-ray impulsive phase.

Harra et al. [41] reported enhancements in non-thermal velocity, that is, excess spectral line broadening beyond thermal expectations, tens of minutes before flare onset, using UV and X-ray spectroscopy. In a subsequent study [45], the same authors used the EUV Imaging Spectrometer (EIS) aboard *Hinode* [29] to spatially localise pre-flare non-thermal velocity enhancements at the base of active regions in four M- and X-class flare events, linking them to possible flux rope activation.

Hudson et al. [46] discovered what they termed "hot X-ray onsets": enhanced isothermal plasma temperatures in the range 10–15 MK appearing during the initial soft X-ray increase, definitively before any detectable hard X-ray emission. This phenomenon occurs regardless of flare classification or configuration. Battaglia et al. [47] confirmed the existence of hot onsets using independent observations from the STIX instrument on Solar Orbiter, finding elevated temperatures of 11–16 MK from the very beginning of the flare, with steadily increasing emission measure indicating continued energy release.

Most directly relevant to this dissertation, To et al. [5] conducted the largest systematic EIS study of pre-flare non-thermal velocities, analysing 1,449 flares observed from 2011 to 2024. They found that non-thermal velocity consistently increases prior to the GOES soft X-ray start time across all flare classes, peaking at the flare peak and then gradually decreasing. For M-class flares, the precursor onset appears 24–45 minutes before the soft X-ray start time, with hotter emission lines (Fe XV 284.16 Å, Fe XVI 262.98 Å) showing enhancements approximately 20 minutes earlier than chromospheric or transition region lines. This dataset forms the basis of the EIS analysis presented in this dissertation, and the machine learning approach adopted here extends the Gaussian-fitting methodology of To et al.

B.2 IRIS Mg ii Formation Physics

The Interface Region Imaging Spectrograph (IRIS) [36] provides high-resolution near-ultraviolet spectra of the Mg II h&k resonance lines, which form at ∼6–10 kK and encode plasma velocities, temperatures, and densities across multiple atmospheric layers [37, 48, 49]. The k line at 2796.35 Å is optically thicker than h at 2803.53 Å, making h sensitive to slightly deeper chromospheric layers; comparison of the two provides height-resolved information used by the classifiers in Chapter 7 and the explainability analysis in Chapter 8.

B.3 Non-Gaussian Flare Line Profiles

EUV emission lines are conventionally modelled as Gaussian profiles, with width from thermal broadening plus a non-thermal component attributed to unresolved plasma motions. If the emitting ion population departs from a Maxwellian distribution (via particle acceleration, turbulence, or wave–particle interactions), the line profile becomes non-Gaussian.

Jeffrey et al. [8] provided the first observational evidence for this effect, analysing Fe XVI 262.976 Å line profiles during an M-class flare observed by *Hinode*/EIS. Using both a higher-order moments analysis and kappa-distribution fitting, they found that the line profiles in loop-top, footpoint, and ribbon regions were consistent with kappa distributions with $\kappa = 2$–3.3, and that many profiles had kurtosis values exceeding the Gaussian value of 3. Jeffrey et al. [9] extended this analysis to a major X-class flare, finding non-Gaussian profiles consistent with kappa distributions ($\kappa = 3$–7) that varied systematically in space and time: lower κ (more non-thermal) at footpoints near hard X-ray emission, and higher κ (closer to thermal) in the corona. Polito et al. [42] reported similar findings for hot flare lines.

A critical observation for this dissertation is that all previous studies of non-Gaussian EIS line profiles focused exclusively on the *impulsive phase* of flares, the period of active energy release. Whether non-Gaussian profiles are present in *pre-flare* spectra, before any detectable flare onset, has not been previously investigated. Chapter 5 of this dissertation addresses this gap by applying higher-order moment analysis (kurtosis and skewness) to pre-flare EIS spectra and demonstrating that non-Gaussianity is a powerful pre-flare diagnostic.

B.4 Kappa-Distribution Validation of the Kurtosis Discriminant

To validate the physical basis of the kurtosis-based classification reported in Chapter 5, kappa distributions [8] were fitted directly to 100,000 spectra per Fe line and compared against Gaussian fits via the Bayesian Information Criterion. For Fe XV 284.16 Å, pre-flare spectra have a median $\kappa \approx 10$ (near-Gaussian) while no-flare spectra have median $\kappa \approx 3$ (strongly non-thermal), with the kappa fit preferred over the Gaussian for 57% of pre-flare and 37% of no-flare spectra. The fitted κ values are flat with time-to-flare ($\rho < 0.01$), confirming that the non-Gaussianity difference is a persistent property. Notably, the direction is opposite to what Jeffrey et al. found during flares: pre-flare active regions have *more* Maxwellian line profiles, consistent with a more ordered, less turbulent corona.

B.5 Physical Interpretation of the Coronal Pre-flare Signal

The Siamese metric-learning analysis (Chapter 4) identifies the line widths of the three Fe lines as the dominant discriminative features. The two hottest lines (Fe XIV and Fe XV, $T \approx 2.0$–2.1 MK) are *narrower* in pre-flare active regions, while the cooler Fe XII ($T \approx 1.3$ MK) is slightly *broader*. In the coronal context, EUV line widths are dominated by non-thermal broadening from unresolved plasma motions [50]. The narrower hot-line profiles could indicate a more ordered, less turbulent corona, consistent with the confined, stressed magnetic geometry of active regions building toward eruption [51]. The broader Fe XII profiles and their slight redshift may reflect enhanced downflows at $\sim$1.3 MK from cooling plasma draining from coronal loops [52]. The poor-observation analysis (Chapter 4) corroborates this: misclassified observations are those whose Fe XII profiles resemble non-flaring spectra, even when their hot-line profiles are partially pre-flare-like.

List of Acronyms

AI Artificial Intelligence
AIA Atmospheric Imaging Assembly (aboard SDO)
API Application Programming Interface
AR Active Region
ARI Adjusted Rand Index
AUC Area Under Curve
BFGS Broyden–Fletcher–Goldfarb–Shanno (optimiser)
BIC Bayesian Information Criterion
BLAS Basic Linear Algebra Subprograms
CCA Canonical Correlation Analysis
CI Confidence Interval
CNN Convolutional Neural Network
CPU Central Processing Unit
CV Cross-Validation
DN Data Number (instrument flux unit)
DNN Deep Neural Network
EIS EUV Imaging Spectrometer (aboard *Hinode*)
ESA European Space Agency
EUV Extreme Ultraviolet
EUVST Extreme Ultraviolet Spectroscopic Telescope (aboard Solar-C)
FITS Flexible Image Transport System (astronomy file format)
FP False Positive
FWHM Full Width at Half Maximum
GELU Gaussian Error Linear Unit
GOES Geostationary Operational Environmental Satellite
GPU Graphics Processing Unit
Grad-CAM Gradient-weighted Class Activation Mapping
HEK Heliophysics Events Knowledgebase
HMI Helioseismic and Magnetic Imager (aboard SDO)
IRIS Interface Region Imaging Spectrograph
kNN k-Nearest Neighbours
KS Kolmogorov–Smirnov (test)
LEAPS Leiden/ESA Astrophysics Programme for Summer Students
LMSAL Lockheed Martin Solar and Astrophysics Laboratory

LR Logistic Regression
MK Megakelvin (10^6 K)
ML Machine Learning
MLP Multi-Layer Perceptron
MMD Maximum Mean Discrepancy
MRR Mean Reciprocal Rank
MSE Mean Squared Error
NF No-Flare (non-flaring active-region spectra)
NMI Normalised Mutual Information
NN Neural Network
NOAA National Oceanic and Atmospheric Administration
NPZ NumPy compressed array file (`.npz`)
OT Optimal Transport
PCA Principal Component Analysis
PF Pre-Flare
PR Precision-Recall
PR-AUC Area Under the Precision-Recall Curve
RAM Random-Access Memory
ROC Receiver Operating Characteristic
ROC-AUC Area Under the Receiver Operating Characteristic Curve
RP Random Projection
SDO Solar Dynamics Observatory
SHAP SHapley Additive exPlanations
SHARP Space-weather HMI Active Region Patches
SMOTE Synthetic Minority Oversampling Technique
STIX Spectrometer/Telescope for Imaging X-rays (aboard Solar Orbiter)
TN True Negative
TNR True Negative Rate
TP True Positive
TPR True Positive Rate
TTF Time-To-Flare
UMAP Uniform Manifold Approximation and Projection
UV Ultraviolet
XGBoost Extreme Gradient Boosting

Glossary of Terms

active region A region of the solar atmosphere with strong, complex magnetic fields, frequently the site of flares and coronal mass ejections

Å (Angstrom) Unit of length equal to 10^{-10} m, the standard wavelength unit in UV and EUV spectroscopy

area normalisation Normalisation of a spectrum by dividing by its integrated area, preserving spectral shape and discarding absolute intensity

chromosphere The cooler ($\sim 10^4$ K) atmospheric layer between the photosphere and the corona, observed in the Mg II h&k lines by IRIS

Cohen's d A standardised measure of the difference between two distributions' means in units of pooled standard deviation; $|d| = 0.8$ is conventionally a large effect

contrastive loss A loss function used to train Siamese networks that pulls embeddings of same-class pairs together and pushes different-class pairs apart by at least a margin

corona The hot, tenuous outer atmosphere of the Sun, with temperatures of ∼1–2 MK in active regions, observed in EUV by EIS

embedding A learned, typically low-dimensional vector representation of an input designed to make downstream tasks (classification, retrieval) easier

enrichment ratio The ratio of pre-flare to no-flare class proportions within a cluster, divided by the global ratio; values > 1 indicate pre-flare enrichment

F1 score The harmonic mean of precision and recall; ranges from 0 (worst) to 1 (perfect), and is a useful single-value classification metric under class imbalance

Fe xii The 11-times ionised iron line at 195.12 Å, formed at ∼1.3 MK

Fe xiv The 13-times ionised iron line at 264.79 Å, formed at ∼2.0 MK

Fe xv The 14-times ionised iron line at 284.16 Å, formed at ∼2.1 MK; the dominant discriminative coronal feature in this work

Gini coefficient A measure of inequality in a distribution; in this work it quantifies how concentrated the pre-flare classifier's spatial detections are within an active region (0 = uniform, 1 = single pixel)

kappa distribution A non-Maxwellian particle velocity distribution parameterised by κ, where $\kappa \to \infty$ recovers the Maxwellian; lower κ values indicate stronger non-thermal tails

kurtosis The fourth standardised moment of a distribution, measuring tail-heaviness; a Gaussian has kurtosis = 3; values > 3 indicate heavier tails (leptokurtic)

Mg ii h&k lines The chromospheric resonance doublet of singly-ionised magnesium at 2796.35 Å (k) and 2803.53 Å (h), formed at ∼6–10 kK

non-thermal velocity The component of EUV emission line broadening beyond thermal

Doppler broadening, attributed to unresolved plasma motions and/or non-Maxwellian particle distributions

observation-disjoint splitting A train/validation/test partitioning scheme where every spatial pixel from a given EIS or IRIS raster is assigned to a single split, preventing within-raster pixel-level information leakage

photosphere The visible surface of the Sun, where most magnetogram observations are made

pre-flare window The time interval before a flare's GOES start time used in this work for classification (typically 0–180 min for EIS, 0–300 min for IRIS)

raster A 2-D spectroscopic observation built up by stepping a slit across the solar surface, where each spatial pixel has a full spectrum at one wavelength range

Siamese network A pair of identical neural-network encoders with shared weights trained to map similar inputs to nearby points in an embedding space

skewness The third standardised moment of a distribution, measuring asymmetry; positive skew indicates a longer right tail

solar flare A sudden release of magnetic energy from the solar corona, classified by GOES soft X-ray peak flux into A, B, C, M, and X classes

Spearman ρ A non-parametric rank-correlation coefficient measuring the monotonic relationship between two variables

Wasserstein-2 distance A distance between probability distributions equal to the minimum cost of transporting one to the other under squared-Euclidean cost, quantifying both magnitude and direction of distributional shift

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